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# UNIT 1 DESCRIPTIVE STATISTICS

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## 1.1 INTRODUCTION

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Most of us associate 'statistics' with the bits of data that appear in news reports: Cricket batting averages, imported car sales, average high temperature on a particular day etc. Advertisements often claim that data show the superiority of the advertiser's product. The word statistics, which is derived from the word 'state', entered the English vocabulary in the eighteenth century. It was used then, and still is used, to mean one or more sets of numerical data on various items like population, taxes, wealth, exports, imports, crop production, etc., which are of interest to state officials. There are two ways to use the word statistics. If we say 'statistics is', we are generally referring to the science of statistics. If we say 'the statistics are', we are referring to numbers such as batting averages, the number of unemployed during the month of October, or the number of deaths from malaria during a given year. It is hard to come up with a concise definition of statistics because it is a broad subject that has many facets. Commonly, it is believed that statistics involves the collection, organisation, analysis, and interpretation of data.

There are several reasons why the scope of statistics and the need to study the subject statistics have grown enormously in the last fifty years. One reason is the increasingly quantitative approach employed in all the sciences, as well as in business and many other activities which directly affect our lives. This includes the use of mathematical techniques in the evaluation of anti-pollution controls, in inventory planning, in the analysis of traffic patterns, in the evaluation of teaching techniques and so forth. The other reasons are that the volume of data that is collected, processed and disseminated to the public for one reason or the other has increased almost beyond comprehension. More and more persons with some knowledge of statistics are therefore needed to take an active part in the collection and analysis of the data, as also in all of the preliminary planning. One question that naturally arise at this stage is why and how such large volume of data should be collected, organised and analysed? We shall address such questions in this unit.

We shall begin by discussing the need for collecting data. We shall then talk about different types of data and the ways of arranging them to obtain any logical conclusions from them. Graphical methods of presenting data are also discussed. Finally, we shall discuss various measures that are commonly used to summarise information contained in a data set.

**Objectives:** After studying this unit you should be able to

- organise the given set of data in a meaningful way;
- construct a frequency distribution of a variable from a given set of data;
- represent graphically a frequency distribution of a variable and interpret the information suggested by it;
- obtain the 'average level' of a given set of data where its frequency distribution is centred by calculating its mean, median or mode;
- obtain the degree to which the individual measurements in a given data vary about this average by calculating its variance and standard deviation.

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## 1.2 COLLECTING DATA

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We start with a few examples which provide you a general idea about situations where we need to handle a large amount of data and where statistics can play a significant role. In these examples, we try to raise issues, which can be handled adequately by the various statistical tools with which we will be introduced to as we go along in this course.

**Example 1:** Suppose we are in the process of drawing up a comprehensive plan for developing public medical facilities in a big city. To work out the rates to be charged for various services proposed to be offered by the hospitals to be set up under the plan, it is necessary to know the economic conditions of the one million households constituting the entire **population** of the city. For this purpose, we need to divide the households which are the **individuals** about which information is sought, into three broad categories - High Income Group (HIG), Middle Income Group (MIG) and Low Income Group (LIG) - according to certain criteria. How do we proceed? Obviously, it would appear as if the task would involve visiting each of the one million households to enquire about, say their monthly incomes. What are the issues and difficulties in implementing this proposal?

There are many; for instance:

- a) visiting one million households - and in many cases, repeat visits may have to be made following non-availability of respondents at the time of visit, is a time consuming affair and we may not have enough time at hand;
- b) to cover such a large number of households, we may have to employ a very large number of investigators and that will mean a lot of expenditure;
- c) even if we are in a position to afford the time and money that we need to cover all the households, two more issues crop up:
  - i) As the survey may take a long time the figures on monthly income of households covered in the beginning of survey and those of the households covered later may refer to different time periods.
  - ii) the data that we will be collecting will be a series of one million figures and obviously such a series as it is will hardly make any sense. We would need proper methodology to properly compile, to analyse and to interpret the data so as to make some sense out of these.

In view of (a) and (b), it seems that the process of complete enumeration of all households will not be really efficient and we will have to think of an alternative procedure. To obviate difficulties mentioned in (a) and (b), we might consider only a subset of the population viz., a **sample** of households and collect information on their earnings only. Then try to use the fractional information thus collected to make a guess about the actual state of affairs pertaining to the totality of households.

Now, of course, this method will reduce the cost and time required for the survey and we will have a shorter series of figures. However, issues enumerated in (c) remain to be handled. Moreover, several other new issues like the following would come up now,

Population is a collection of all the individuals we are studying

Sample is a collection of some, but not all, of the individuals of the population under study, used to describe the population

- d) how do we select our sample?
- e) how do we relate the sample findings to the state of affairs in respect of the entire population of one million households?
- f) also, even if we are in a position to make such an assessment on the basis of the information contained in the sample, how much reliability should we attach to it, since after all the assessment is based on our observations in a single sample which is perhaps comprised of a small portion of the totality and the same could be quite different if we had a different sample. This aspect surely introduces an element of 'uncertainty' in our conclusion as different samples may differ in respect of their information contents and thus may lead to different conclusions relating to the same population. How do we handle the situation then?

\* \* \*

Thus, though the problem initially looked rather simple, the issues involved are not so by any standard. Solutions of such issues come under the purview of statistics.

The population i.e. the totality of households in the above example is finite and observable. However, often the population is neither finite nor physically existent; rather, it may only be conceivable or hypothetical.

**Example 2:** Consider a system where 'customers' arrive at a 'counter' for 'service'. 'Customers' may be patients coming to a clinic for medical attention or may be aircrafts waiting for clearance from the air traffic control to take off or even broken down machines in a factory waiting for the attention of an operator and so on. Our objective is to prescribe a policy so that congestions can be avoided. However, neither the number of arrivals is fixed on all occasions nor is the service time the same for all customers - these are usually uncertain and thus subject to chance factors. How do we then propose to proceed?

\* \* \*

**Example 3:** Suppose a new brand of pain reliever has been marketed recently. The manufacturer claims that it relieves pain 25% time faster than any of the comparable brands already available in the market. How do we propose to verify this claim? Obviously, we have to administer these drugs to a sample of individuals. But then how should the sample be chosen? Also, individuals may react differently to the same drug. How do we take this into account? How do we process the sample data? Finally, the good old question - to what extent can we generalise our sample findings so as to be able to come up with a conclusion pertaining to the entirety? How reliable is the sample finding in this case?

\* \* \*

**Example 4:** The yield of a certain crop is dependent on the location of the plot, amount of fertilisers applied, amount of rainfall, availability of irrigation facilities etc. However, it is also known that even when all these factors are applied equally, the yields may still vary. This is so because the factors we listed may not exhaust all the factors that influence the yield. Given such a situation can we build up an appropriate forecasting formula involving the identifiable factors so that the yield can be predicted adequately? Even if we can come up with such a formula, can we judge how good the formula is?

\* \* \*

Many such examples may be cited to illustrate the areas of application of statistics. However, it will be gradually apparent that the basic issues involved in these illustrations are similar and can be discussed within a broad framework and this framework is provided by statistics.

To study a given phenomenon, the basic raw material would be data (information) relating to it. For example to study the growth in the use of telephones, we may collect

the number of telephones that several workers install on a given day or that one worker installs per day over a period of several days. These figures then constitute our data for studying growth in the use of telephone.

Observation before it is arranged and analysed is called raw data. For data to be useful, our observations need to be organised so that we can pick out trends and come to logical conclusions. We shall thus discuss the techniques of arranging data in tabular and graphical forms to be able to make genuine sense out of it. First, we shall start with describing different kinds of data.

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E1) Cite at least two examples from your own experience illustrating the application of statistics

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### 1.2.1 Kinds of Data

The operation of collection of relevant data comes in the initial phase of any statistical study. Data relevant for a study can be obtained either from published works or from the collections of the government or research organisations or through direct fieldwork. The mode of collection of data and the methodology for analysing the same comprise the core of statistical discipline. Statisticians gather data from a sample. They use this information to make inferences about the population that the sample represents. Thus sample and population are relative forms. A population is a whole, and a sample is a fraction or segment of that whole.

Whenever the data are numerical, then the corresponding characters are called **variables**. Thus, the number of items failing to meet specifications in a lot of 100 items, the daily number of customers visiting a particular shop, the hourly number of telephone calls received by an operator, the life (in hours) of an electric bulb etc. are all examples of variables.

In some cases, however, the data may not be numerical in nature. This will be the case when, for example, one is examining a lot of manufactured items and classifying them as either 'good' or 'bad'. Similarly, if each member of a group of individuals is asked to say 'yes' or 'no' according as his/her monthly income is at least Rs.5000/- or less than Rs.500/- a month, the resulting data will not be numerical. This exercise will produce only a series of "yes"es and "no"s. However, in both these instances, the data are easily convertible to numerical terms. For instance, in the last example, we may code a "yes" as 1 and a "no" as 0. Such characteristics (e.g. quality of manufactured items or the salary position) are called **attributes**.

You may note here that in the first three examples given above, the variables take on only some isolated values; for instance, the number of items failing to meet specifications in a lot of 100 items can be 0 or 1... or at most 100, i.e., a whole number between 0 and 100, and never a figure like 3.7, say. Such variables which take on only isolated values (which are often integers, but need not be such always) are called **discrete** or **discontinuous** variables. This kind of data result from counts and therefore, values jump from one point to the next with no possible measures in between. On the other hand, certain variables are such that they may take on any value along a suitable scale. For instance, the distance (in meters) between two points, in the interval between 210.5 and 320.6 can take any value like 210.56, 210.687, 315.685 and so on. Such variables are called **continuous variables**. Variables representing height, weight, age, time to complete a journey, temperature etc. are all of this variety. Although continuous data can take on a theoretically infinite number of possible measures, the values that we use in practice are determined by

- i) the precision of our measuring instrument and;
- ii) how precise we need to be.

For example, if you measure the length of a new born baby, you would record it as 48



cm., 49 cm., etc. For most practical purposes we will not be required to measure a baby's length to the fifth decimal place.

We now present some data to illustrate the concepts of attributes and variables:

**Example 5:** The following table is a summarised version of data relating to the educational background at graduation level of students admitted in the MBA Programme of a certain college. Here the educational background of students admitted is an attribute which can be either arts, science, commerce, engineering or any other branch.

**Table 1: Educational Background of Students Admitted to the MBA Programme in a College**

| Graduation Background | Year          |              |               |              |                |
|-----------------------|---------------|--------------|---------------|--------------|----------------|
|                       | 1994-95       | 1995-96      | 1996-97       | 1997-98      | 1998-99        |
| Arts                  | 5<br>(4.0)    | 7<br>(7.1)   | 10<br>(10.7)  | 15<br>(10.1) | 18<br>(11.25)  |
| Science               | 10<br>(8.0)   | 5<br>(5.1)   | 12<br>(8.1)   | 17<br>(11.5) | 10<br>(6.25)   |
| Commerce              | 5<br>(4.0)    | 3<br>(3.1)   | 12<br>(8.1)   | 16<br>(10.8) | 12<br>(7.5)    |
| Engineering           | 103<br>(84.0) | 83<br>(84.7) | 109<br>(73.1) | 98<br>(66.2) | 118<br>(73.75) |
| Others                | -             | -            | -             | 2<br>(1.4)   | 2<br>(1.25)    |
| Total enrolled        | 123           | 98           | 149           | 148          | 160            |

(Figures in parentheses are percentages of the total number of students enrolled)

\* \* \*

**Example 6:** The following table gives the raw data relating to the marks (out of 10) of 100 students in a statistics examination.

**Table 2: Marks of 100 Students in a Statistics Examination**

|   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 5 | 0 | 5 | 7 | 6 | 6 | 7 | 4 | 8 |
| 4 | 6 | 7 | 3 | 6 | 6 | 5 | 6 | 2 | 6 |
| 6 | 4 | 5 | 7 | 4 | 4 | 7 | 4 | 6 | 4 |
| 3 | 4 | 8 | 1 | 5 | 8 | 7 | 5 | 7 | 7 |
| 7 | 6 | 5 | 7 | 4 | 5 | 5 | 3 | 6 | 6 |
| 5 | 8 | 6 | 6 | 7 | 7 | 3 | 4 | 3 | 5 |
| 9 | 4 | 8 | 5 | 3 | 5 | 9 | 5 | 5 | 7 |
| 1 | 9 | 3 | 5 | 5 | 7 | 6 | 8 | 8 | 2 |
| 5 | 4 | 4 | 4 | 6 | 3 | 5 | 6 | 4 | 4 |
| 8 | 2 | 8 | 5 | 5 | 6 | 7 | 3 | 6 | 9 |

Here the marks in the test is a variable, which varies from one student to another. Since the marks are in whole numbers between 0 and 10, it is a discrete variable. The data in the given format is called **ungrouped data**.

\* \* \*

The way the marks have been reported does not help us in knowing who scored how much. Perhaps that is not important for the purpose of the study at this stage. We are interested in knowing how the performance was in general? Was the test very simple in the sense that a large proportion of students scored high marks? Was the test too difficult in the sense that a large proportion of students scored very low marks? It may not be easy for you at this moment to answer such questions with the data available.

To respond to such questions, we have to present the data in an organised manner. We shall take up this example again in the next section and try to find answers to the above questions but before that, let us look at the data corresponding to continuous variables.

**Example 7:** Consider the following data, which relate to life (in hours) of 100 electric bulbs.

**Table 3: Lives of 100 Electric Bulbs**

|        |        |        |        |        |
|--------|--------|--------|--------|--------|
| 511.6  | 977.7  | 600.2  | 1099.7 | 803.7  |
| 923.4  | 1108.3 | 906.7  | 759.6  | 1111.9 |
| 918.3  | 1051.1 | 992.5  | 817.2  | 665.3  |
| 1143.6 | 948.4  | 939.8  | 1163.0 | 715.2  |
| 936.1  | 750.5  | 991.2  | 1199.5 | 950.2  |
| 1061.7 | 1027.7 | 995.1  | 966.5  | 1146.5 |
| 848.0  | 956.8  | 1100.0 | 955.2  | 1023.0 |
| 900.5  | 982.3  | 699.2  | 1069.8 | 1245.3 |
| 1059.5 | 1091.0 | 850.7  | 1219.3 | 1012.6 |
| 1053.2 | 939.5  | 777.8  | 749.6  | 980.8  |
| 991.3  | 1016.3 | 930.4  | 1242.2 | 1131.4 |
| 1314.7 | 1137.2 | 763.1  | 1394.4 | 117.3  |
| 1204.1 | 980.1  | 922.3  | 1057.7 | 907.2  |
| 808.0  | 857.7  | 1127.1 | 934.3  | 1262.3 |
| 965.4  | 873.4  | 955.1  | 806.5  | 1033.0 |
| 1068.3 | 950.3  | 930.6  | 1000.1 | 898.5  |
| 1293.1 | 940.9  | 1293.8 | 1035.2 | 706.0  |
| 880.9  | 912.2  | 803.5  | 922.6  | 846.1  |
| 1092.3 | 1182.0 | 985.2  | 945.3  | 835.0  |
| 1001.5 | 1048.8 | 895.1  | 1067.2 | 1062.8 |

Here the life of a bulb is a variable which varies from one bulb to another. In this example, the figures have been recorded correct to 1/10th of an hour. All values larger than 511.55 but not larger than 511.65 have been approximated and represented by 511.6. Although, conceivably the life could be any value on the continuous scale from zero to infinity, the limitation on the part of the measuring instrument invariably imposes an artificial discreteness in the data. This is due to limitations of measuring instruments. For instance, in this case the scale could measure only up to the first place of decimal; but, if we had a scale which could measure up to, say five places of decimals, the accuracy would have increased. No matter how fine the scale is, one would have to stop after a finite number of places after the decimal point and as such a discreteness would any way creep in. But, theoretically, the variable under consideration is continuous in character as it can adopt any value over a specified interval, finite or otherwise.

\* \* \*

You may now try the following exercises

- 
- E2) Identify each of the following as a population or as a sample.
- All the adult males residing in India.
  - Twenty cancer patients chosen to participate in a program to test a new drug.
  - All the AIDS patients who could conceivably be given a new treatment for the disease.
- E3) Data on the variables below were recorded for a study in a school in Delhi. Which of these are continuous, and which are discrete?
- age
  - year of birth
  - height
  - number of students admitted to the school in a calendar year

E4) Give two examples of situations that would yield continuous data.

Let us now see how the raw data in Examples 6 and 7 can be arranged to obtain any logical conclusion from them and obtain an answer to the questions asked earlier.

### 1.2.2 Frequency Distribution of a Variable

From the data in Example 6 for instance, it is not possible to readily answer questions like how many students scored 3 and above or how many of them scored between 5 and 7 or what is the score obtained by most of the students. It will be easier to answer such questions if we compress the data in the form of a Table 4 given below. Here we put a tally mark against a score as soon as we come across the same as we go along the list (data) in any given order. For instance, if we read the data column wise then first entry in first column is 2, so we put a tally mark against 2 as shown in Table 4. Number of tally marks against 2 indicate the number of students getting 2 marks. You may observe here in this table that every fifth tally is a slash over the first four tally marks.

**Table 4: Tally marks against marks scored**

| Scores | Tally Marks                                                       |
|--------|-------------------------------------------------------------------|
| 0      | /                                                                 |
| 1      | //                                                                |
| 2      | ////                                                              |
| 3      | /// <del>///</del> ////                                           |
| 4      | /// <del>///</del> <del>///</del> <del>///</del>                  |
| 5      | /// <del>///</del> <del>///</del> <del>///</del> <del>///</del> / |
| 6      | /// <del>///</del> <del>///</del> <del>///</del> <del>///</del>   |
| 7      | /// <del>///</del> <del>///</del>                                 |
| 8      | /// <del>///</del> ////                                           |
| 9      | ////                                                              |

Finally we count the tallies for each variable value (i.e. score) to obtain the number of times it appears in the data set. We call this number the **frequency** of the variable.

**Definition:** The frequency of any value occurring in a data set is the number of times the value occurs in the set.

Once we know the frequency with which the values occur in a data set, we can construct a table showing the frequency of each value against it. We call this table a **frequency distribution**. Table-5 gives the frequency distribution for the data set in Example 6.

**Table-5: Frequency Distribution of marks of 100 students**

| Scores | Frequency | Relative Frequency |
|--------|-----------|--------------------|
| 0      | 1         | .01                |
| 1      | 2         | .02                |
| 2      | 4         | .04                |
| 3      | 9         | .09                |
| 4      | 15        | .15                |
| 5      | 21        | .21                |
| 6      | 20        | .20                |
| 7      | 15        | .15                |
| 8      | 9         | .09                |
| 9      | 4         | .04                |
| Total  | 100       | 1.00               |

Table 5 above gives a **grouped version** of the data in Table 2. An entry in the second column gives the number of students receiving the corresponding score that is depicted in the same row under the first column. Thus, there are 10 groups (classes or

categories) in the table above. Each group corresponds to a single value of the underlying variable. Note that the third column gives the proportion of students receiving the various scores called the **relative frequency** of the value.

**Definition:** Relative frequency of a value occurring in a data set is the frequency of a value as a fraction or a percentage of the total number of observations.

Now suppose we want to see how many students got less than 6 marks. Then we have to add up the frequencies of 0, 1, 2, 3, 4 and 5 marks to obtain the number of students getting marks less than 6. This gives us the **cumulative frequency** of the 'less than' type of 6. The cumulative totals of the frequencies obtained by proceeding from the top class of the table downwards are called the 'less than' type cumulative frequencies. Similarly, if we add up the frequencies from the bottom class upwards, we get the cumulative frequencies of the 'more than' type. We have given both these types in Table 6 for the data under consideration.

**Table 6: Cumulative Frequency of the marks of 100 students:**

| Score | Cumulative Frequency | Cumulative Frequency |
|-------|----------------------|----------------------|
|       | (Less than type)     | (More than type)     |
| 0     | 1                    | 100                  |
| 1     | 3                    | 99                   |
| 2     | 7                    | 97                   |
| 3     | 16                   | 93                   |
| 4     | 31                   | 84                   |
| 5     | 52                   | 69                   |
| 6     | 72                   | 48                   |
| 7     | 87                   | 28                   |
| 8     | 96                   | 13                   |
| 9     | 100                  | 4                    |

For instance, the less than type cumulative frequency of 5 is 53. This means that 53 students scored 5 or less marks. Also, the greater than type cumulative frequency of 5 is 68. Thus, there are 68 students who scored 5 or more marks. From the table, we can also see that the number of students scoring above 5 but not above 8 is  $96 - 53 = 43$ . We call the data presented in Table 6, grouped. This is because we have grouped together all the observations having the same value.

We would like to mention here that the operation of grouping the observations is not unique. For example, consider the following grouping of the set of observations with the values in a group as given in Table 7 below.

**Table 7: Frequency Distribution of marks of 100 students**

| Scores | Frequency | Cumulative Frequency |                |
|--------|-----------|----------------------|----------------|
|        |           | Less than type       | More than type |
| 0-1    | 3         | 3                    | 100            |
| 2-3    | 13        | 16                   | 97             |
| 4-5    | 36        | 52                   | 84             |
| 6-7    | 35        | 87                   | 48             |
| 8-9    | 13        | 100                  | 13             |
| Total  | 100       |                      |                |

The grouping in Table 7 informs us that 35 students scored 6 or 7 marks in the test, but it does not give us the exact number of students receiving each of these marks. This shows that some information has been lost in this process of classifying observations into classes. Of course, as we have seen, this problem is avoidable in case of discrete variables by making each class correspond to a single value of the variable provided, the number of distinct values that the variable assumes is not too large as is the case with Example 6 discussed above.

But, summarisation of data at the expense of loss of such information becomes almost unavoidable in the case of a continuous variable. In this case if we take a class corresponding to each different value taken by the variable, then the resulting number of classes will be unduly large. This approach will then look artificial since a continuous variable, as you know, by definition is capable of assuming any of the values represented in a relevant interval. In such a case, we decide on the mode of classification depending on the nature of the data and the purpose of the study. Additionally, the following points are of help and can be taken care while deciding on the classes;

- i) each class should correspond to an interval of values of the variable;
- ii) the classes should be non-overlapping and exhaustive i.e. an observation must be included in exactly one of the classes;
- iii) the number of classes should not be too small since otherwise the actual nature of the distribution may be difficult to visualise and thus the summarisation will fail to bring out the actual characteristics of the distribution;
- iv) the number of classes should also not be too large;
- v) classes should preferably be of equal width, since otherwise the frequencies of various classes may not be comparable.

As a rule of thumb, the number of classes should be between 10 and 20 wherever the total frequency is more than 1000. Let us now take up the data in Table 3, corresponding to Example 7 to illustrate the points made above. You may observe here that the smallest and the largest values are 511.6 and 1394.4 respectively. Here we take classes as given in Table 8 below.

**Table 8: Tally marks of lives of 100 electric bulbs**

| Life (hours)<br>(inclusive of end point) | Tally marks           |
|------------------------------------------|-----------------------|
| 510.6-590.5                              | /                     |
| 590.6-670.5                              | //                    |
| 670.6-750.5                              | ///                   |
| 750.6-830.5                              | /// ///               |
| 830.6-910.5                              | /// /// ///           |
| 910.6-990.5                              | /// /// /// /// /// / |
| 990.6-1070.5                             | /// /// /// ///       |
| 1070.6-1150.5                            | /// /// //            |
| 1150.6-1230.5                            | /// /                 |
| 1230.6-1310.5                            | /// /                 |
| 1310.6-1390.5                            | /                     |

**Note** that here the values have been recorded correct to 1/10 th of an hour. The life  $x$  hours of a bulb with  $510.55 < x \leq 510.65$  has been recorded as 510.6; the class 510.6 - 590.5 actually includes all bulbs with life  $x$  hours satisfying  $510.55 < x \leq 590.55$ . While 510.6 and 590.5 are respectively called the lower and upper **class limits** 510.55 and 590.55 are referred to as the lower and upper **class boundaries**. Similar is the case with the other classes. Once the classes have been determined, the frequency distribution can now be obtained exactly as before.

The difference between the upper and lower boundaries of a class is called the **class interval**. The mid point of a class is called its **class mark**. Thus  $(510.55 + 590.55)/2 = 550.55$  is the class mark of the first class. The last column of Table 9 gives **frequency density** of a class. It is the frequency per unit length of the class interval. For the first class, frequency density is  $\frac{1}{590.55 - 510.55} = 0.0125$  and so on.

**Table 9: Frequency Distribution of Lives of 100 Electric Bulbs**

| Life (hours): Class boundaries<br>(exclusive of left boundaries but inclusive of right boundaries) | Frequency | Relative frequency | Cumulative frequency |                  | Frequency density |
|----------------------------------------------------------------------------------------------------|-----------|--------------------|----------------------|------------------|-------------------|
|                                                                                                    |           |                    | 'Less than' type     | 'More than' type |                   |
| 510.55-590.55                                                                                      | 1         | .01                | 1                    | 100              | 0.0125            |
| 590.55-670.55                                                                                      | 2         | .02                | 3                    | 99               | 0.025             |
| 670.55-750.55                                                                                      | 5         | .05                | 8                    | 97               | 0.0625            |
| 750.55-830.55                                                                                      | 8         | .08                | 16                   | 92               | 0.1               |
| 830.55-910.55                                                                                      | 13        | .13                | 29                   | 84               | 0.1625            |
| 910.55-990.55                                                                                      | 26        | .26                | 55                   | 71               | 0.325             |
| 990.55-1070.55                                                                                     | 20        | .20                | 75                   | 45               | 0.25              |
| 1070.55-1150.55                                                                                    | 12        | .12                | 87                   | 25               | 0.15              |
| 1150.55-1230.55                                                                                    | 6         | .06                | 93                   | 13               | 0.075             |
| 1230.55-1310.55                                                                                    | 6         | .06                | 99                   | 7                | 0.075             |
| 1310.55-1390.55                                                                                    | 1         | .01                | 100                  | 1                | 0.0125            |
| Total                                                                                              | 100       | 1.00               |                      |                  |                   |

And now some exercises for you.

- 
- E5) The number of nurses on duty each day at a hospital are grouped into a distribution having the classes 20-34, 35-49, 50-64, 65-79 and 80-94. Find
- the class limits
  - the class boundaries
  - the class marks
  - the class interval of the distribution.
- E6) The total scores  $X$  obtained by 50 students in a psychology test of 100 marks are given below.

|    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|
| 75 | 89 | 66 | 52 | 90 | 68 | 83 | 94 | 77 | 60 |
| 38 | 47 | 87 | 65 | 97 | 49 | 65 | 70 | 73 | 81 |
| 85 | 77 | 83 | 56 | 63 | 79 | 69 | 82 | 84 | 70 |
| 62 | 75 | 29 | 88 | 74 | 37 | 81 | 76 | 74 | 63 |
| 69 | 73 | 91 | 87 | 76 | 58 | 63 | 60 | 71 | 82 |

Answer the following question on the basis of the data given above.

- Is the random variable  $X$  = Score of a student, discrete or continuous? What are the minimum and maximum scores?
- Using the classes 20 – 29, 30 – 39, 40 – 49, ... and 90 – 99 draw up the frequency distribution of  $X$ .
- What percentage of the students score above the pass marks of 50?
- How many of the students score between 50 and 79?

Consider the following frequency distribution of income of 1000 individuals belonging to a particular section of the population:

| Income (Rs.) | Frequency |
|--------------|-----------|
| $\leq 1000$  | 40        |
| 1000-2000    | 55        |
| 2000-4000    | 141       |
| 4000-6000    | 152       |
| 6000-10000   | 275       |
| 10000-15000  | 199       |
| 15000-25000  | 103       |
| $\geq 25000$ | 35        |

- What percentage of people earn more than Rs.4,500?
- What percentage of people earn at least Rs.1,500?
- What percentage of people have earnings between Rs.2,000 & Rs.5000?

We can thus say that frequency distributions condense large sets of data and display them in an 'easy to understand' form. Graphical methods are also used for presenting data and ideas. Graphical methods provide an effective way to present data as they give an idea of the pattern of variation of the random variable at a glance. We shall now discuss some of the more common forms of graphical presentation and see how they help us to illustrate, clarify and interpret the information contained in the data.

### 1.2.3 Graphical Representation of Frequency Distributions

A frequency distribution relating to a discrete variable is commonly represented through either of the following two diagrams:

- Bar diagrams:** Fig.1 gives the bar diagram for the data given in Table-7.

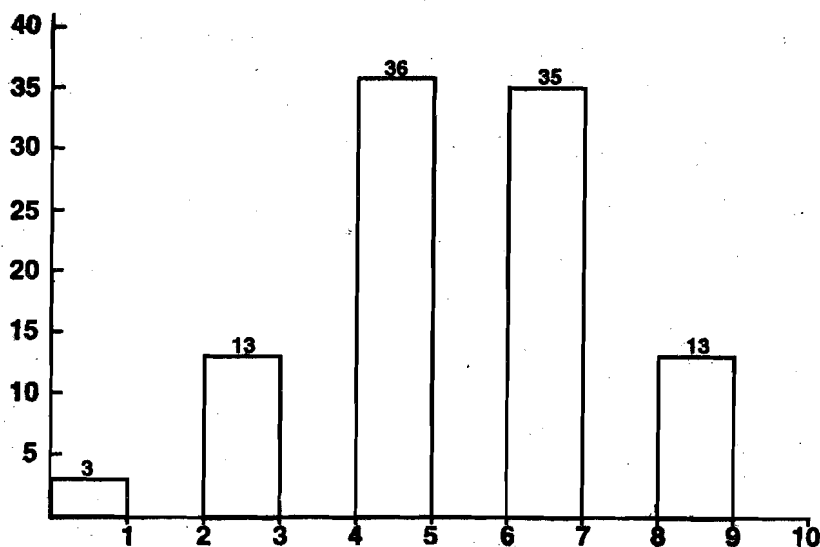


Fig.1

We can easily construct the bar diagram. Here the marks scored by the students are located on the horizontal, x-axis, and the frequencies of their occurrence on the vertical, y-axis. Note that the height of the rectangles, or bars, represent the class frequencies. From the heights of these rectangles you can easily deduce at a glance that a maximum number of students scored between 4-5 marks. For convenience, we have at the top of each rectangle given the corresponding frequency.

- Frequency Polygon:** Another form of graphical presentation is the frequency polygon. Frequency polygon for the data in Table 7 for which the bar diagram is drawn in Fig.1 is shown in Fig.2.

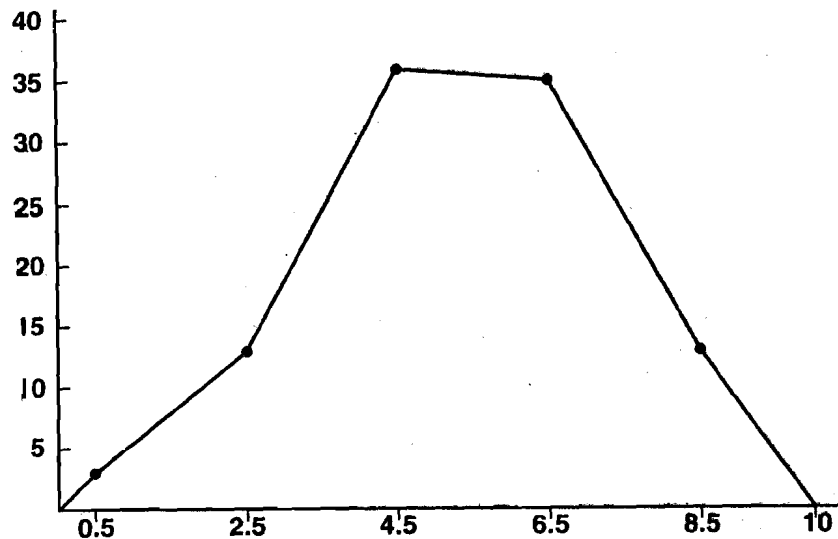


Fig.2

Here the class frequencies are plotted at the class marks and the successive points are connected by straight lines. It is thus tacitly assumed that the frequency of a class interval is concentrated at its mid-point. **Note** that we added classes with zero frequencies at both ends of the distribution to “tie down” the graph to the horizontal scale.

In the case of a continuous variable, one often uses another diagram called a **histogram**. To draw the histogram of a set of given data, take a graph paper with rectangular coordinate axes and plot the class boundaries along the x-axis. Now over each class interval erect a rectangle the height of which equals the relative frequency of this class. The resulting figure is the histogram of the given data. Here we use class boundaries to demarcate the class intervals and not class limits. This ensures that there is no gap left between the rectangles. Thus, the area of these rectangles represent the frequencies of the corresponding classes.

Fig.3 gives a frequency polygon and a histogram for the frequency distribution of lives of 100 electric bulbs given by Table-9.

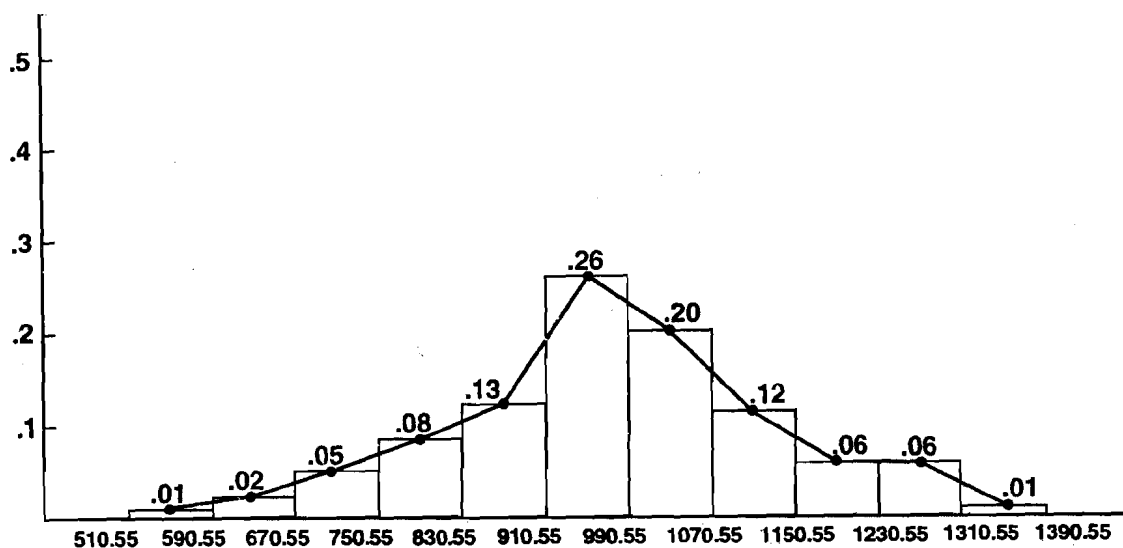


Fig.3

**Note** that in Fig.3 above we have plotted the rectangles by taking relative frequency of a class along the y-axis. This histogram has the same shape as an absolute frequency



histogram made from the same data set. This is because in both the relative size of each rectangle is the frequency of that class compared to the total number of observations. Presenting the data in terms of the relative rather than the absolute frequency of observations in each case is useful because, while the absolute numbers may change, the relationship among the classes may remain stable. Also note that if the width of the class intervals are the same, then the heights of the rectangles are proportional to their areas.

Histograms and frequency polygons are similar. However, each one has some advantages. In the case of histogram

- i) The rectangle clearly shows each separate class in the distribution
- ii) The area of each rectangle, relative to all the other rectangles, shows the proportion of the total number of observations that occur in that class. For instance, looking at Fig.3 one can easily conclude that normally, the life span of an electric bulb is between 910-990 hrs.

Similarly, frequency polygon has certain advantages viz.,

- i) The frequency polygon is simpler than its histogram counterpart
- ii) It sketches an outline of the data pattern more clearly
- iii) The polygon becomes increasingly smooth and curve like as we increase the number of classes and the number of observations.

Representation of a frequency distribution graphically on the basis of cumulative frequencies is also quite common. The diagram that one gets by plotting cumulative frequencies (less than type) against the upperclass boundaries and joining the points by line segments is called the **less than type ogive** of the frequency distribution.

Similarly one gets the **more than type ogive** by plotting the more than type cumulative frequencies against lower class boundaries. Less than type ogive for the frequency distribution of lives of 100 electric bulb given by Table-9 is given in Fig.4. Such figures are useful for finding how many measurements are located above or below a given point.

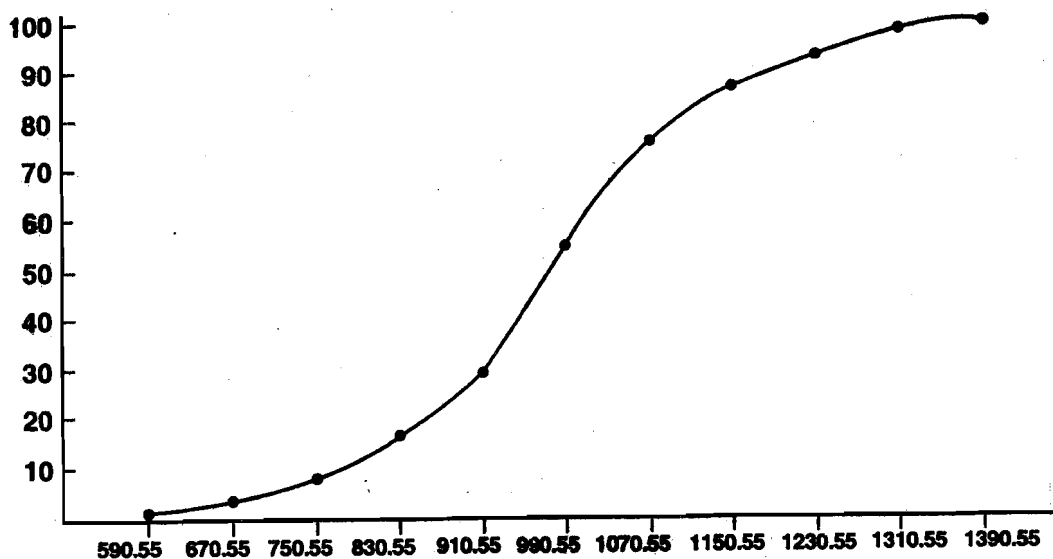


Fig.4

Frequency distributions are often presented graphically as **pie charts**, where a circle is divided into sectors, pie-shaped pieces, which are proportional in size to the corresponding frequencies or percentages. To construct a pie chart, we first convert the class frequencies into percentages of the total number of observations. Then since a complete circle corresponds to 360 degrees, we obtain the central angles of the various sectors by multiplying the percentages by 3.6. Fig.5 gives the pie chart for the data in

Table 7 giving the marks scored by 100 students in a statistics test. In this table you may observe that the total frequency is 100. Hence, the frequencies given by column 2 of this table also equals the percentage frequencies. Regions shaded differently in Fig.5 represent these percentages.

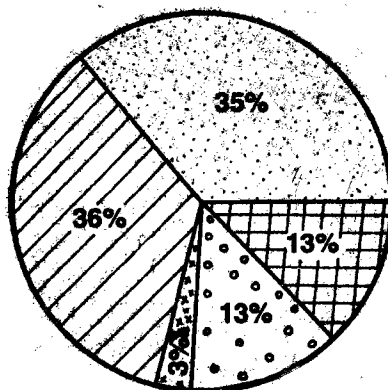


Fig.5

You may now try the following exercises:

- E8) In a sample of 60 families in a certain locality, the number of children per family are recorded as follows:

|   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 4 | 3 | 1 | 2 | 6 | 3 | 1 | 3 | 4 | 2 | 2 |
| 0 | 2 | 2 | 3 | 0 | 3 | 5 | 3 | 1 | 2 | 4 | 3 |
| 3 | 3 | 1 | 2 | 3 | 4 | 3 | 2 | 4 | 0 | 3 | 1 |
| 6 | 1 | 3 | 5 | 3 | 7 | 1 | 5 | 2 | 3 | 1 | 4 |
| 3 | 3 | 4 | 3 | 5 | 2 | 4 | 1 | 2 | 3 | 5 | 3 |

Obtain a frequency distribution of the these observations and represent it in a suitable diagram. Also draw the cumulative frequency graph of the above data. What proportion of families will have at least 2 children? Also compute the percentage of families having not less than 2 and not more than 4 children.

- E9) The yields (in quintals) of a grain from 500 small plots grouped in classes with a common width of the class interval are given below.

| Class boundaries | Frequency |
|------------------|-----------|
| 2.7-2.9          | 4         |
| 2.9-3.1          | 15        |
| 3.1-3.3          | 20        |
| 3.3-3.5          | 47        |
| 3.5-3.7          | 63        |
| 3.7-3.9          | 78        |
| 3.9-4.1          | 88        |
| 4.1-4.3          | 69        |
| 4.3-4.5          | 59        |
| 4.5-4.7          | 35        |
| 4.7-4.9          | 10        |
| 4.9-5.1          | 8         |
| 5.1-5.3          | 4         |

Represent the data graphically. Also draw the cumulative frequency graph.

- E10) Data given below show the areas of the various continents of the world in million square miles.

| Continent                                | Area (million square miles) |
|------------------------------------------|-----------------------------|
| Africa                                   | 11.7                        |
| Asia                                     | 10.4                        |
| Europe                                   | 1.9                         |
| North America                            | 9.4                         |
| Oceania                                  | 3.3                         |
| South America                            | 6.9                         |
| Russia and other former Soviet Republics | 7.9                         |

Represent the above data by means of a suitable diagram.

So far, we learned to construct tables and graphs using raw data. We see that the frequency distribution of a variable summarises the statistical data on the variable and brings out the pattern and feature of its variation. But what if we need more exact measures of a data set? In that case, we can use single numbers, called summary statistics to describe certain characteristics of a data set. For instance, how do you compare the performance of two batsmen in the game of cricket? You must have seen people saying that Batsman-1 is better because his average score is better; hardly, you'll hear anybody making a match to match comparison. Thus the average number of runs scored is a summary of all his scores in all the matches he has played so far. So in this case average is the summary statistics that can be used to rate the performance of the batsmen.

We shall now discuss the summary measures that are commonly used to summarise information contained in a data set.

### 1.3 SUMMARISATION OF DATA

The choice of a single number or summary statistics that we choose to summarise a given data depends on the particular characteristics we want to describe. In one study we may be interested in the value which is exceeded by only 25 percent of the data; in another, in the value which exceeds the lowest 10 percent of the data; and in still another, in a value which describes the centre or middle of the data. The statistical measures which describe such characteristics are called **measure of location or measures of tendency**. Some of such important measures are:

- i) measures of central tendency
- ii) measures of dispersion

We shall now discuss these measures one-by-one.

#### 1.3.1 Measures of Central Tendency

If you take a close look at any data set, you would notice that though the manifestation of the variable is different for different observational units, the values tend to cluster around a central value. This property is referred to as **central tendency**. For instance, look at the data sets in Table 2 and Table 3. In these cases, at a first glance, the corresponding observations seem to be clustering around 6 and 1000 or thereabouts respectively.

A representative value around which a given set of observations tends to cluster (or equivalently be located) is a measure of central tendency or location or is simply an average. **Arithmetic mean (a.m.)**, **median** and **mode** are the three commonly used averages. Other averages are **geometric mean** and **harmonic mean**.

##### Arithmetic Mean

The most popular measure of central location is what the layman calls an 'average' and what the statistician calls an **arithmetic mean**, or simply a **mean**. The arithmetic mean

is the sum of the numbers included in the relevant set of data divided by the number of such numbers.

### Mean from Ungrouped Data

Let there be  $N$  items or units in a population. In Table-2,  $N=100$ . If we order these numbers from 1 to  $N$ ,  $x_1$  being the first number,  $x_2$  being the second number, and so on up to  $x_N$ , which is the  $N$ th number, then the population mean is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i \quad (1)$$

In particular, if we have  $N=4$ , where  $x_1 = 30, x_2 = 50, x_3 = 70, x_4 = 90$ , then

$$\bar{x} = \frac{30 + 50 + 70 + 90}{4} = 60$$

Thus, arithmetic mean in this case turns out to be 60.

E11) The average weekly wage of 60 workers in a factory was calculated as Rs.80.50. Later, it was found that for one worker the wage was recorded as Rs.92.00 whereas the actual figure should have been Rs.80.00. What is the corrected average weekly wage?

E12) In a construction project, 60% of the labourers work on a daily rate of Rs.60.00 a day. The rest are paid at a weekly rate of Rs.450.00. If the rate for the first group is increased by 15% and the rate of the second group is reduced by 20%, will the average income of the labourers increase or decrease? (Sundays are holidays).

### Mean from Grouped Data

The computation of the mean is easy whenever data are grouped in such a way that each class corresponds to single observed value of the variable (see Table 5). For example, suppose that the distinct values in the data are  $x_1, x_2, \dots, x_k$  with  $f_1, f_2, \dots, f_k$  as the corresponding frequencies (and thus  $k$  is the number of classes). Since each  $x_i$  appears  $f_i$  times,  $i = 1, 2, \dots, k$  in the data, the sum of all observations is equal to

$$\begin{aligned} & \underbrace{x_1 + x_1 + \dots + x_1}_{\leftarrow f_1 \text{ times} \rightarrow} + \underbrace{x_2 + x_2 + \dots + x_2}_{\leftarrow f_2 \text{ times} \rightarrow} + \dots + \underbrace{x_k + x_k + \dots + x_k}_{\leftarrow f_k \text{ times} \rightarrow} \\ &= x_1 f_1 + x_2 f_2 + \dots + x_k f_k \end{aligned}$$

Now the arithmetic mean will be given by

$$\bar{x} = \frac{1}{n} \sum_{j=1}^k x_j f_j \quad (2)$$

Let us illustrate the above formula through Example 6.

**Example 8:** Consider the data of Example 6 in the following computational format.

**Table 10: Frequency Distribution of Scores of 100 Students in Statistics test**

| Score ( $x_j$ ) | Frequency ( $f_j$ ) | $x_j f_j$ |
|-----------------|---------------------|-----------|
| 0               | 1                   | 0         |
| 1               | 2                   | 2         |
| 2               | 4                   | 8         |
| 3               | 9                   | 27        |
| 4               | 15                  | 60        |
| 5               | 21                  | 105       |
| 6               | 20                  | 120       |
| 7               | 15                  | 105       |
| 8               | 9                   | 72        |
| 9               | 4                   | 36        |
| Total           | 100                 | 535       |

Thus,  $\bar{x} = 535/100 = 5.35$  is the mean marks scored by a student in the test.

How do we calculate the mean of the data given in tables 7 or 9? In either of these tables, it is not possible to calculate the sum of all the observations exactly because from these tables it is not clear which value appeared how many times in the respective data sets. For instance, Table 7 merely tells us that 3 of the observations are either 0 or 1, 13 of the observations are either 2 or 3 etc.. Similarly, Table 9 informs us that there is just one observation which is larger than 590.55 but not larger than 670.55 etc., but it does not inform us the exact numerical values of these observations. Thus, computation of the exact mean is impossible whenever the relevant data is available in the grouped form with more than one variate value in each class or category (e.g. each class in Table 7 represents two variate values and each class in table 9 represents uncountably many values). However, one cannot really complain much, because grouping the observations always leads to loss of such information anyway. In any case, in such situations, only an estimate of the mean can be obtained by making use of Formula (2) above, but with  $x_j$  interpreted as the class mark of the  $j$ -th class. Observe that here one would implicitly assume that all the observations falling in a given class are located at the class mark. This assumption is not as unrealistic as it may seem. Although all observations cannot be expected to be located at the class marks, some will fall above and some below and then the error of estimation of the mean would be expected to average out. Let us compute the mean of the data given in Table 9.

**Example 9:** Consider the data of Example 7 in the following computational format.

**Table 11: Frequency Distribution of Lives of 100 Electric Bulbs**

| Life (hours):<br>class boundaries<br>(exclusive of left<br>boundaries but<br>inclusive of right<br>boundaries) | Frequency<br>$f_j$ | Class<br>marks<br>$x_j$ | $x_j f_j$ |
|----------------------------------------------------------------------------------------------------------------|--------------------|-------------------------|-----------|
| 510.55-590.55                                                                                                  | 1                  | 550.55                  | 550.55    |
| 590.55-670.55                                                                                                  | 2                  | 630.55                  | 1261.10   |
| 670.55-750.55                                                                                                  | 5                  | 710.55                  | 3552.75   |
| 750.55-830.55                                                                                                  | 8                  | 790.55                  | 6324.40   |
| 830.55-910.55                                                                                                  | 13                 | 870.55                  | 11317.15  |
| 910.55-990.55                                                                                                  | 26                 | 950.55                  | 24714.30  |
| 990.55-1070.55                                                                                                 | 20                 | 1030.55                 | 20611.00  |
| 1070.55-1150.55                                                                                                | 12                 | 1110.55                 | 13326.60  |
| 1150.55-1230.55                                                                                                | 6                  | 1190.55                 | 7143.30   |
| 1230.55-1310.55                                                                                                | 6                  | 1270.55                 | 7623.30   |
| 1310.55-1390.55                                                                                                | 1                  | 1350.55                 | 1350.55   |
| Total                                                                                                          | 100                |                         | 97775     |

Thus the estimated mean  $\bar{x} = 97775/100 = 977.75$  hours.

If the mean of the measurements in each class interval is close to the midpoint of the class interval i.e. the class marks, then the error in approximating  $\bar{x}$  by Formula 2 is very small.

You will observe in some cases that the measurements in a sample or a population need not be weighted equally, as in Eqn.(1). For example, suppose that you want to calculate the mean profit rate for a group of firms. Since some firms are much bigger than others, a firm's profit rate should be weighted according to its size in determining the average level of profit rates. If  $w_i$  is the weight attached to the  $i$ th measurement in a sample, the **weighted arithmetic mean** is

$$\bar{x}_w = \frac{\sum_{i=1}^n x_i w_i}{\sum_{i=1}^n w_i} \quad (3)$$

For example, suppose that we have a sample of three firms' profit rates as 10%, 12% and 15%. The firm with the 10% profit rate has assets of 2 crores, whereas the other two firms have assets of 1 crore each. If a firm's assets are used to weight its profit rate, the weighted arithmetic mean of the profit rates of these three firms is

$$\bar{x}_w = \frac{2(10) + 1(12) + 1(15)}{2 + 1 + 1} = \frac{47}{4} = 11.75$$

Thus, the weighted mean is 11.75%.

If you compare Formula (2) with Formula (3) you would notice that the arithmetic mean based on grouped data is a type of weighted arithmetic mean. It is a weighted mean of the midpoints of the class intervals, the weight attached to each particular midpoint being the number of measurements falling within that class interval.

---

E13) Compute the mean for the data given in E6).

---

### The Median

Another measure of central tendency is the **median**. The median of a given data set is defined to be the middlemost observation or the mean of the two middle observations depending on whether the total number of observations  $n$  is odd or even, once the observations are sorted in the increasing or decreasing order of magnitude. Thus there are as many values above the median as there are below.

#### Median from Ungrouped Data

To calculate the median of a set of  $n$  observations, the  $n$  observations are arranged in the increasing or decreasing order of magnitude and then  $(n + 1)/2$ -th observation is identified as the median whenever  $n$  is odd; when  $n$  is even, then the median is computed as the mean of the  $n/2$ -th and  $(n/2+1)$ -th observations.

For example, suppose the number of road accidents on the first five days (i.e. Monday through Friday) of a week in a city was 2,7,4,1 and 5. We arrange these observations in the increasing order as 1,2,4,5,7 so that the  $(5+1)/2$ -th i.e. the 3rd value which is 4 is the median in this case. On the other hand, suppose we also know that 3 accidents took place on Saturday in the same week so that now we will have six observations which when arranged in the increasing order would look like 1,2,3,4,5,7 and the median will be the average of the  $(6/2)$ -th i.e. the 3rd and the  $(6/2+1)$ -th i.e. the 4th values i.e.  $(3+4)/2 = 3.5$ .

Now let us discuss the procedure for computing the median in the case of grouped data.

#### Median for Grouped Data

For calculating the median from the frequency distribution, consider the data set as represented in the frequency distribution of Table 9. What are the exact observations in the class 510.55-590.55 or, in the next class 590.55-670.55? These are not known from the table. In fact, the individual observations have lost their identities because of grouping which has led to loss of such information. But, unless the observations are known, we cannot really arrange them in order of magnitude. Thus exact calculation of the median in such cases will not be possible and we have to find some way of estimating the same. In such a situation, one proceeds as follows:

The first step in calculating the median from a frequency distribution is to find the class interval that contains the median. To do this, we start with the lowest class interval, cumulate the number of measurements in one, two, three and subsequent class intervals and stop with the interval where the cumulated number of measurements first exceeds or equals  $n/2$ . This particular class interval contains the median. Let us illustrate what we have said above with the help of the data given in Table 9. In this case  $n/2 = \frac{100}{2} = 50$ . If you look in the column with cumulative frequency 'less than type'

then 50 lies in the sixth class interval. Since the cumulative frequency for the fifth class interval is 29 and for the sixth class interval is 55, it is clear that sixth class interval is the first where the cumulative number exceeds 50. Thus this is the class interval in which the median is located. Now to find the median  $M$  for the grouped data of this sort, we use the following expression

$$M = \left( \frac{n/2 - c}{f_m} \right) l + L_m \quad (4)$$

where  $c$  = the number of measurements in class intervals below the one containing the median;

$f_m$  = the number of measurements in the class interval containing the median;

$l$  = the width of the class interval containing the median

$L_m$  = the lower boundary of the class interval containing the median.

For the data in Table 9,  $n=100$ ,  $c=29$ ,  $f_m = 26$ ,  $l=80$  and  $L_m = 910.55$  and hence using Formula 4, the median is  $M = 975.17$ .

E14) For the following data, calculate an estimate of the median

| Class     | 0-24.9 | 25-49.9 | 50-74.9 | 75-99.9 | 100-124.9 | 125-149.9 |
|-----------|--------|---------|---------|---------|-----------|-----------|
| Frequency | 6      | 11      | 14      | 16      | 13        | 10        |

Before proceeding further with the discussion of another measure of central tendency, let us compare the uses of the mean and the median.

#### Uses of the Mean and the Median

Both the mean and the median are important and useful measures of central tendency. In some circumstances the mean is a better measure than the median, and in others the converse is true. The following factors contribute to the determination of whether the mean or the median should be used.

- i) **Sensitivity to Extreme Observations:** The median is often preferred over the mean when the latter can be influenced strongly by extreme observations. Consider for instance, the computation of an average income of the families in an apartment building containing 14 families, 3 of which earn Rs.10,000 per month, 5 of which earn Rs.12,000 per month, 5 earn Rs.15,000 per month and 1 of which earns Rs.1 lakh per month. Then the mean income per month in rupees of the 14 families equals

$$\frac{3(10,000) + 5(12,000) + 5(15,000) + 1,00,000}{14} = \frac{2,65,000}{13} = \text{Rs.18,929 approx.}$$

However, this figure is not a very good description of the monthly income level of the majority of the families in the building. A better measure might be the median, which in this case is Rs.12,000 per month. The median is much less affected by the one extreme value.

- ii) **Open Ended Class Intervals:** It may happen that in a frequency distribution some intervals do not have finite upper or lower limits. For example, in a frequency distribution of the monthly income of families, two class intervals might be "less than Rs.15,000" and "Rs.30,000 and more". Each of these class intervals is open-ended. With such class intervals, there may be no alternative but to use the median, since calculation of the mean requires a knowledge of the sum of the measurements in the open-ended classes.
- iii) **Mathematical Convenience:** The mean rather than the median is often the preferred measure of central tendency because it possesses convenient mathematical properties that the median lacks. For instance, the mean of two

combined populations or samples is a weighted mean of the means of the individual populations or samples. On the other hand, given the medians of two populations or samples, there is no way to determine what the median of the two populations combined or two samples combined would be.

- iv) **Extent of Sampling Variation:** Sample statistics such as the sample mean or the sample median are often used to estimate the population mean. A major reason for preferring the mean to the median is that the sample mean tends to be more reliable than the sample median in estimating the population mean. In other words, the sample mean is less likely than the sample median to depart considerably from the population mean. You will be able to appreciate this consideration more when you study estimation later in Block 2.

We shall now move forward and discuss the third measure of central tendency, the mode.

### The Mode

Mode is defined as the most frequent observed value of the measurements in the relevant set of data. In a set of observations, if all the observations are distinct so that each of these occur with frequency 1, then it will be meaningless to say each of them is a mode; as such, in such a situation we say that the mode does not exist. However, from the definition, it is clear that a given data set may have more than one mode. For instance, if there are two modes then the set of observations is referred to as a **bimodal** data set. In an interval-grouped data set, the mode is estimated by the class mark of the class depicting highest frequency. In Table 10, the score 5 appears with the highest frequency so that 5 is the mode of the scores of the 100 students. On the other hand, the mode of the data represented in Table 9 is estimated as the mean of the two end boundaries of the class 910.55-990.55 as this class has highest frequency. The class interval containing the largest number of measurements is called the **modal class**. Thus the modal class in Table 9 is "910.55 to 990.55".

For distributions which are symmetrical (that is, where the corresponding frequency polygons or histograms are symmetrical), mean, mode and median coincide. For slightly asymmetric distributions, it has been empirically found that

$$\begin{aligned}(\text{Mean} - \text{Median}) &= \frac{1}{3}(\text{Mean} - \text{Mode}) \\ \therefore \text{Mode} &= \text{Mean} - 3(\text{Mean} - \text{Median})\end{aligned}$$

### Uses of the Mode

The mode, like the median, can be used as a central location for quantitative as well as qualitative data. If a printing press turns out 5 impressions, which we rate "Very sharp", "Sharp", "Sharp", "Sharp" and "blurred", then the modal value is "Sharp". Like the median the mode is not unduly affected by extreme values. Even if the high values are very high and the low values very low, we choose the most frequent value of the data set to be the modal value. Also, mode can be used even when one or more of the classes are open-ended. However, the mode is not used as often to measure central tendency as are the mean and median. Too often, there is no modal value because the data set contains no values that are repeated more than once. There may be cases when every value occurs the same number of times, or the data sets contain two, three, or many modes. In such cases the mode is not a useful measure.

---

E15) Classify the following statements as True or False

- The value of every observation in the data set is taken into account when we calculate its median.
- Measure of central tendency in a data set refer to the extent to which the observations are scattered.
- We can compute a mean for any data set, once we are given its frequency



- d) The mode is always found at the highest point of a graph of a data distribution

A measure of central tendency, as has been mentioned above, gives us a general idea about the average value or the magnitude of the observations. However, two distributions though may have the same mean, say, may differ in respect of several other characteristics. For instance, consider the following data which relate to performances of two suppliers: a manufacturer of a certain electrical equipment purchases 100 cardboard boxes for packing purposes every week from each of two suppliers A and B and the following are the two distributions of defectives in the weekly lots:

**Table 12:** Distributions of Number of Defective Boxes

| Week  | No. of Defective Boxes Supplied by |    |
|-------|------------------------------------|----|
|       | A                                  | B  |
| 1     | 12                                 | 6  |
| 2     | 3                                  | 7  |
| 3     | 7                                  | 6  |
| 4     | 11                                 | 5  |
| 5     | 0                                  | 7  |
| 6     | 2                                  | 5  |
| 7     | 9                                  | 6  |
| 8     | 1                                  | 5  |
| 9     | 1                                  | 6  |
| 10    | 14                                 | 7  |
| 11    | 2                                  | 4  |
| 12    | 10                                 | 8  |
| Total | 72                                 | 72 |

**Observe** that both the suppliers have supplied 6 defective boxes on an average per week. Judging from this aspect alone, we may be tempted to conclude that both the suppliers have performed equally. However, a closer look at the distributions reveals that this is not really so. While the number of defectives supplied by A varied widely between 0 and 14 over the weeks, the performance of B has been more or less stable and consistent. The number of defectives supplied by him being more or less around 6. This suggests that the extent of **variation or dispersion** of the given sets of observations from the respective averages together with the averages themselves can give us a much wider scope for comparing the performances.

Similarly, suppose that the average score obtained by students in 10-marks class test is 5 and suppose it is further known that the scores varied between 3 and 6. With such information in view, one can predict his performance with much more confidence than when the scores are known to have varied between 2 and 8.

These examples suggest that the average along with an idea about the scatter or spread of variation or dispersion of the observations about the average give us a more complete picture about the state of affairs than the average alone. The less is the range, the more will be the concentration of observations around the mean. We shall now discuss as to how the dispersion of the observations about the average can be measured.

### 1.3.2 Measures of Dispersion or Variation

There are two types of summary measures of dispersion; **distance measures** and **measures of average deviation**.

Distance measures describe the variation in the data in terms of the distance between selected measurements. The most frequently used distance measure is the range.

**Range:** It is defined to be the difference between the largest and the smallest

observations.

In the above table, the range for A is  $14 - 0 = 14$  and that of B is  $8 - 4 = 4$ .

The range, because of its complete dependence on two extreme values, is a quick but not a very accurate measure of dispersion. For instance, both the sets of observations

Set I: 1, 1, 1, 1, 1, 1, 1, 1, 10

Set II: 1, 10, 10, 10, 10, 10, 10, 10, 10

have the same range namely  $10 - 1 = 9$ ; but, evidently, the distributions are very different.

Significant measures of average deviation are **variance** and the **standard deviation**. Both of these tell us an average distance of any observation in the data set from the mean of the distribution.

### Mean Deviation and Standard Deviation:

We start with considering the case of **ungrouped data**. Let  $x_1, x_2, \dots, x_N$  be  $N$  observations and let  $\mu$  be the mean of these observations. Consider the deviations  $(x_j - \mu), j = 1, 2, \dots, N$  of the observations from the mean. The variation or dispersion is small if the observations  $x_1, x_2, \dots, x_N$  are bunched together in the close neighbourhood of  $\mu$  and it is large if these are scattered widely away from  $\mu$ . Thus, higher the dispersion, higher will be the deviations in magnitude. These distances should then be suitably combined to give rise to a consolidated measure of dispersion. However, the arithmetic mean of these deviations, which may initially seem to be a reasonable measure, does not serve our purpose since sum of these deviations is always zero (how?) i.e.

$$\sum_{j=1}^N (x_j - \mu) = 0 \quad (5)$$

---

E16) Verify result (5) for the data 1, 2, 4, 6, 8.

---

In any case, at this stage, let us appreciate that to measure dispersion, we need to know the extent of absolute deviations of the observations from the mean, and not the direction of these deviations. As such, instead of considering the deviations  $(x_j - \mu)$ , we should deal with either  $|x_j - \mu|$  or  $(x_j - \mu)^2$ . Following this logic, we define two measures

$$MD(\mu) = \frac{\sum_{j=1}^N |x_j - \mu|}{N} \quad (6)$$

$$SD = \sigma = + \left\{ \sum_{j=1}^N \frac{(x_j - \mu)^2}{N} \right\}^{\frac{1}{2}} \quad (7)$$

In the above, the measure  $MD(\mu)$  is called the **mean deviation** (about the mean); the measure  $SD$  denoted by  $\sigma$  is called the **standard deviation**. Both the measures are expressed in the same unit in which the observations are measured. The square of the standard deviation i.e  $SD^2$  is called the **variance** of the data and is denoted by  $\sigma^2$

For **grouped data** with one variate value for each class, the formula for the mean deviation and the standard deviation are given by

$$MD(\bar{x}) = \left\{ \sum_{j=1}^k \frac{|x_j - \mu| f_j}{N} \right\} \quad (8)$$

$$SD = \sigma = \left\{ \sum_{j=1}^k \frac{(x_j - \mu)^2 f_j}{N} \right\}^{\frac{1}{2}} \quad (9)$$

where  $x_1, x_2, \dots, x_k$  are the distinct observations,  $f_j$  is the frequency of  $x_j$  and  $f_1 + f_2 + \dots + f_k = N$ , the total number of observations.

Formula (9) is used to compute the variance of observations presented in the form of a frequency distribution with  $k$  classes with  $x_j$  as the class mark of the  $j$ -th class,  $j = 1, 2, \dots, k$ . As discussed earlier, this will give us only an estimate of the actual variance.

The computation of the variance can be drastically simplified through some simple algebraic treatments as follows:

$$\begin{aligned} \sum_{j=1}^k (x_j - \mu)^2 f_j &= \sum_{j=1}^k (x_j^2 - 2\mu x_j + \mu^2) f_j = \sum_{j=1}^k x_j^2 f_j - 2\mu \sum_{j=1}^k x_j f_j + \mu^2 \sum_{j=1}^k f_j \\ &= \sum_{j=1}^k x_j^2 f_j - 2\mu (N\mu) + \mu^2 N \\ &= \sum_{j=1}^k x_j^2 f_j - N\mu^2 \end{aligned} \quad (10)$$

Let us now illustrate these formulas through some examples. You may recall that we computed the mean for the data of Table 5 to be 5.36. Now let us compute the variance and standard deviation of the same data.

**Example 10:** Consider the data of Table 5 giving the marks of 100 students in a Statistics test in the form given by the following table

**Table 13: Calculation of Variance of the Frequency Distribution of marks of 100 Students in Statistics**

| Score ( $x_j$ ) | Frequency ( $f_j$ ) | $x_j f_j$ | $x_j^2 f_j$ |
|-----------------|---------------------|-----------|-------------|
| 0               | 1                   | 0         | 0           |
| 1               | 2                   | 2         | 2           |
| 2               | 4                   | 8         | 16          |
| 3               | 9                   | 27        | 81          |
| 4               | 15                  | 60        | 240         |
| 5               | 21                  | 105       | 525         |
| 6               | 20                  | 120       | 720         |
| 7               | 15                  | 105       | 735         |
| 8               | 9                   | 72        | 576         |
| 9               | 4                   | 36        | 324         |
| Total           | 100                 | 535       | 3219        |

Thus, the variance using Formula (10) can be obtained as

$$\begin{aligned} \sigma^2 &= (1/100) \{ 3219 - (100)(5.35)^2 \} \\ &= 3.5675 \end{aligned}$$

$$\text{so that } SD = (3.5675)^{\frac{1}{2}} = 1.8888.$$

In the same way you can try the following exercise.

**E17) Given the following sample of 20 numbers:**

12 41 48 58 14 43 50 59 15 45  
52 72 18 45 54 78 41 47 56 79

a) Compute the mean, the variance, and the standard deviation.

- b) If the largest value in the above set of number is changed to 500 to what degree are the mean and variance affected by the change?
- 

While doing these exercises you must have realised that whenever the  $x$ -observations for a given data are large in magnitude, the computations for the calculation of the variance become lengthy. But we can make these computations a bit more manageable. Let us see how this can be done.

As before, let the distinct observations be  $x_j, j = 1, 2, \dots, k$  and  $f_j$  be the frequency of  $x_j$ . Let us write

$y_j = (x_j - a), j = 1, 2, \dots, k$ , where  $a$  and  $b$  are two pre-specified constants.

Thus,  $x_j = a + by_j, j = 1, 2, \dots, k$ .

Hence,

$$\sum_{j=1}^k x_j f_j = \sum_{j=1}^k (a + by_j) f_j = a \sum_{j=1}^k f_j + b \sum_{j=1}^k y_j f_j$$

so that, by dividing both sides by  $N$ , we note that

$$\mu = a + b\mu' \text{ where } \mu' = \frac{\sum_{j=1}^k y_j f_j}{N}.$$

Also, we can write

$$\sum_{j=1}^k (x_j - \mu)^2 f_j = \sum_{j=1}^k (a + by_j - a - b\mu')^2 f_j = b^2 \sum_{j=1}^k (y_j - \mu')^2 f_j$$

so that, we obtain the following result.

$$\text{Variance of } x\text{-observations} = b^2(\text{Variance of } y\text{-observations}). \quad (11)$$

This formula is extremely useful from the computational point of view. Basically, the idea is that whenever the  $x$ -observations are large in magnitude, we can convert them into  $y$ -observations which can be smaller in magnitude if the constants  $a$  and  $b$  are chosen suitably. We then calculate the variance of the  $y$ -observations and simply multiply the same by  $b^2$  to arrive at the variance of  $x$ -observations. **Usually  $a$  is chosen to be a value in the middle of the range of the observations and  $b$  is the class interval.**

Let us now take up an example to see how Result (11) helps us in simplifying the computations for a given set of data.

**Example 11:** Consider the data of Table 11 as shown in Table 14 on the next page.

The estimated mean  $\mu'$  of the  $y$ -observations is

$$\begin{aligned} \mu' &= \frac{\sum_{i=1}^{11} y_i f_i}{N} \\ &= \frac{34}{100} = 0.34 \end{aligned}$$

and the estimated mean  $\mu$  of the  $x$ -observations is

$$\begin{aligned} \mu &= a + b\mu' \\ &= 950.55 + 80(0.34) \\ &= 977.75 \text{ hours.} \end{aligned}$$

**Table 14: Frequency Distribution of Life of 100 Electric Bulbs**

| Life (hours);<br>class boundaries<br>(exclusive of left<br>boundaries but<br>inclusive of right<br>boundaries) | Frequency<br>$f_j$ | Class<br>marks<br>$x_j$ | $y_j = \frac{((x_j - 950.55))}{80}$ | $y_j f_j$ | $y_j^2 f_j$ |
|----------------------------------------------------------------------------------------------------------------|--------------------|-------------------------|-------------------------------------|-----------|-------------|
| 510.55 - 590.55                                                                                                | 1                  | 550.55                  | -5                                  | -5        | 25          |
| 590.55 - 670.55                                                                                                | 2                  | 630.55                  | -4                                  | -8        | 32          |
| 670.55 - 750.55                                                                                                | 5                  | 710.55                  | -3                                  | -15       | 45          |
| 750 - 830.55                                                                                                   | 8                  | 790.55                  | -2                                  | -16       | 32          |
| 830.55 - 910.55                                                                                                | 13                 | 870.55                  | -1                                  | -13       | 13          |
| 910.55 - 990.55                                                                                                | 26                 | 950.55                  | 0                                   | 0         | 0           |
| 990.55 - 1070.55                                                                                               | 20                 | 1030.55                 | 1                                   | 20        | 1           |
| 1070.55 - 1150.55                                                                                              | 12                 | 1110.55                 | 2                                   | 24        | 48          |
| 1150.55 - 1230.55                                                                                              | 6                  | 1190.55                 | 3                                   | 18        | 54          |
| 1230.55 - 1310.55                                                                                              | 6                  | 1270.55                 | 4                                   | 24        | 96          |
| 1310.55 - 1390.55                                                                                              | 1                  | 1350.55                 | 5                                   | 5         | 5           |
| Total                                                                                                          | 100                |                         |                                     | 34        | 351         |

**Note** that here we have taken  $a=950.55$  and  $b=80$ .

Using Formula (10), the estimated variance of the  $y$ -observations is

$$\begin{aligned} \frac{1}{N} \sum_{j=1}^{11} y_j^2 f_j - \mu^2 &= \frac{351}{100} - (0.34)^2 \\ &= 3.39 \end{aligned}$$

Now using Formula (11),

$$\begin{aligned} \text{the estimated variance of } x\text{-observations} = \sigma^2 &= b^2(\text{variance of } y\text{-observations}) \\ &= (80)^2(3.39) = 21696(\text{hours})^2 \end{aligned}$$

$$\therefore \text{the estimated SD of the } x\text{-observations} = \sigma = +\sqrt{21696} = 147.2956 \text{ hours.}$$

E18) The following table gives the frequency distribution for the heights (in cms.) of 75 individuals.

| Heights (cms.) | Frequency |
|----------------|-----------|
| 150.6 - 152.5  | 6         |
| 152.6 - 154.5  | 7         |
| 154.6 - 156.5  | 9         |
| 156.6 - 158.5  | 13        |
| 158.6 - 160.5  | 17        |
| 160.6 - 162.5  | 8         |
| 162.6 - 164.5  | 9         |
| 164.6 - 166.5  | 6         |
|                | 75        |

Find the mean and the s.d. of the above data. Also, estimate the median.

E19) The mean and standard deviation of a set of  $n$  observations  $x_1, x_2, \dots, x_n$  are  $\bar{x}$  and  $\sigma_x$  respectively. The mean and the standard deviation for another set of  $m$  observations  $y_1, y_2, \dots, y_m$  are  $\bar{y}$  and  $\sigma_y$  respectively. Show that the standard deviation of the pooled set  $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m$  of  $m+n$  observations is

$$\sqrt{\frac{n\sigma_x^2 + m\sigma_y^2}{n+m} + \frac{nm}{(n+m)^2}(\bar{x} - \bar{y})^2}$$

---

We now end this unit by giving the summary of whatever we have covered here.

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## 1.4 SUMMARY

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In this unit we have covered the following points

- 1) To study a given phenomenon/any physical situation, the basic raw material is the **data** (information) relating to it.
- 2) **Statistics** involves the collection, organisation, analysis and interpretation of data.
- 3) A collection of data is a **data set** and a single observation from a data set is the **data point**.
- 4) For a data set to be **useful**, observations need to be organised in order to pick out trends and come to logical conclusions.
- 5) The number of times any value occurs in a data set is the **frequency** of that value.
- 6) An organised display of data showing the frequency of each value in a data set against its value or against mutually exclusive classes into which these values fall is a **frequency distribution**.
- 7) A tabular display of data showing how many observations lie above, or below, certain values is a **cumulative frequency distribution**.
- 8) **Graphical methods** of presenting data provide an effective way to present data as it gives an idea of the pattern of variation of the random variable (character corresponding to numerical value) at a glance.
- 9) A **Histogram** is a graph of a data set, composed of a series of rectangles, each proportional in width to the range of values in a class and proportional in height to the number of items falling in the class.
- 10) A line graph connecting the midpoints of each class in a data set, plotted at a height corresponding to the frequency of the class is a **frequency polygon**.
- 11) **Ogive** is a graph of a cumulative frequency distribution.
- 12) Single numbers that describe certain characteristic of a data set are **summary measures**.
- 13) Summary measures that are commonly used are **measures of central tendency** and **measures of dispersion or variation**.
- 14) A measure indicating the value to be expected of a typical or **middle data point** is a measure of central tendency. Mean, mode, median are such measures.
- 15) A measure describing how **scattered** or **spread** out the observations in a data set are is a measure of dispersion. Mean deviation and variance are such measures.
- 16) **Mean** is a central tendency measure representing the arithmetic average of a set of observations.
- 17) **Median** is the middle point of a data set, a measure of location that divides the data set into values.
- 18) The value most often repeated in the data set is the **mode**. It is represented by the highest point in the distribution curve of a data set.
- 19) In a data set, the average distance of the observations from the mean is the **mean deviation**.

## 1.5 SOLUTIONS/ANSWERS

E1) Examples could be taken from situations arising from your daily life experience.

E2) a) population      b) sample      c) population

E3) b) and d) discrete; a), c) and e) are continuous.

E4) For example data collected to find the average height of females in the age group 15-20 years in Delhi.

Think of other similar examples.

E5) a) 20 and 34, 35 and 49, 50 and 64, 65 and 79, and 80 and 94;

b) 19.5, 34.5, 49.5, 64.5, 79.5, 94.5;

c) 27, 42, 57, 72 and 87;

d) 15.

E6) a) X is a discrete variable; min. score = 29, max. score = 97.

b)

| Scores | Frequency | Cumulative frequency |                  |
|--------|-----------|----------------------|------------------|
|        |           | 'less than type'     | 'more than type' |
| 20-29  | 1         | 1                    | 50               |
| 30-39  | 2         | 3                    | 49               |
| 40-49  | 2         | 5                    | 47               |
| 50-59  | 3         | 8                    | 45               |
| 60-69  | 12        | 20                   | 42               |
| 70-79  | 14        | 34                   | 30               |
| 80-89  | 12        | 46                   | 16               |
| 90-99  | 4         | 50                   | 4                |

c) 45

d) 29

E7) i) Because of the nature of partition on the range of income, the exact percentage of people earning more than Rs.4500/- cannot be found out; however, an estimate can be tried assuming that the frequency of 152 in the class 4000 - 6000 is uniformly distributed over the interval (4000, 6000). The estimate is obtained by simple linear interpolation as follows:

$$\begin{aligned} & \frac{100}{1000} \left\{ \text{Total frequency} - \text{frequency in } (0, 4000) \right. \\ & \quad \left. - \frac{152}{6000 - 4000} (4500 - 4000) \right\} \\ & = \frac{100}{1000} \{ 1000 - 236 - 38 \} = 72.6\% \end{aligned}$$

ii) As above, an estimate is

$$\frac{100}{1000} \left\{ 1000 - 40 - \frac{55}{2000 - 1000} (500) \right\} = 93.25\%$$

iii)  $100 \{ 141 + (1000)152/2000 \} / 1000 = 21.7\%$

E8)

| No. of Children | Frequency |
|-----------------|-----------|
| 0               | 3         |
| 1               | 9         |
| 2               | 12        |
| 3               | 20        |
| 4               | 8         |
| 5               | 5         |
| 6               | 2         |
| 7               | 1         |
| Total           | 60        |

Frequency polygon can be drawn for the above observation as given in Fig.6.

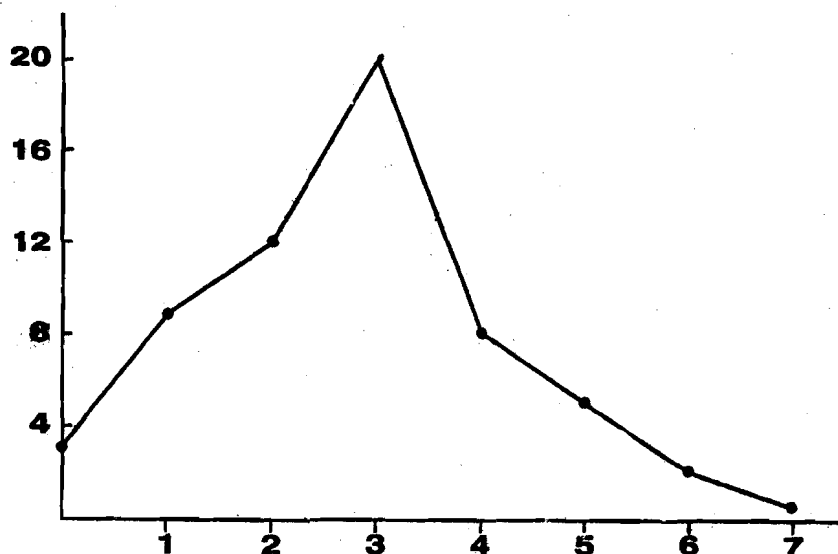


Fig.6

Similarly draw cumulative frequency graph for the above data.

Proportion of families having at least 2 children =  $\frac{48}{60} = 80\%$

Proportion of families with not less than 2 children, but not greater than 4 children  
 $= \frac{(12 + 20 + 8)}{60} = \frac{40}{60} = 66.67\%$ .

E9) Use a histogram.

E10) Use pie diagram

E11) Correct average weekly wage

$$= \{80.50(60) - 92.00 + 80.00\} / 60 = 80.30$$

E12) Suppose there are 100 workers; thus 60 of them are paid @Rs.60/- per day each and 40 of them are paid @ Rs.450/6 = 75/- per day each. The average income then =  $\{60(60) + 75(40)\} / 100 = 66/-$

After modification, the average =  $\{60(1.15)(60) + 40(.80)(75)\} / 100 = 65.40$

E13) Mean =  $\frac{2909.5}{50} = 58.19$

E14) For the given data  $\frac{n}{2} = \frac{70}{2} = 35$

$c = 31, f_m = 16, l = 24.9, L_m = 75$ , using Formula (4), median  $M = 81.23$

E15) a) T,      b) F,      c) F,      d) T

E16)  $\sum_{i=1}^5 (x_i - \bar{x}) = -3.2 - 2.2 - 0.2 + 1.8 + 3.8 = 0$

E17) a)  $\bar{x} = 46.35, \sigma^2 = 384.56, \sigma = 19.61$



- b) Changing the largest value 79, in the given set to 500,  
 $\bar{x} = 67.40, \sigma^2 = 1484.18$

E18)

| Class mark (x) | $y = (x - 157.55)/2$ | Freq. (f) | yf  | $y^2f$ |
|----------------|----------------------|-----------|-----|--------|
| 151.55         | -3                   | 6         | -18 | 54     |
| 153.55         | -2                   | 7         | -14 | 28     |
| 155.55         | -1                   | 9         | -9  | 9      |
| 157.55         | 0                    | 13        | 0   | 0      |
| 159.55         | 1                    | 17        | 17  | 17     |
| 161.55         | 2                    | 8         | 16  | 32     |
| 163.55         | 3                    | 9         | 27  | 81     |
| 165.55         | 4                    | 6         | 24  | 96     |
| Total          |                      | 75        | 43  | 317    |

Thus,

$$\bar{x} = 157.55 + 2\bar{y} = 157.55 + 2(43/75) = 158.70;$$

$$\sigma_x = 2, \sigma_y = 2\sqrt{\left(317 - \frac{(43)^2}{75}\right)/75} = 7.80$$

$$M = 158.55 + \{(0.5 - 35/75)(75)(2)\} / 75 = 158.62$$

E19) The pooled set has mean  $\bar{u}$  and variance  $\sigma^2$ , say. Then

$$\bar{u} = \frac{n\bar{x} + m\bar{y}}{n + m}$$

$$\begin{aligned}
 \text{Also, } (n + m)\sigma^2 &= \sum_{j=1}^n (x_j - \bar{u})^2 + \sum_{j=1}^m (y_j - \bar{u})^2 \\
 &= \sum_{j=1}^n x_j^2 + \sum_{j=1}^m y_j^2 - (n + m)\bar{u}^2 \\
 &= (n\sigma_x^2 + n\bar{x}^2) + (m\sigma_y^2 + m\bar{y}^2) - (n + m)\bar{u}^2 \\
 &= (n\sigma_x^2 + m\sigma_y^2) + \frac{nm}{n + m}(\bar{x} - \bar{y})^2
 \end{aligned}$$

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## UNIT 2 PROBABILITY CONCEPTS

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### 2.1 INTRODUCTION

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Sunil is an enterprising class VIII student who wants to use his free hours more fruitfully. A news paper agent agrees to employ him for one hour in the morning between 5.30 AM to 6.30 AM for distributing news papers in a residential colony where there are 85 regular subscribers. In addition, Sunil finds that there are about 10 irregular customers who may buy the paper from him on a day to day basis. On every additional news paper Sunil sells, he makes an extra income of 30 paise. But on every unsold news paper that he takes back to the agent he loses 10 paise. Sunil has to decide how many newspapers he should collect from the agent each morning so that he makes the maximum possible gain. His dilemma is about the 10 irregular customers as he has to decide how many of these 10 will actually buy from him on any given day. If you ask him whether he knows probability theory, he may be surprised and say he knows nothing. Actually he may already be using some of the ideas of probability theory while not being aware of or articulate about them. This is a very commonly occurring situation.

When asked, "Do you know anything about probability?" most people are quick to answer, "No!" Usually that is not the case at all. The words probable and probably are used commonly in everyday language. We say, "It will probably rain tomorrow" or "there is 0% chance of rain today." Such statements often have a vague, subjective quality and based sometimes on certain information and at other times on intuition only.

We live in an uncertain world. When we get up in the morning we cannot say exactly who we are going to meet, what the weather will be like or what event will be on the television news during the day.

In our everyday lives, we cope with this uncertainty by making hundreds of guesses, calculated risks and some gambles. We don't take a coat with us for the weekend because it is unlikely to be cold. We allow a particular length of time to travel to an important interview because it will probably be enough.

All these decisions are made by assessing the relative probability (chance) of all the possible outcomes — even if we do this unconsciously and intuitively. Business decisions are made in a similar climate of uncertainty. A publisher must decide how large the print run of a new book should be to avoid unusual storage of unsold copies and yet ensure availability. A stock market dealer decides to sell a particular share because a financial model tells her that the price is likely to fall.

The penalties of estimating chances inaccurately and hence making a wrong decision vary from minor inconvenience, to loss of income to bankruptcy. So, in business (and other fields) we endeavour to measure uncertainty using some scientific method. Rather than make vague statements containing 'likely', 'may be' or 'probably'; we need to be more precise.

Historically, the oldest way of measuring uncertainties is the probability concept. Probability theory had its beginnings over 300 years ago, when gamblers of that period asked mathematicians to develop a system for predicting outcomes of a turn of the roulette wheel or a roll of a pair of dice. The word probability is associated with a quantitative approach to predicting the outcome of an event (the outcome of a presidential election, the side effects of a new medication, etc.).

In this unit we shall see how uncertainties can actually be measured, how they can be assigned numbers (called probabilities) and how these numbers are to be interpreted. After starting with some preliminaries of the probabilities we shall concentrate on the rules which probabilities must obey. This includes the basic postulates, the relationship between probabilities and odds, the addition rules, the definition of conditional probability, the multiplication rules etc.

### Objectives

After reading this unit, you should be able to

- describe trials, events, sample spaces associated with an experiment;
- express the union, intersection, complement of two or more events in terms of a new event;
- ] define the probability of the occurrence of an event and obtain it;
- obtain the conditional probability of an event.

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## 2.2 PRELIMINARIES

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What is an experiment? Many of you will relate experiments with all that you were expected to do in physics, chemistry or biology laboratories in your schools and colleges. For example, you may perhaps recollect that in the chemistry laboratory, one of the experiments you performed was to explore as to what would happen if sulfuric acid is poured in a jar containing zinc. Yet in another experiment in the physics laboratory, you might have performed the act of inserting a battery of a certain specification in a given circuit with a view to finding out the quantum of the flow of electricity through the said circuit. Thus, generally speaking, an **experiment** is merely the performance of an act for generating an observation on the phenomenon under study. In fact, when you pick up an item from a lot consisting of a number of items coming out of a manufacturing process, say, to decide whether the picked up item is defective or not, then also you are performing an experiment. Whenever you are investing a certain sum of money in the share market to see to what extent your money grows over a certain specified time interval, you are performing an experiment too. When you are stocking a number of units of a particular brand of a consumer good in your store in anticipation of sale, that is also an experiment. Tossing a coin, rolling a die, observing the number of road accidents on a given day in a city etc. are all experiments. In each such case, a certain act is performed and its outcome is observed. Is there then any difference between the former type of laboratory experiments and the latter type of experiments that we presumably perform in some form or the other in our daily lives? The answer to this question is 'Yes'. The difference is in terms of the outcomes that you associate with the experiments. **Notice** that in the former type of experiments performed under the controlled conditions of a laboratory, the outcomes

are known a priori from the conditions under which these are carried out. It is known a priori that sulfuric acid and zinc together will yield hydrogen gas and zinc sulfate. The quantum of electricity flow through the circuit is known from the specifications of the inserted battery through the celebrated Ohm's Law. Such experiments are **deterministic** in the sense that the conditions under which these are carried out would inform us what the result is going to be even before the experiment is performed. However, when you are blindly picking up an item from a manufactured lot, you have no way of knowing a priori whether the item being picked up would be defective or not. When in your role as a shopkeeper, you stock certain units of a particular commodity, you will have no prior knowledge what your sales will be like over a specified period. When you toss a coin (it is equivalent to picking up an item blindly from a lot containing both defective and non-defective items - how?), you do not know the result of the experiment beforehand. Experiments whose outcomes cannot be precisely predicted primarily because the totality of the factors influencing the outcomes are either not precisely identifiable or not controllable at the time of experimentation, even if known, are called **random or stochastic** experiments. If you look at the dilemma of Sunil, the news paper boy, do you think what Sunil observes regarding the buying behaviour of the ten irregular customers on a given day can be modelled as the outcome of a random experiment? As Sunil does not know whether any of the specified irregular customer will actually buy from him, it is possible to think of this as a random experiment.

For the purposes of this block, by an experiment, we shall always mean a random experiment only. We now introduce some of the commonly occurring terms of the probability theory.

### 2.2.1 Trials, Sample Space, Events

You must have often observed that a random experiment may comprise of a series of smaller sub-experiments. These are called **trials**. Consider for instance the following situations.

**Example 1:** Suppose the experiment consists of observing the results of three successive tosses of a coin. Each toss is a trial and the experiment consists of three trials so that it is completed only after the third toss (trial) is over.

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**Example 2:** Suppose from a lot of manufactured items, ten items are chosen successively following a certain mechanism for checking. The underlying experiment is completed only after the selection of the tenth item is completed; the experiment obviously comprises of 10 **trials**.

\* \* \*

**Example 3:** If you consider Example 1 once again you would notice that each toss (trial) results into either a head (H) or a tail (T). In all there are 8 possible outcomes of the experiment viz.,  $s_1 = (H,H,H)$ ,  $s_2 = (H,H,T)$ ,  $s_3 = (H,T,H)$ ,  $s_4 = (T,H,H)$ ,  $s_5 = (T,T,H)$ ,  $s_6 = (T,H,T)$ ,  $s_7 = (H,T,T)$  and  $s_8 = (T,T,T)$ .

Each of the above outcome  $s_1, s_2, \dots, s_8$  is called a **sample point**. Notice that each sample point has three entries separated by commas. For example, the sample point  $s_2$  indicates that the first toss results in a head, the second also produces head while the third one leads to a tail. Thus the sequence in which H's and T's appear is important. The set  $\zeta = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8\}$  of all possible outcomes is called the **sample space** of the experiment. Sample spaces are classified as **finite** or **infinite** according to the number of sample points (finite/infinite) they contain. For instance, an infinite sample space arises when we throw a dart at a target and there is a continuum of points we may hit. Sample space that arises in the case of Sunil, the news paper boy is an example of a finite sample space. What do you think is the sample space in this case? In this case the random experiment is to observe the 10 irregular customers on any day

and note down whether each one “buys” or “does not buy” the newspaper. Therefore the sample space contains  $2^{10}$  simple points which are sequences of the form (B, NB, NB, B, B, B, NB ...) etc. The sample space is, of course, finite. A specific collection or subset of sample points, say  $E_1 = \{s_2, s_3, s_4\}$  is called an **event**. Event is a **simple event** if it consists of a single sample point. In this case  $\{s_i\}$  is a simple event for  $i = 1, 2, \dots, 8$ . A **word of caution** here that not every subset of a sample space is an event. We shall not explain it here as it is beyond the scope of this course.

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The following examples will further strengthen your understanding of various terms defined.

**Example 4:** Suppose our experiment consists in observing the number of road accidents in a given city on a given day. Obviously,  $\zeta = \{0, 1, 2, \dots, b\}$ , where  $b$  is the maximum possible number of accidents in a day and this can very well be infinity, in which case the sample space is infinite. The event  $E$  that there are five or less number of accidents on that day can be described by  $E = \{0, 1, 2, 3, 4, 5\}$ .

\* \* \*

**Example 5:** Suppose, we are interested in noting down the time (in hours) to failure of an equipment. Here, the sample space  $\zeta = \{x | 0 < x \leq b\} = ]0, b]$ , the half open interval between 0 and  $b$ , where  $b$  is the maximum possible life of the equipment. Here also  $b$  may be infinite. In any case,  $\zeta$  is not only infinite, but also uncountable i.e. the elements of  $\zeta$  cannot be put into one-to-one correspondence with the set of natural numbers. The event  $E$  that the equipment survives for **at least** 500 hours of operation can be described as  $E = \{x | 500 \leq x \leq b\} = [500, b]$ .

\* \* \*

We shall now consider an example which shows the importance of order of selection of items/objects under consideration.

**Example 6:** Suppose an urn contains three marbles which are identical in all respect excepting that two of them are red in colour while the third one is white. The experiment is to pick up two marbles, one after the other, blindly and observing their colour. This experiment comprises of two trials. How does the experimenter report the result of the experiment? For this purpose, we may identify the two red marbles as  $r_1, r_2$  and the white marble may be identified as  $w$ . Thus, a typical outcome, for instance, can be  $(w, r_2)$ ; this means that the white marble was picked up at the first draw while the second red marble appeared in the next draw. **Note** that the order in which  $w$  and  $r_2$  appeared in  $(w, r_2)$  is important; the first letter corresponds to the first selection while the second letter corresponds to the second. Thus the sample space  $\zeta$  of this random experiment is the following set:

$$\zeta = \{(r_1, r_2), (r_1, w), (r_2, r_1), (r_2, w), (w, r_1), (w, r_2)\}$$

How do we describe the ‘event  $E$  that’ of the two marbles picked up, one is red and the other is white? Well, this event materialises if one of the two red marbles is picked up in one of the two trials while the white marble is picked up in the other trial. Thus,  $E$  materialises if the outcome of the experiment is either  $(r_1, w)$  or  $(r_2, w)$  or  $(w, r_1)$  or  $(w, r_2)$ ; thus,  $E = \{(r_1, w), (r_2, w), (w, r_1), (w, r_2)\}$ .

Let us now modify the conditions of the experiment a little. Suppose, we return the first marble selected back to the urn before the second marble is selected. Then, there will be some new possibilities. Specifically, now it becomes possible for the same marble to be selected twice. Hence, the sample space gets modified to

$$\zeta = \{(r_1, r_1), (r_1, r_2), (r_1, w), (r_2, r_1), (r_2, r_2), (r_2, w), (w, r_1), (w, r_2), (w, w)\}$$

but, the above event  $E$  remains unaltered.

Incidentally, in our initial experiment, marbles were being drawn **without replacement** while in the modified experiment, the marbles were being drawn **with replacement**.

## Probability and Statistics

Selection is said to be done **with/without replacement** if each object chosen is **returned/not returned** to the population before the next object is drawn

The former experiment is equivalent to drawing two marbles in succession without the previously drawn marble being returned to the urn before the next draw. In the latter experiment the previously drawn marble was being replaced before the next draw, so that the number of marbles in the urn was always the same and it is possible that same marble is chosen more than once.

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You may now try the following exercises:

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E1) In each of the following exercises, an experiment is described. Specify the relevant sample spaces:

- a) A machine manufactures a certain item. An item produced by the machine is tested to determine whether or not it is defective.
- b) An urn contains six balls, which are coloured differently. A ball is drawn from the urn and its colour is noted.
- c) A person is asked to which of the following categories he belongs: unemployed, self-employed, employed by the Central Government, employed by a state or local Government, employed by a private organisation, employed by an employer who does not fulfil any of the above descriptions. The person's answer is recorded.
- d) An urn contains ten cards numbered 1 through 10. A card is drawn, its number noted and the card is replaced. Another card is drawn and its number is noted.

E2) Suppose a six-faced die is thrown twice. Describe each of the following events:

- i) The maximum score is 6.
- ii) The total score is 9.
- iii) Each throw results in an even score.
- iv) Each throw results in an even score larger than 2.
- v) The scores on the two throws differ by at least 2.

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It must have been quite clear to you by now that events associated with a random experiment can always be represented by the collection of sample points each of which leads to the occurrence of the said event. The phrase 'event  $E$  occurs' is an alternative way of saying that the outcome of the experiment has been one of those that lead to the materialisation of said event. Let us now see how an event can be expressed in terms of two or more events by forming unions, intersections and complements.

### 2.2.2 Algebra of Events

Let  $\zeta$  be a fixed sample space. We have already defined an event as a collection of sample points from  $\zeta$ . Imagine that the (conceptual) experiment underlying  $\zeta$  is being performed. The phrase "the event  $E$  occurs" would mean that the experiment results in an outcome that is included in the event  $E$ . Similarly, non-occurrence of the event  $E$  would mean that the experiment results into an outcome that is not an element of the event  $E$ . Thus, the collection of all sample points that are not included in the event  $E$  is also an event which is complementary to  $E$  and is denoted as  $E^c$ . The event  $E^c$  is therefore the event which contains all those sample points of  $\zeta$  which are not in  $E$ . As such, it is easy to see that the event  $E$  occurs if and only if the event  $E^c$  does not take place. The events  $E$  and  $E^c$  are **complementary events** and taken together they comprise the entire sample space, i.e.,  $E \cup E^c = \zeta$ .

You may recall that  $\zeta$  is an event which consists of all the sample points. Hence, its complement is an empty set in the sense that it does not contain any sample point and is called the **null event**, usually denoted as  $\phi$  so that  $\zeta^c = \phi$ .

Let us once again consider Example 3. Consider the event  $E$  that the three tosses produce at least one head. Thus,  $E = \{s_1, s_2, s_3, s_4, s_5, s_6, s_7\}$  so that the complementary event  $E^c = \{s_8\}$ , which is the event of not scoring a head at all.

Again in Example 6 in the case of selection without replacement, event that the white marble is picked up at least once is defined as  $E = \{(r_1, w), (r_2, w), (w, r_2), (w, r_1)\}$ . Hence,  $E^c = \{(r_1, r_2), (r_2, r_1)\}$  i.e. the event of not picking the white marble at all.

Let us now consider two events  $E$  and  $F$ . We write  $E \cup F$ , read as  $E$  "**union**"  $F$ , to denote the collection of sample points, which are responsible for occurrence of either  $E$  or  $F$  or both. Thus,  $E \cup F$  is a new event and it occurs if and only if either  $E$  or  $F$  or both occur i.e. if and only if at least one of the events  $E$  or  $F$  occurs. Generalising this idea, we can define a new event  $\cup_{j=1}^k E_j$ , read as "**union**" of the  $k$  events  $E_1, E_2, \dots, E_k$ , as the event which consists of all sample points that are in at least one of the events  $E_1, E_2, \dots, E_k$  and it occurs if and only if at least one of the events  $E_1, E_2, \dots, E_k$  occurs.

**Example 7:** Consider Example 3 and let  $E_j$  be the event that the three tosses produce  $j$  heads,  $j = 0, 1, 2, 3$ . Hence,  $\cup_{j=0}^2 E_j$  is the event that the three tosses produce at most 2 heads i.e. not all the three tosses produce heads.

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Again, let  $E$  and  $F$  be two given events. We write  $E \cap F$ , read as  $E$  "**intersection**"  $F$ , to denote the collection of sample points any of whose occurrence implies the occurrence of both  $E$  and  $F$ . Thus,  $E \cap F$  is a new event and it occurs if and only if both the events  $E$  and  $F$  occur. Generalising this idea, we can define a new event  $\cap_{j=1}^k E_j$  read as "**intersection**" of the  $k$  events  $E_1, E_2, \dots, E_k$ , as the event which consists of sample points that are common to each of the events  $E_1, E_2, \dots, E_k$ , and it occurs only if all the  $k$  events  $E_1, E_2, \dots, E_k$  occur simultaneously.

Further, two events  $E$  and  $F$  are said to be **mutually exclusive** or **disjoint** if they do not have a common sample point i.e.  $E \cap F = \phi$ . Two mutually exclusive events then cannot occur simultaneously. In the coin-tossing experiment for instance, the two events, heads and tails, are mutually exclusive: if one occurs, the other cannot occur. To have a better understanding of these events let us once again look at Example 3.

Let  $E$  be the event of scoring an odd number of heads and  $F$  be the event that tail appears in the first two tosses, so that  $E = \{s_1, s_5, s_6, s_7\}$  and  $F = \{s_5, s_8\}$ . Now  $E \cap F = \{s_5\}$ , the event that only the third toss yields a head. Thus events  $E$  and  $F$  are not mutually exclusive.

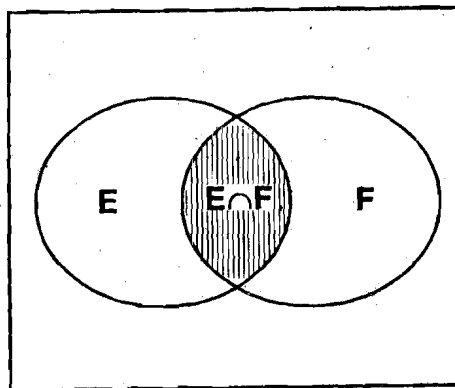
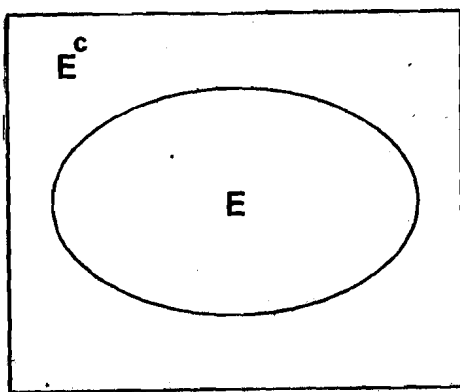


Fig. 1(a)

Fig. 1(b)

The above relations between events can be best viewed through a Venn diagram. A rectangle is drawn to represent the sample space  $\zeta$ . All the sample points are represented within the rectangle by means of points. An event is represented by the region enclosed by a closed curve containing all the sample points leading to that event. The space inside the rectangle but outside the closed curve representing E represents the complementary event  $E^c$  (See Fig.1(a) in the previous page.) Similarly, in Fig.1(b), the space inside the curve represented by the broken line represent the event  $E \cup F$  and the shaded portion represents  $E \cap F$ .

Why don't you try the following exercises now.

- E3) Suppose A, B and C are events in a sample space  $\zeta$ . Specify whether the following relations are true or false.:  
a)  $(A \cap B) \cap (A \cup B) = \phi$   
b)  $(A^c \cap B^c)^c = A \cup B$   
c)  $A^c \cap (A \cap B) = A \cup B^c$   
d)  $A \cap (B \cup B^c) = A \cap (A \cup B) = A$

After the above preliminaries, we are now ready to bring in the concept of probability of events.

2.3 PROBABILITY CONCEPTS

As is clear by now, the outcome of a random experiment being uncertain, none of the various events associated with a sample space can be predicted with certainty before the underlying experiment is performed and the outcome of it is noted. However, some events may intuitively seem to be more likely than the rest. For example, talking about human beings, the event that a person will live 20 years seems to be more likely compared to the event that the person will live 200 years. Such thoughts motivate us to explore if one can construct a scale of measurement to distinguish between likelihoods of various events. Towards this, a small but extremely significant fact comes to our help. Before we elaborate on this, we need a couple of definitions.

Consider an event E associated with a random experiment; suppose the experiment is repeated n times under identical conditions and suppose the event E (which is not likely to occur with every performance of the experiment) occurs  $f_n(E)$  times in these n repetitions. Then,  $f_n(E)$  is called the **frequency** of the event E in n repetitions of the experiment and  $r_n(E) = f_n(E)/n$  is called the **relative frequency** of the event E in n repetitions of the experiment. Let us consider the following example.

**Example 8:** Consider the experiment of throwing a coin. Suppose we repeat the process of throwing a coin 5 times and suppose the following are the frequencies of a head:

| Table 1                |                                   |                                        |
|------------------------|-----------------------------------|----------------------------------------|
| No. of repetitions (n) | Frequency of head<br>( $f_n(H)$ ) | Relative frequency of<br>head $r_n(H)$ |
| 1                      | 0                                 | 0                                      |
| 2                      | 1                                 | 1/2                                    |
| 3                      | 2                                 | 2/3                                    |
| 4                      | 3                                 | 3/4                                    |
| 5                      | 3                                 | 3/5                                    |

Notice that the third column in Table-1 gives the relative frequencies  $r_n(H)$  of heads. We can keep on increasing the number of repetitions n and continue calculating the values of  $r_n(H)$  in Table 1.



Merely to fix ideas regarding the concept of probability of an event, we present below a very naive approach which in no way is rigorous, but it helps to see things better at this stage.

### 2.3.1 Probability of an Event

While the outcome of a random experiment and hence the occurrence of any event  $E$  cannot be predicted beforehand, it is interesting to know that the relative frequency  $r_n(E)$  of  $E$  though may initially fluctuate significantly, would settle down around a constant eventually i.e.

$$r_n(E) \approx r_{n+1}(E) \approx r_{n+2}(E) \approx \dots$$

for all large values of  $n$ . We shall not give a formal mathematically rigorous proof of this phenomenon at this stage but we shall try to explain it through a simple argument (we caution you at this stage that this argument is not strictly mathematically rigorous. It merely helps you to visualise that such a result is a possibility).

**Observe** that the frequency  $f_{n+1}(E)$  of  $E$  in  $n+1$  repetitions must be equal to either  $f_n(E)$  or  $f_n(E) + 1$  depending on whether  $E$  did not or did occur at the  $(n+1)$ -th repetition. Thus,

$$\begin{aligned} r_{n+1}(E) - r_n(E) &= f_{n+1}(E)/(n+1) - f_n(E)/n \\ &= \{nf_{n+1}(E) - (n+1)f_n(E)\} / \{(n+1)n\} \\ &= \{nf_n(E) - (n+1)f_n(E)\} / \{(n+1)n\}, \text{ or} \\ &\quad \{n(f_n(E) + 1) - (n+1)f_n(E)\} / \{(n+1)n\} \\ &= -f_n(E)/n(n+1), \text{ or} \\ &\quad (n - f_n(E))/(n+1)n, \end{aligned}$$

so that

$$\begin{aligned} |r_{n+1}(E) - r_n(E)| &= \left| -\frac{f_n(E)/n}{n+1} \right| \leq \frac{1}{n+1}, \text{ or} \\ &= \left| \frac{1 - f_n(E)/n}{n+1} \right| \leq \frac{1}{n+1} \end{aligned}$$

as  $f_n(E)/n$  being the relative frequency of  $E$  in  $n$  repetitions can never exceed 1. Thus, the difference between  $r_{n+1}(E)$  and  $r_n(E)$  can be made as small as we please by increasing the number of repetitions  $n$ .

In any case, the constant around which the relative frequency of an event  $E$  settles down as the number of repetitions becomes large (i.e.  $\lim_{n \rightarrow \infty} r_n(E)$ ) is called the **probability of  $E$**  and is denoted as  $P(E)$ . Thus,  $P(E)$  can be interpreted to be the proportion of times one would expect the event  $E$  to take place when the underlying random experiment is repeated a large number of times under identical conditions. How large should the number of repetitions be in order for the relative frequency to settle down is another matter. Let us now illustrate through examples what we have discussed above.

**Example 9:** When we say that the probability of scoring a head when it is tossed is 0.5, we merely mean that 50% of a large number of tosses should result in heads. We do not mean that in 10 tosses, 5 will turn heads and 5 tails. However if the coin is tossed  $N$  times and  $N$  is sufficiently large, then we may expect nearly  $N/2$  heads and  $N/2$  tails.

\* \* \*

**Example 10:** When we say that the probability of a certain machine failing before 500 hours of operation is 2%, we merely mean that out of a large number of such machines, about 2% of them will fail before 500 hours of operation.

\* \* \*

Evidently, then such a numerical assessment of likelihood of the event in a given situation helps make an appropriate decision. For example, suppose we have two

brands A and B of an equipment available in the market such that the probability of failure before 500 hours of operation is 0.02 for an equipment of brand A and 0.10 for an equipment of brand B. We will obviously choose the former variety as the long term proportion of these failing before 500 hours of operation is less.

**Notice** that the relative frequency of any event E in n repetitions,  $r_n(E)$ , satisfies the inequalities  $0 \leq r_n(E) \leq 1$  and therefore  $P(E)$  the limit of  $r_n(E)$  as  $n \rightarrow \infty$ , must satisfy

$$0 \leq P(E) \leq 1 \quad (1)$$

Further, since  $\zeta$  denotes the event that the experiment will result into one of the outcomes listed in  $\zeta$  and since by definition, it consists of all possible outcomes,  $r_n(\zeta) = 1$ , for every n, so that

$$P(\zeta) = 1 \quad (2)$$

Consider two **mutually exclusive** events E and F. Then, to calculate  $r_n(E \cup F)$ , i.e., the proportion of times either E or F is observed, we must add the proportion of times E is observed and the proportion of times F is observed so that  $r_n(E \cup F) = r_n(E) + r_n(F)$ , and hence in the limit,

$$P(E \cup F) = P(E) + P(F), \quad (3)$$

**Remember** that E and F cannot occur together as they are mutually exclusive. Arguing in a similar manner, one observes that if  $E_1, E_2, \dots, E_k$  are k mutually exclusive events, then

$$P(\cup_{j=1}^k E_j) = P(E_1) + P(E_2) + \dots + P(E_k) \quad (4)$$

i.e., the probability of observing at least one of the k mutually exclusive events  $E_1, E_2, \dots, E_k$  (and hence exactly one) is equal to the sum of their individual probabilities.

From the above properties (1)-(4) of probability, the following useful results follow immediately:

**PROPOSITION 1:** In view of relations (1) and (2), for any event E,

$$1 = P(\zeta) = P(E \cup E^c)$$

so that for any event E,

$$P(E^c) = 1 - P(E). \quad (5)$$

You will realise later that this result is particularly useful in many problems where it is easier to compute  $P(A^c)$  than  $P(A)$ . So whenever we wish to evaluate  $P(A)$ , we compute  $P(A^c)$  and get the desired result by subtraction from unity

**PROPOSITION 2:** In Eqn.(5) above, substituting  $\zeta$  for E, so that  $\zeta^c = \phi$ , we obtain

$$P(\phi) = 1 - P(\zeta) = 1 - 1 = 0 \quad (6)$$

**Notice** that you may come across situations where the converse of result (6) is not true. That is, if  $P(A) = 0$ , we cannot in general conclude that  $A = \phi$ , for, there are situations in which we assign probability zero to an event that can occur.

**PROPOSITION 3:** Consider a finite sample space  $\zeta = \{s_1, s_2, \dots, s_m\}$ . Suppose, from the nature of the experiment, it is reasonable to assume that all the possible outcomes  $s_1, s_2, \dots, s_m$  are equally likely i.e. after taking into consideration all the relevant information pertaining to the experiment, none of the m outcomes seems to be more likely than the rest. For  $i = 1, 2, \dots, m$ , let  $E_i$  be the simple event defined as

$$E_i = \{s_i\},$$

Since, all these simple events are equally likely,

$$P(E_1) = P(E_2) = \dots = P(E_m) = p, \text{ say}$$

Now,

$$1 = P(\zeta) = P(\cup_{j=1}^m E_j) = P(E_1) + P(E_2) + P(E_3) + \dots + P(E_m)$$

so that for  $i = 1, 2, \dots, m$ ,

$$P(E_i) = p = 1/m$$

Also, any event  $E$  that consists of  $k$  of the  $m$  sample points ( $k \leq m$ ) can be represented as the union of the  $k$  simple events (which are trivially mutually exclusive). Arguing as above, we conclude that

$$P(E) = kp \quad (7)$$

$$\left. \begin{aligned} &= k/m \\ &= \frac{(\text{Total no. of sample points in } E)}{(\text{Total no. of sample points in } \zeta)} \\ &= \frac{(\text{Total no. of outcomes favouring } E)}{(\text{Total no. of the outcomes of the experiment})} \\ &= \frac{(\text{Total no. of ways } E \text{ can occur})}{(\text{Total no. of the outcomes of the experiment})} \end{aligned} \right\} \quad (8)$$

**Example 11:** Let us once again consider Example 3. Whenever the coin is unbiased,  $H$  and  $T$  are equally likely and then all the eight simple events are equally likely. Then for an event  $E$  that the three tosses produce at least one head, Formula (8) gives  $P(E) = 7/8$ , since  $E$  consists of 7 sample points and the sample space  $S$  consists of 8 sample points.

\* \* \*

**Example 12:** Suppose a fair die is rolled once and the score on the top face is recorded. Obviously, the sample space  $\zeta = \{1, 2, 3, 4, 5, 6\}$ . Since the die is fair, all the six outcomes are equally probable so that the probability of scoring  $i$  is  $1/6$ , for  $i = 1, 2, \dots, 6$ . Now, consider the event  $E$  that the top face shows an odd score; clearly,  $E = \{1, 3, 5\}$ . Thus,  $P(E) = 3/6 = 1/2$ .

\* \* \*

**Example 13:** Consider two fair dice which are rolled simultaneously and the scores are recorded. Hence, the sample space consisting of  $6 \times 6 = 36$  points is

$$\begin{aligned} \zeta &= \{(1, 1), (1, 2), \dots, (6, 6)\} \\ &= \{(i, j) : i = 1, 2, \dots, 6; j = 1, 2, \dots, 6\} \end{aligned}$$

Here, a typical sample point is denoted by  $(i, j)$  where  $i$  is the score on the first die and  $j$  is the score on the second die. Now, consider the event  $E$  that the sum of the two scores is 7 or more. Then,

$E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (3, 6), (4, 5), (5, 4), (6, 3), (4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)\}$ . Since the dice are fair, all the 36 outcomes listed in  $\zeta$  are equally likely and since  $E$  comprises of 21 sample points,  $P(E) = 21/36 = 7/12$ .

Now consider the event  $F$  that none of the two scores is even. To calculate the probability of  $F$ , we can surely list out all the outcomes, which favour this event. However, to economise on computations, we may like to use a simple argument as follows. Since none of the scores is even, then, both must be odd. Now, on each die, the possible odd scores are 1, 3 or 5. Also with each odd score on the first die, any of the three possible odd scores on the second die may be associated. Thus with each odd score on the first die, there will then be three possible results of the complete experiment and since there are 3 possible odd scores on the first die, the total number of sample points in  $F$  is  $3 \times 3 = 9$ . Thus,  $P(F) = 9/36 = 1/4$ .

Let us consider another event  $F'$  described by the condition that the only possible scores on the two dice is either 1 or 3. Then the number of possible sample points in  $F'$  are  $2 \times 2 = 4$  and  $P(F') = 4/36 = 1/9$ .

You may notice here that  $E \cap F' = \phi$ , since the sum of the two scores corresponding to any sample point in  $F'$  can be at most 6 so that  $P(E \cap F') = 0$ .

\* \* \*

**Example 14:** Consider the case of drawing marbles without replacement as in

$k$  events are mutually exclusive if no two of them have any element in common

**Example 6.** Suppose, we are interested in working out the probability of the event  $E$  that the two marbles drawn are of different colour. When we are reaching out for marbles blindly and the marbles are of identical size, we are not favouring any particular marble so that all the possible pairs are equally likely to be picked up. Now, the total number of sample points in  $S$  is 6 and the number of sample points favouring the occurrence of the event  $E$  is 4, so that  $P(E) = 4/6 = 2/3$ .

\* \* \*

And now some exercises for you

- 
- E4) In Example 14 above find the probability of the event  $E$  when drawing is done, with replacement.
- E5) The manager of a shoe store sells from 0 to 4 pairs of shoes of a recently introduced design every day. Based on experience, the following probabilities are assigned to daily sales of 0, 1, 2, 3, or 4 pairs:
- |        |   |      |
|--------|---|------|
| $P(0)$ | = | 0.08 |
| $P(1)$ | = | 0.18 |
| $P(2)$ | = | 0.32 |
| $P(3)$ | = | 0.30 |
| $P(4)$ | = | 0.12 |
|        |   | 1.00 |
- a) Are these valid probability assignments? Why or why not?
- b) Let  $A$  be the event that 2 or fewer are sold in a day. Find  $P(A)$ .
- c) Let  $B$  be the event that 3 or more are sold in a day. Find  $P(B)$
- 

Before proceeding further we have the following remarks

**Remark 1:** For a population of  $N$  objects  $a_1, a_2, \dots, a_N$ , choosing one object at random from these  $N$  objects would mean that the selection mechanism is such that each of the  $N$  objects has probability  $1/N$  of being chosen.

**Remark 2:** For a population of  $N$  objects  $a_1, a_2, \dots, a_N$ , choosing  $n$  ( $n \leq N$ ) objects at random from these  $N$  objects would mean that all the combinations of  $n$  objects out of  $N$  objects are equally likely. This can be achieved provided the selection mechanism is such that objects are selected one after the other and at each draw all the objects still remaining in the population are equally likely to be chosen.

Let us now take up some more examples to illustrate the use of Formula (8) for calculating probabilities in the cases of selections with/without replacement.

**Problem 1:** Two balls are drawn at random from a bowl containing 6 balls, of which four balls are white and two are red. What is the probability that (a) both balls are white, (b) both balls are of the same colour, and (c) at least one of the balls is white?

**Solution: Case 1: Balls drawn with replacement:** Let us number the balls as 1, 2, 3, 4, 5 and 6 such that the balls numbered as 1-4 are white while the balls numbered as 5 and 6 are red. We are required to draw 2 balls randomly with replacement. For the first draw, we have 6 choices; since the ball chosen in the first draw is sent back to the bowl before the second ball is drawn, for the second draw also, we have 6 choices. Thus, the sample space  $\zeta = \{(i, j) | i = 1, 2, \dots, 6; j = 1, 2, \dots, 6\}$  has  $6 \times 6 = 36$  sample points. Let  $E_j$  be the event that the  $j$  of the two balls selected are white,  $j = 0, 1, 2$ . Then

- a) The required probability of both the balls being white will be  $P(E_2)$ . In order for  $E_2$  to take place, the outcome  $(i, j)$  must be such that both  $i$  and  $j$  are numbers between 1 and 4. Since the ball chosen in the first draw is sent back to the bowl before the second ball is drawn, the number of sample points belonging to  $E_2$  is  $4 \times 4 = 16$ . Hence  $P(E_2) = 16/36 = 4/9$ .

b) In this case the event we are interested in is  $E_2 \cup E_0$  because  $E_2$  is the event that both selected balls are white while  $E_0$  is the event that none of the two selected ball is white implying that both must be red. Then the required probability is  $P(E_2 \cup E_0) = P(E_2) + P(E_0)$ , since these two events are mutually exclusive. In view of an argument of the previous case, the number of sample points belonging to  $E_0$  is  $2 \times 2 = 4$ . Hence the required probability is  $16/36 + 4/36 = 20/36 = 5/9$ .

c) Now the event that at least one of the two balls drawn is white is complementary to  $E_0$ . Thus the probability that at least one of the two balls drawn is white is  $1 - P(E_0) = 1 - 4/36 = 32/36 = 8/9$ .

We shall now consider the case of finding probabilities without replacement.

**Case 2: Balls drawn without replacement:** Here, we are required to draw 2 balls randomly without replacement. For the first draw, we have 6 choices; since the ball chosen in the first draw is not being sent back to the bowl before the second ball is drawn, for the second draw, we have 5 choices. Thus, the sample space  $\zeta = \{(i, j) | i = 1, 2, \dots, 6; j = 1, 2, \dots, 6; i \neq j\}$  has  $6 \times 5 = 30$  sample points (combinations of 2 out of 6 balls, keeping in mind the order in which the balls are drawn). Let  $E_j$  be the event that the  $j$  of the two balls selected are white,  $j=0, 1, 2$ .

a) The required probability will be  $P(E_2)$ . In order for  $E_2$  to take place, the outcome  $(i, j)$  must be such that both  $i$  and  $j$  are numbers between 1 and 4. Since the ball chosen in the first draw is not sent back to the bowl before the second ball is drawn, the number of sample points belonging to  $E_2$  is  $4 \times 3 = 12$ . Hence  $P(E_2) = 12/30 = 2/5$ .

b) Here the event we are interested in is  $E_2 \cup E_0$  and as such the required probability is  $P(E_2 \cup E_0) = P(E_2) + P(E_0)$ , since these two events are mutually exclusive. In view of an argument of the previous case, the number of sample points belonging to  $E_0$  is  $2 \times 1 = 2$ . Hence the required probability is  $12/30 + 2/30 = 14/30 = 7/15$ .

c) The event that at least one of the two balls drawn is white is complementary to  $E_0$ . Thus the probability that at least one of the two balls drawn is white is  $1 - P(E_0) = 1 - 2/30 = 28/30 = 14/15$ .

————— × —————

**Problem 2:** Consider a workshop that employs three mechanics in shifts to repair the machines as and when these fail. During a given shift, 4 machines failed and the repair duties were assigned to the three mechanics through a system of lottery such that for each failed machine, all the three mechanics were equally likely to be chosen. What is the probability that none of the mechanics as well as machine remained idle in that shift?

**Solution:** Suppose we name the mechanics as 1, 2 and 3 and designate the four machines as I, II, III and IV. The result of an allocation of mechanics to the four machines can be represented as  $(i, j, k, l)$  meaning that mechanic  $i$  is assigned to machine I, mechanic  $j$  is assigned to machine II, mechanic  $k$  is assigned to machine III and mechanic  $l$  is assigned to machine IV, where  $i, j, k$  and  $l$  are each numbers between 1 and 3 designating the concerned mechanic. Note that there being only three mechanics and four machines,  $i, j, k, l$  cannot be all different.

Thus,  $\zeta = \{(i, j, k, l) | i = 1, 2, 3; j = 1, 2, 3; k = 1, 2, 3; l = 1, 2, 3\}$ . Since the allocation of the same mechanic to more than one machine is feasible and conversely one or two mechanics being idle is also feasible, each possible value of  $i$  can be combined with each possible value of  $j, k$  and  $l$  and since each of  $i, j, k$  and  $l$  can adopt three values, the total number of sample points in  $\zeta$  must be  $3 \times 3 \times 3 \times 3 = 3^4 = 81$ .

Now if all the machines have to have work on that day then one of the mechanics must have been assigned two machines and the other two mechanics one each. This is the

only way the event "no machine and no mechanics is idle" can occur. In terms of the sample point  $(i,j,k,l)$  this would mean that two of the numbers  $i,j,k$  and  $l$  must be equal and the other two must be different and different from each other. For example, a typical allocation leading to all the mechanics being engaged and no machine is idle on that day can be  $(1,1,2,3)$ . To count all the possible allocations so that everybody is engaged, we have to choose two of the indices  $i,j,k$  and  $l$  that are to be identical and this can be done in  ${}^4C_2 = 6$  ways. While these two indices are identical the other two indices can be switched around in 2 ways. Thus, for example, if we allocate machines I and II to mechanic 1, then machines III and IV can be allocated to mechanics 2 and 3 respectively or to mechanics 3 and 2 respectively. Since there are altogether 3 mechanics, the total number of ways the machines can be allocated to them so that all the three mechanics are engaged and no machine is idle will be  $3 \times {}^4C_2 \times 2 = 36$ . Thus the required probability =  $36/81 = 4/9$ .

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Why don't you try the following exercises now?

- 
- E6) Suppose that in a library, two of the six copies of a book are damaged. If the library assistant selects 2 out of the six copies at random, what is the probability that he will select (i) the two damaged copies? (ii) at least one of the two damaged copies?
- E7) A student takes a multiple-choice test composed of 100 questions, each with six possible answers. If, for each question, he rolls a fair die to determine the answer to be marked, what is the probability that he answers 20 questions rightly?
- 

We now discuss the rule for finding the probability of the union of any two events, disjoint or not.

### 2.3.2 Probability of Compound Events

We have seen that for two mutually exclusive events  $E$  and  $F$ ,  $P(E \cup F) = P(E) + P(F)$ . But this formula cannot be used, for example, to find the probability that at least one of two friends will pass a language examination or the probability that a customer will buy a shirt, a sweater, a belt, or a tie at a departmental store. Both friends can pass the examination and customer of the departmental store can buy any number of these items. To find a formula for  $P(E \cup F)$  which holds regardless of whether  $E$  and  $F$  are mutually exclusive or not, let us consider the following proposition

**PROPOSITION 4:** For any two events  $E$  and  $F$ ,

$$P(E \cup F) = P(E) + P(F) - P(E \cap F). \quad (9)$$

Let us draw the Venn diagram as in Fig.2.

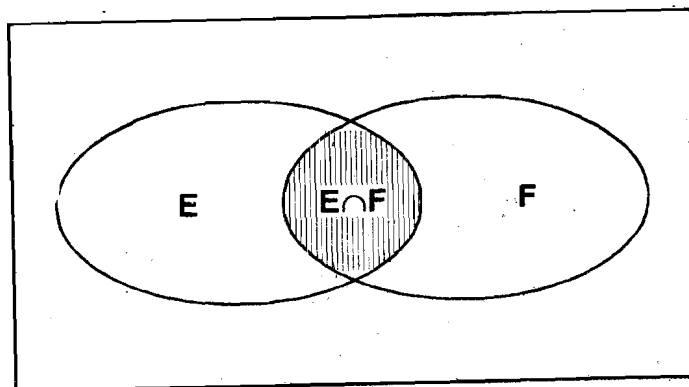


Fig.2

**Note** that each of the events E and F can be decomposed into mutually exclusive components as

$$E = (E \cap F^c) \cup (E \cap F)$$

$$F = (E^c \cap F) \cup (E \cap F)$$

Then by Formula (3)

$$P(E) = P(E \cap F^c) + P(E \cap F) \quad (10)$$

$$P(F) = P(E^c \cap F) + P(E \cap F) \quad (11)$$

Hence, by summing the two Eqns. (10) and (11), we have

$$\begin{aligned} P(E) + P(F) &= \{P(E \cap F^c) + P(E \cap F) + P(E^c \cap F)\} + P(E \cap F); \\ &= P(E \cup F) + P(E \cap F), \end{aligned} \quad (12)$$

since  $(E \cap F^c) \cup (E \cap F) \cup (E^c \cap F) = E \cup F$ , our result (9) follows from Eqn.(12).

**Observe** that the above result reduces to Formula 3 whenever E and F are mutually exclusive; this is so because if E and F are mutually exclusive, then  $E \cap F = \phi$  and  $P(\phi) = 0$ .

**PROPOSITION 5:** For any k events  $E_i, i = 1, 2, \dots, k$ , the above formula can be further generalised for calculation of the probability of occurrence of at least one of these events as follows:

$$P(\cup_{i=1}^k E_i) = S_1 - S_2 + S_3 - \dots + (-1)^{k-1} S_k \quad (13)$$

where,

$$S_1 = P(E_1) + P(E_2) + \dots + P(E_k)$$

$$\begin{aligned} S_2 &= P(E_1 \cap E_2) + P(E_1 \cap E_3) + P(E_1 \cap E_4) + \dots + P(E_2 \cap E_3) \\ &\quad + P(E_2 \cap E_4) + \dots + P(E_{k-1} \cap E_k), \\ &\quad \dots + P(E_{k-2} \cap E_{k-1} \cap E_k), \end{aligned}$$

$$\vdots$$

$$S_k = P(E_1 \cap E_2 \cap E_3 \cap \dots \cap E_k).$$

For example, whenever  $k = 4$ , Formula (13) will be as follows:

$$P(\cup_{i=1}^4 E_i) = S_1 - S_2 + S_3 - S_4$$

where,

$$S_1 = P(E_1) + P(E_2) + P(E_3) + P(E_4)$$

$$\begin{aligned} S_2 &= P(E_1 \cap E_2) + P(E_1 \cap E_3) \\ &\quad + P(E_1 \cap E_4) + P(E_2 \cap E_3) + P(E_2 \cap E_4) + P(E_3 \cap E_4), \end{aligned}$$

$$S_3 = P(E_1 \cap E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_4) + P(E_2 \cap E_3 \cap E_4),$$

$$S_4 = P(E_1 \cap E_2 \cap E_3 \cap E_4).$$

Given Proposition 4, Formula (13) can be proved by induction. We, however, do not discuss the proof here but shall illustrate it through a problem.

**Problem 3:** Suppose that a candidate has applied for admission to three management schools A, B and C. It is known that his chances of being selected by A, B, C, A and B both, A and C both, B and C both are 0.47, 0.29, 0.22, 0.08, 0.06, 0.07 respectively. It is also known that the chance of his being selected by all the three schools is 0.03. What is the probability that he will be selected by none?

**Solution:** Suppose A represents the event that he will be selected by the school A; events B and C are similarly defined. From above, we know that

$$P(A) = 0.47, P(B) = 0.29, P(C) = 0.22, P(A \cap B) = 0.08,$$

$$P(A \cap C) = 0.06, P(B \cap C) = 0.07, P(A \cap B \cap C) = 0.03. \text{ Hence,}$$

$$S_1 = P(A) + P(B) + P(C) = 0.47 + 0.29 + 0.22 = 0.98;$$

$$S_2 = P(A \cap B) + P(A \cap C) + P(B \cap C) = 0.08 + 0.06 + 0.07 = 0.21; S_3 = 0.03.$$

Thus, as per Proposition 5, the probability that he will be selected by at least one of the

schools A, B and C will be  $P(A \cup B \cup C) = S_1 - S_2 + S_3 = 0.98 - 0.21 + 0.03 = 0.80$ . Since the event that he will be selected by none is complementary to the event that he will be selected by at least one of the schools A, B and C i.e.  $A \cup B \cup C$ , the required probability  $P(A^c \cap B^c \cap C^c) = 1 - P(A \cup B \cup C) = 1 - 0.80 = 0.20$ .

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Based on Proposition 5, we may further develop an inequality, called Boole's Inequality which provides an upper bound for the probability of occurrence of at least one out of  $k$  events in terms of the probabilities of these individual events. This inequality obviously provides a way to check probability computations in a situation where it would be computationally involved to apply Formula 13.

**PROPOSITION 6:** For any  $k$  events  $E_i, i = 1, 2, \dots, k$  the above formula can be further generalised for calculating the probability of occurrence of at least one of these events as follows:

$$P(\cup_{i=1}^k E_i) \leq P(E_1) + P(E_2) + \dots + P(E_k) \quad (14)$$

Let us define  $B_j = \cup_{i=1}^j E_i, j = 1, 2, \dots, k$ . Then,

$$\begin{aligned} P(\cup_{i=1}^k E_i) &= P(B_k) = P(B_{k-1} \cup E_k) \\ &= P(B_{k-1}) + P(E_k) - P(B_{k-1} \cap E_k) \\ &\leq P(B_{k-1}) + P(E_k) \quad (\text{since probability of any event} \\ &\quad \text{is non-negative}) \\ &\leq P(B_{k-2}) + P(E_{k-1}) + P(E_k) \\ &\leq P(B_{k-3}) + P(E_{k-2}) + P(E_{k-1}) + P(E_k) \\ &\vdots \\ &\leq P(E_k) + P(E_2) + \dots + P(E_{k-2}) + P(E_{k-1}) + P(E_k) \end{aligned}$$

And now a few exercises for you.

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E8) Two dice are thrown  $n$  times simultaneously. Find the probability that each of the six combinations  $(1, 1), (2, 2), \dots, (6, 6)$  appears at least once?

E9) How many times should an unbiased coin be tossed in order that the probability of observing at least one head be equal to or greater than 0.9?

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We now introduce another concept which is important in the study of probability theory.

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## 2.4 CONDITIONAL PROBABILITY AND INDEPENDENT EVENTS

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Let  $\zeta$  be the sample space corresponding to an experiment and  $E$  and  $F$  are two events of  $\zeta$ . Suppose the experiment is performed and the outcome is known only partially to the effect that the event  $F$  has taken place. Thus there still remains a scope for speculation about the occurrence of the other event  $E$ . Keeping this additional piece of information confirming the occurrence of  $F$  in view, it would be appropriate to modify the probability of occurrence of  $E$  suitably. That such modifications would be necessary can be readily appreciated through two simple instances as follows:

**Example 15:** Suppose,  $E$  and  $F$  are such that  $F \subset E$  so that occurrence of  $F$  would automatically imply the occurrence of  $E$ . Thus with the information that the event  $F$  has taken place in view, it is plausible to assign probability 1 to the occurrence of  $E$  irrespective of its original probability.



**Example 16:** Suppose, E and F are two mutually exclusive events and thus they cannot occur together. Thus whenever we come to know that the event F has taken place, we can rule out the occurrence of E. Therefore, in such a situation, it will be appropriate to assign probability 0 to the occurrence of E.

\* \* \*

**Example 17:** Suppose a pair of balanced dice A and B are rolled simultaneously so that each of the 36 possible outcomes is equally likely to occur and hence has probability  $\frac{1}{36}$ . Let E be the event that the sum of the two scores is 10 or more and F be the event that exactly one of the two scores is 5.

Then  $E = \{(4, 6), (5, 5), (5, 6), (6, 4), (6, 5), (6, 6)\}$  so that  $P(E) = 6/36 = 1/6$ .

Also,  $F = \{(1, 5), (2, 5), (3, 5), (4, 5), (6, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 6)\}$ .

Now suppose we are told that the event F has taken place (note that this is only partial information relating to the outcome of the experiment). Since each of the outcome originally had the same probability of occurring, they should still have equal probabilities. Thus given that exactly one of the two scores is 5 each of the 10 outcomes of event F has probability  $\frac{1}{10}$ , while the probability of remaining 26 points in the sample space is 0. In the light of the information that the event F has taken place the sample points (4, 6), (6, 4), (5, 5) and (6, 6) in the event E must not have materialised. One of the two sample points (5, 6) or (6, 5) must have materialised. Therefore probability of E would no longer be  $1/6$ . Since all the 10 sample points in F are equally likely, the revised probability of E which given the occurrence of F, which occur through the materialization of one of the two sample points (6, 5) or (5, 6) should be  $2/10 = 1/5$ .

\* \* \*

The probability just obtained is called the conditional probability that E occurs given that F has occurred and is denoted by  $P(E|F)$ . We shall now derive a general formula for calculating  $P(E|F)$ .

Consider the following probability table:

**Table 2**

| Events | E | $E^c$ |
|--------|---|-------|
| F      | p | q     |
| $F^c$  | r | s     |

In Table 2,  $P(E \cap F) = p$ ,  $P(E^c \cap F) = q$ ,  $P(E \cap F^c) = r$  and  $P(E^c \cap F^c) = s$  and hence,  $P(E) = P((E \cap F) \cup (E \cap F^c)) = P(E \cap F) + P(E \cap F^c) = p + r$  and similarly,  $P(F) = q + s$ .

Now suppose that the underlying random experiment is being repeated a large number of times, say N times. Thus, taking a cue from the long term relative frequency interpretation of probability, the approximate number of times the event F is expected to take place will be  $NP(F) = N(q + s)$ . **Under the condition that the event F has taken place**, the number of times the event E is expected to take place would be  $NP(E \cap F)$  as both E and F must occur simultaneously. Thus, the long term relative frequency of E under the condition of occurrence of F, i.e., the probability of occurrence of E under the condition of occurrence of F, should be  $NP(E \cap F)/NP(F) = P(E \cap F)/P(F)$ . This is the proportion of times E occurs out of the repetitions where F takes place.

With the above background, we are now ready to define formally the conditional probability of an event given another.

**Definition:** Let E and F be two events from a sample space  $\zeta$ . The **conditional probability** of the event E given the event F, denoted by  $P(E|F)$ , is defined as

$$P(E|F) = P(E \cap F)/P(F), \text{ whenever } P(F) > 0. \quad (15)$$

When  $P(F) = 0$ , we say that  $P(E|F)$  is undefined. We can also write from Eqn.(15)

$$P(E \cap F) = P(E|F)P(F). \quad (16)$$

Referring back to Example 17, we see that  $P(E) = 6/36$ ,  $P(F) = 10/36$ ; since,  $E \cap F = \{(5, 6), (6, 5)\}$ ,  $P(E \cap F) = 2/36$ .

From Result (15),  $P(E|F) = (2/36)/(10/36) = 2/10 = 1/5$ , which is same as that obtained in Example 17.

Result (16) can be generalised to  $k$  events  $E_1, E_2, \dots, E_k$ , where  $k \geq 2$ . And now an exercise for you.

E10) Two fair dice are rolled simultaneously. What is the conditional probability that the sum of the scores on the two dice will be 7 given that (i) the sum is odd (ii) the sum is greater than 6, (iii) the outcome of the first die was odd, (iv) the outcome of the second die was even (v) the outcome of at least one of the dice was odd?

Let us now consider some more applications of Formula (16).

**Problem 4:** A blood disease is present in 12% of a population and is not present in the remaining 88%. An imperfect clinical test successfully detects the disease and with probability 0.90. Thus, if a person has the disease in the serious form, the probability is 0.9 that the test will be positive and it is 0.1, if the test is negative. Moreover, among the unaffected persons, the probability that the test will be positive is 0.05.

(a) A person selected at random from the population is given the test and the result is positive. What is the probability that this person has the disease?

(b) What is the probability that the test correctly detects the disease?

**Solution:** Let  $E$  be the event that a person has the disease and  $F$  be the event that the blood test is positive. From the given data, we note that

$$P(E) = 0.12, P(F|E) = 0.90, P(F^c|E) = 0.10, P(F|E^c) = 0.05$$

(a) We are required to compute  $P(E|F)$ .

$$\text{By definition, } P(E|F) = \frac{P(E \cap F)}{P(F)}.$$

$$\text{We have, } P(E \cap F) = P(F|E)P(E) = (0.90)(0.12) = 0.108$$

Also,

$$\begin{aligned} P(F) &= P(F \cap E) + P(F \cap E^c) \\ &= P(F|E)P(E) + P(F|E^c)P(E^c) \\ &= (0.90)(0.12) + (0.05)(0.88) \\ &= 0.108 + 0.044 \\ &= 0.152 \end{aligned}$$

$$\text{Hence the required conditional probability } P(E|F) = 0.108/0.152 = 0.7105.$$

b) Let  $G$  be the event that the test correctly detects the disease. Then  $G$  will consist of all those who actually have the disease and their blood test is positive and also those who do not have the disease and their blood test is negative.

Then we can write  $G = (E \cap F) \cup (E^c \cap F^c)$

so that  $P(G) = P(E \cap F) + P(E^c \cap F^c)$

$$\text{Now } P(E^c \cap F^c) = P(F^c|E^c)P(E^c) = \{1 - P(F|E^c)\}P(E^c) = (0.95)(0.88) = 0.836$$

$$\text{Thus } P(G) = 0.108 + 0.836 = 0.944$$

Before we go further you may try these exercises.

E11) Consider a family with two children. Assume that each child is as likely to be a boy as it is to be a girl. What is the conditional probability that both children are boys given that (i) the elder child is a boy (ii) at least one of the children is a boy?

From the definition of conditional probability, you may observe that whenever E and F are two mutually exclusive events then  $P(E|F) = 0$ ; similarly, if  $F \subset E$ , then  $P(E|F) = 1$ . In each of these cases, the knowledge of occurrence of F gives us a definite idea about the probability of occurrence of E. However, there are many situations where the knowledge of occurrence of some event F hardly have any bearing whatsoever on the occurrence or non-occurrence of another event E. To understand this let us once again consider Problem 1.

Suppose E is the event that the first ball drawn is white and F be the event that the second ball drawn is red. Let us consider the following two cases:

**Case 1: Drawing balls with replacement:** Arguing as in Problem 1,

$$P(E \cap F) = (4 \times 2)/36, P(E) = (4 \times 6)/36 = 4/6 \text{ and } P(F) = (6 \times 2)/36 = 2/6.$$

Hence,  $P(E|F) = P(E \cap F)/P(F) = 4/6$  which is the same as the unconditional probability of E.

In other words, the occurrence of F had no influence on the occurrence or non-occurrence of E. In this sense, therefore, the two events E and F are **independent**.

**Case 2: Drawing balls without replacement:** Arguing as in Problem 1,

$$P(E \cap F) = (4 \times 2)/30, P(E) = (4 \times 5)/30 = 4/6 \text{ and } P(F) = (5 \times 2)/30 = 2/6.$$

Hence,  $P(E|F) = P(E \cap F)/P(F) = 2/5$  which is different from the unconditional probability of E.

In other words, the occurrence of F had an influence on the occurrence or non-occurrence of E. In this sense, therefore, the two events E and F are **dependent**.

**Definition** Two events E and F from a given sample space are said to be **independent** if and only if

$$P(E \cap F) = P(E)P(F);$$

otherwise, they are said to be **dependent**.

Again, using the definition of conditional probability, one can equivalently say that two events E and F from a given sample space are independent if and only if

$$P(E|F) = P(E), \text{ and}$$

$$P(F|E) = P(F).$$

Consider the following example.

**Example 18:** Suppose an unbiased coin is tossed twice. Let F be the event that the first toss results in a head and E be the event that the second toss produces a head. The sample space in this case is  $\zeta = \{(H, H), (H, T), (T, H), (T, T)\}$  and the coin being unbiased, all these outcomes are equally likely so that each has probability 0.25.

Here,  $E = \{(H, H), (T, H)\}$  and  $F = \{(H, H), (H, T)\}$  so that  $E \cap F = \{(H, H)\}$ .

As such,  $P(E) = 2/4 = 0.50$ ,  $P(F) = 2/4 = 0.50$ ,  $P(E \cap F) = 0.25$ .

Thus,  $P(E \cap F) = 0.25 = (0.50)(0.50) = P(E)P(F)$ .

Hence E and F are independent events.

\* \* \*

And now some exercises for you.

**E12)** Kalpana (K) and Rahul (R) are taking a statistics course which has only 3 grades A, B and C. The probability that K gets a B is 0.3, the probability that R gets a B is 0.4. The probability that neither gets an A, but at least one gets a B is .42. The probability that K gets an A and R gets a B is 0.08. What is the probability that atleast one gets a B but neither gets a C, (assume the grades of the two students to be independent).

**E13)** The probability that at least one of two independent events occur is 0.5.

Probability that the first event occurs but not the second is  $3/25$ . Also the probability that the second event occurs but not the first is  $8/25$ . Find the probability that none of the two events occur.

- E14) Suppose that A and B are independent events associated with an experiment. If the probability that A or B occurs equals 0.6 while the probability that A occurs equals 0.4, determine the probability that B occurs.

We now end this unit by giving a summary of what we have covered in it.

## 2.5 SUMMARY

In this unit we have covered the following.

- 1) Experiments whose outcomes cannot be precisely predicted primarily because the totality of the factors influencing the outcomes are either not identifiable or not controllable at the time of experimentation, even if known, are called **random experiments**.
- 2) Random experiment may comprise of a series of smaller sub-experiments called **trials**
- 3) Each outcome of an experiment is a **sample point** and a set of all possible sample points constitute the **sample space** of the experiment.
- 4) A specific collection or subset of sample points is called an **event**.
- 5) Two events E and F are **mutually exclusive** or disjoint if they do not have a common sample point i.e.  $E \cap F = \phi$ . Two mutually exclusive events cannot occur simultaneously.
- 6) If an experiment is repeated n number of times under identical conditions and an event E occurs  $f_n(E)$  times in these n repetitions then  $f_n(E)/n$  is the **relative frequency** of the event E in n repetitions of the experiment.
- 7) The **probability** P(E) of the event E is the proportion of times an event E takes place when the underlying random experiment is repeated a large number of times under identical conditions.
- 8) For two events E and F, the probability of an event E under the condition that F has already occurred is the **conditional probability** P(E|F) of E.
- 9) Two events for which the occurrence of one has no influence on the occurrence or non-occurrence of the other are **independent** events, otherwise, they are **dependent**.

## 2.6 SOLUTIONS/ANSWERS

- E1) a)  $\zeta = \{g, d\}$ , where g stands for "good" or "non-defective" and d stands for "defective"
- b) Suppose we code the six colours by the numerals 1, 2, 3, 4, 5 and 6 and only one ball is drawn. Hence  $\zeta = \{1, 2, 3, 4, 5, 6\}$ .
- c) Let the six categories be coded as 1, 2, 3, 4, 5 and 6. Here,  $\zeta = \{1, 2, 3, 4, 5, 6\}$ .
- d) Suppose we code the ten cards by the numerals 1, 2, ..., 10.  
 $\zeta = \{(x, y) | 1 \leq x \leq 10, 1 \leq y \leq 10, x, y \text{ integers.}\}$
- E2) Suppose i is the result on the first throw and j is the result on the second throw,  $i = 1, 2, \dots, 6$  and  $j = 1, 2, \dots, 6$ . Thus, the sample space is as follows:  
 $S = \{(i, j) : i = 1, 2, \dots, 6 \text{ and } j = 1, 2, \dots, 6\}$ .

- (i) Event that the maximum score is 6  
 $= \{(6, j) : j = 1, 2, \dots, 6\} \cup \{(i, 6) : i = 1, 2, \dots, 6\}.$
- (ii) Event the total score is 9  
 $= \{(i, j) : i = 1, 2, \dots, 6, j = 1, 2, \dots, 6 \text{ and } i + j = 9\}$
- (iii) Event that each throw results in an even score  
 $= \{(i, j) : i = 2, 4, 6 \text{ and } j = 2, 4, 6\}$
- (iv) Event that each throw results in an even score larger than 2  
 $= \{(i, j) : i = 4, 6 \text{ and } j = 4, 6\}$
- v) Event that the scores on the two throws differ by at least 2  
 $= \{(i, j) : i = 1, 2, \dots, 6; j = 1, 2, \dots, 6 \text{ and } |i - j| > 2\}.$
- E3) (a) Consider  $S = \{i : i = 1, 2, \dots, 6\}$   
 Take  $A = \{1, 2, 3, 4\}, B = \{3, 4, 5, 6\}$ . Then  
 $(A \cap B) \cap (A \cup B) = \{3, 4\} \cap \{1, 2, 3, 4, 5, 6\} = \{3, 4\} \neq \phi$   
 so that the statement is false.
- (b)  $\omega \in (A^c \cap B^c)$  if and only if  $\omega \notin (A^c \cap B^c)^c$   
 ie if and only if  $\omega$  is neither in  $A^c$  nor in  $B^c$   
 i.e. if and only if  $\omega$  is in  $A$  or in  $B$   
 i.e. if and only if  $\omega \in A \cup B$ . Thus, the statement is true.
- (c) Take the example of (a) above.  
 $A^c \cap (A \cap B) = \{5, 6\} \cap \{3, 4\} = \phi \neq \{1, 2, 3, 4\} \cup \{1, 2\} = A \cup B^c$ . So the statement is false.
- (d) The statement is true, because  $B \cup B^c = S$ .
- E4) In case of drawing with replacement, total number of sample points in the sample space is given by  
 $\zeta = \{(r_1, r_1)(r_1, r_2), (r_2, r_2)(r_1, w), (w, r_1), (r_2, w), (w, r_2), (w, w), (r_2, r_1)\}$   
 No. of sample points favouring the occurrence of event E is 4, so that  
 $P(E) = 4/9$ .
- E5) (a) Yes, because the non-negative quantities associated with the mutually exclusive and collectively exhaustive simple events add up to 1.
- (b)  $P(A) = P(0) + P(1) + P(2) = 0.58$ .
- (c)  $P(B) = P(4) + P(3) = 0.42$ ; Note also that  $B = A^c$ . Hence  
 $P(B) = 1 - P(A)$ .
- E6) Let us code the books as 1, 2, 3, 4, 5 and 6. say, the damaged books coded as 1 and 2. Thus,  $P(\text{he will elect the two damaged copies})$   
 $\frac{2}{6} \times \frac{1}{5} = \frac{1}{15}$ . Also,  
 $P(\text{he will elect at least one of the two damaged copies}) = 1 - P(\text{both the books he picks up are the undamaged ones}) = 1 - \frac{4}{6} \times \frac{3}{5} = \frac{3}{5}$ .
- E7) Obviously, the total number of ways in which he can answer all the 100 questions is  $6 \times 6 \times 6 \times \dots \times 6 = 6^{100}$ . He can choose the 20 questions to be answered correctly in  ${}^{100}C_{20}$  ways. For wrong answers, he can answer each of the 80 questions in 5 ways while each of the 20 questions has one correct answer so that each such question can be answered correctly in only one way. Thus exactly 20 questions can be answered correctly and hence the remaining 80 are answered wrongly in  
 ${}^{100}C_{20} 1^{20} 5^{80}$  ways. Therefore,  
 $P(20 \text{ questions are answered correctly}) = {}^{100}C_{20} 1^{20} 5^{80}/6^{100}$   
 $= {}^{100}C_{20} (1/6)^{20} (5/6)^{80}$ .

E8) There are 36 different results possible for each throw of a pair of dice. Thus, if the pair is thrown  $n$  times, the total number of possible results is  $36^n$ . Again, there are 6 identical results on both dice each time the pair is thrown and thus these do not occur in a single throw in 30 different ways; therefore, these do not occur at all in  $n$  throws in  $30^n$  ways. So,  $P(\text{any of the six combinations } (1,1), (2,2), \dots, (6,6) \text{ appears at least once}) = 1 - (30/36)^n$ .

E9) Suppose  $n$  is the required number. The probability of no head at all in these  $n$  tosses is  $(1/2)^n$ , so that the probability of at least one head is  $1 - (1/2)^n$ . We need to determine  $n$  such that  $1 - (1/2)^n \geq 0.9$  i.e.  $(1/2)^n \leq 0.1$  i.e.  $n \ln(1/2) \leq \ln(1/10)$  i.e.  $-n \ln 2 \leq -1$  i.e.  $n \geq (1/\ln 2) = 3.32$ ; Thus, we shall take  $n$  as 4.

E10) (i) Because, the events connected with different tosses are independent

$P(\text{sum is odd})$

$= P(\text{score on one die is odd and the score on the other is even})$

$= 2 P(\text{score on die I is odd and the score on die 2 is even})$

$= 2 P(\text{score on die I is odd}) P(\text{the score on die II is even})$

$= 2 (3/6)(3/6) = 3/6$ .

$P(\text{sum of scores is } 7 | \text{sum is odd})$

$= P(\text{sum of scores is } 7 \cap \text{sum is odd}) / P(\text{sum is odd})$

$= P(\text{sum is } 7) / P(\text{sum is odd})$

$= (6/36) / (3/6) = 1/3$ .

(ii)  $P(\text{the sum is greater than } 6) = 1 - P(\text{sum is } 6 \text{ or less})$

$= 1 - (15/36) = 21/36 = 7/12$ .

$P(\text{sum of scores is } 7 | \text{the sum is greater than } 6)$

$= P(\text{sum is } 7) / P(\text{the sum is greater than } 6) = (6/36) / (21/36) = 6/21 = 2/7$ .

(iii)  $P(\text{the outcome of the first die was odd}) = 3/6$ .

$P(\text{sum of scores is } 7 | \text{the outcome of the first die was odd})$

$= P(\text{scores of } (1, 6) \text{ or } (3, 4) \text{ or } (5, 2)) / P(\text{the outcome of the first die was odd})$

$= (3/36) / (3/6) = 1/6$ .

(iv)  $P(\text{the outcome of the second die was even}) = (3/6)$

$P(\text{sum of scores is } 7 | \text{the outcome of the second die was even})$

$= P(\text{scores of } (5, 2) \text{ or } (3, 4) \text{ or } (1, 6)) / P(\text{outcome of the second die was even}) = (3/36) / (3/6) = 1/6$ .

(v)  $P(\text{the outcome of at least one of the dice was odd})$

$= 1 - P(\text{outcome of both even}) = 1 - (3 \times 3 / 6 \times 6) = \frac{3}{4}$ .

$P(\text{sum of scores is } 7 | \text{the outcome of at least one of the dice was odd})$

$= P(\text{scores of } (1, 6) \text{ or } (2, 5) \text{ or } (3, 4) \text{ or } (4, 3) \text{ or } (5, 2) \text{ or } (6, 1)) / (3/4)$   
 $= (6/36) / (3/4) = 2/9$ .

E11) (i)  $P(\text{the elder child is a boy}) = \frac{1}{2}$ .

$P(\text{both children are boys} | \text{the elder child is a boy})$

$= P(b, b) / \{P(b, b) + P(b, g)\} = (1/4) / (1/4 + 1/4) = \frac{1}{2}$ .

(ii)  $P(\text{at least one of the children is a boy}) = 1 - P(\text{both girls}) = 3/4$ .

$P(\text{both children are boys} | \text{at least one of the children is a boy})$

$= P(\text{both children are boys}) / P(\text{at least one of the children is a boy})$   
 $= (1/4) / (3/4) = 1/3$ .

E12) Let  $K_A$  be the event that Kalpana gets an A grade; the events  $K_B, K_C, R_A, R_B$  and  $R_C$  are similarly defined. Given

(a)  $P(K_B) = 0.3$ ,

(b)  $P(R_B) = 0.4$ ,

$$\begin{aligned}
 \text{(c)} \quad & P((K_B \cap R_B) \cup (K_B \cap R_C) \cup (K_C \cap R_B)) = 0.42 \\
 & \text{i.e. } P(K_B \cap R_B) + P(K_B \cap R_C) + P(K_C \cap R_B) = 0.42 \\
 & \text{i.e. } P(K_B)P(R_B) + P(K_B)P(R_C) + P(K_C)P(R_B) = 0.42 \\
 & \text{i.e. } 0.12 + .3 P(R_C) + .4 P(K_C) = 0.42 \\
 & \text{i.e. } .3 P(R_C) + .4 P(K_C) = .30 \quad (17)
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad & P(K_A \cap R_B) = .08 \\
 & \text{i.e. } P(K_A) P(R_B) = .08 \\
 & \text{i.e. } P(K_A) = .08/.4 = 0.2. \\
 & \text{From (17), } .3 P(R_C) = .30 - .4 P(K_C) = .30 - (.4)(1 - P(K_A) - P(K_B)) \\
 & = .30 - (.4)(1 - 0.2 - 0.3) = 0.10 \\
 & \text{so that } P(R_C) = 1/3.
 \end{aligned}$$

Therefore, from (a), (b), (c) & (d).

$$\begin{aligned}
 & P(\text{at least one gets a B, but neither gets a C}) \\
 & = P((K_B \cap R_B) \cup (K_B \cap R_A) \cup (K_A \cap R_B)) \\
 & = P(K_B)P(R_B) + P(K_B)P(R_A) + P(K_A)P(R_B) \\
 & = (.3)(.4) + (.3)(.2) + (.2)(1/3) = 0.2467.
 \end{aligned}$$

E13) Let A and B be the two events. Given:

$$\text{(a) } P(A \cup B) = 0.5, \quad \text{(b) } P(A \cap B^c) = 3/25, \quad \text{(c) } P(A^c \cap B) = 8/25.$$

$$\text{From (a), we have } P(A) + P(B) - P(A)P(B) = 0.5; \quad (18)$$

$$\text{from (b) we have, } P(A)\{1 - P(B)\} = 3/25, \text{ i.e. } P(A)P(B) = P(A) - 3/25;$$

$$\text{from (c), we have, } \{1 - P(A)\}P(B) = 8/25, \text{ i.e. } P(A)P(B) = P(B) - 8/25. \quad (19)$$

Hence,  $P(A) - 3/25 = P(B) - 8/25$ , so that  $P(A) = P(B) - 5/25$ . Now, from

$$\text{(18) and (19), } P(B) - 5/25 + P(B) - \{P(B) - 8/25\} = 0.5,$$

$$\text{i.e. } P(B) = 0.5 - 3/25 = 0.38 \text{ which implies } P(A) = P(B) - 5/25 = 0.38 - 0.2 = 0.18.$$

$$\text{Then } P(\text{none of the two events occur}) = P(A^c \cap B^c) = P(A^c)P(B^c)$$

$$= \{1 - P(A)\}\{1 - P(B)\} = (.82)(.62) = 0.5084.$$

E14) Given :  $0.6 = P(A \cup B) = P(A) + P(B) - P(A)P(B)$

$$= 0.4 + P(B) - 0.4 P(B), \text{ so that } P(B) = .2/.6 = 1/3.$$

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## UNIT 3 PROBABILITY DISTRIBUTIONS

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### 3.1 INTRODUCTION

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In the first two units of this block you learnt a few methods for exploring and describing numerical data. The examples you saw there showed that numerical observations on a characteristic might need careful exploration for any useful and correct inference. In general, when you decide to collect data on a characteristic, you have a specific purpose; you want to either verify a hypothesis or want to estimate a quantity for certain purpose. For example, you may want to know if the district of Jalandhar produces more wheat in a year than the district of Faridkot in the state of Punjab. You may have a hypothesis that in general, Jalandhar district produces more wheat per year than Faridkot district. When you have such a specific question or hypothesis, you will decide to collect the appropriate information or data, in this particular instance, on the annual wheat production in the two districts of Punjab. You may find a secondary source, such as a publication from an agency or a department of the government which has already collected this information and has tabulated the data. Of course, you need to be sure about the reliability of the source from where you have obtained the required data. When you actually look at the data on the amount of annual wheat production in the two districts, for say, the past ten years, what do you think you will find? Do you think the amount produced in any district will be the same from year to year? This is very unlikely as there are a very large number of factors that influence the amount of wheat produced and these factors do not remain constant over the 10 years. Also it is very unlikely that you will find Jalandhar district producing more wheat than the district of Faridkot in each of the 10 years. In such a situation how does one compare the two districts in respect of wheat production?

Firstly, it is clear from the above discussion that a realistic model will be to assume that the annual production of wheat in a district is a random variable (result of a random experiment introduced in the previous unit). The ten year observations may be thought of as the outcome of ten repetitions of a random experiment for a district. Secondly, we need to develop measures to compare the two districts with respect to the annual production of wheat when this is thought of as outcome of a random experiment.



In certain situations or for certain problems, you may find that there is no reliable secondary source where data on the appropriate characteristic are available, and that you have to take measurements or make observations, if necessary after conducting a controlled experiment. As in practice, it is never possible to control or eliminate the influence of all the factors, this experiment may also have to be modelled as a random experiment. This need arises, for instance, when a scientist claims to have developed a variety of wheat which yields more per acre than the existing varieties and you want to verify this claim. To summarise, you notice the following:

- 1) Given a specific purpose, data are collected on an appropriate characteristic.
- 2) In most real-life problem situations, the value of the characteristic of interest may depend on a large number of factors which are not constant over space or time. It is better, then, to model the situation as a random experiment and the value of the characteristic of interest as an outcome of a random experiment.
- 3) Methods have to be developed to draw the correct conclusion or inference from the data collected on a characteristic, which is the outcome of a random experiment.
- 4) If there is no secondary source of data, you may need to either conduct a controlled experiment, or conduct a survey, to obtain data on the appropriate characteristic.

We start this unit by defining the notion of a random variable and its probability distribution in Section 3.2. The notions of discrete and continuous random variables are introduced next, followed by the notions of expectation and variance. You will see that to compare random variables or to draw inferences about them in a practical application, their probability distributions are important. Also important are certain measures of the distribution such as the mean and variance. These are discussed in Sec.3.2. In the next four sections we have discussed the most commonly used four probability distributions in detail.

Here is a list of what you should be able to do by the end of this unit.

### Objectives

After reading this unit, you should be able to

- specify when a variable is a random variable and classify it as discrete or continuous.
- find the probability distribution of discrete and continuous random variables and calculate the mean and variance of these distributions and use these measures to make judgements about the real-life situation.
- describe the following distributions
  - a) binomial distribution
  - b) Poisson distribution
  - c) uniform distribution
  - d) normal distribution

and calculate the mean and variance associated with these distributions.

- distinguish between the real-life situations which can be modelled (or studied) by these distributions - when to use binomial or when to use Poisson and likewise other distributions.

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## 3.2 RANDOM VARIABLE

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Let us start with the following examples:

- 1) The number of telephone calls received by an operator in a specified interval of time.
- 2) The amount of rainfall on a day.

- 3) The outcome of throwing a die.
- 4) The scores of students in an examination
- 5) The number of misprints on a randomly chosen page of a book.
- 6) The volume of sale of a certain manufactured item in a given year.
- 7) Time to failure of a machine.

All these examples have one common feature. They describe a characteristic which associates a numerical value or number to each outcomes of a random experiment. Recall from Unit 2 that **random experiments are experiments the outcomes of which cannot be predicted in advance**. We call this characteristic a variable. This characteristic depends on the outcome of the experiment and its value cannot be predicted in advance. This variable is called a random variable. *"When a variable takes different values according to chance, (which can't be predicted before hand), it is called a random variable."* In order to make this idea clear, consider the following example.

**Example 1:** Suppose we are interested in the number  $X$  of heads obtained in three tosses of a coin. Let us see what is the variable in this experiment.

For that we first find the set of possible outcomes of the experiments i.e., the sample space  $S$ . The outcomes are

$$\begin{aligned} a_1 &= HHH, & a_2 &= HHT, & a_3 &= HTH, \\ a_4 &= THH, & a_5 &= TTH, & a_6 &= THT, \\ a_7 &= HTT, & a_8 &= TTT. \end{aligned}$$

Then

$$S = \{a_1, a_2, \dots, a_7, a_8\}$$

Let us denote by  $X(a_j)$  the number of heads obtained when the outcome of our experiment is  $a_j$ , where  $j = 1, 2, \dots, 8$ . You can easily check that

$$X(a_1) = 3, X(a_2) = X(a_3) = X(a_4) = 2, \quad (1)$$

$$X(a_5) = X(a_6) = X(a_7) = 1, X(a_8) = 0 \quad (2)$$

Then do you agree that the  $X$  maps elements of the sample space  $S$  to the values 0, 1, 2, 3? i.e.  $X$  is a function from the space  $S$  to the set  $N = \{0, 1, 2, 3\}$ .

Also note that corresponding to each value, there is always some sample point or a set of sample points. For example, the set of sample points corresponding to the value '0' is the single point  $a_8$ , whereas for 1 the set is:  $\{a_5, a_6, a_7\}$ . That means corresponding to each value of  $X$ , there is a subset of the sample space  $S$ .

Now you again recall from Unit 2 that an event is a subset of a sample space. Thus we note that each value of  $X$  is associated with an event.

You can, therefore make the following identification of events corresponding to the values associated by  $X$ . Denote the event corresponding to '0' as  $[X = 0]$ , likewise for other values. Then

$$[X = 0] = \{a_8\}, [X = 1] = \{a_5, a_6, a_7\}$$

$$[X = 2] = \{a_2, a_3, a_4\}, [X = 3] = \{a_1\}$$

Assuming that all the sample points are equally likely, we assign probabilities of  $1/8$  to each of the sample points.

From here, using the law of probability you can calculate the probabilities as follows:

$$P[X = 0] = P\{a_8\} = 1/8,$$

$$P[X = 1] = P\{a_5, a_6, a_7\} = P\{a_5\} + P\{a_6\} + P\{a_7\} = 3/8,$$

$$P[X = 2] = 3/8 \text{ and } P[X = 3] = 1/8,$$

where we read  $P[X = j]$  as "probability that  $X$  equals  $j$ ."

$$P[X = 0] + P[X = 1] + P[X = 2] + P[X = 3] = 1.$$

\* \* \*

Summing up our discussion above, we find that a random variable  $X$  is a function defined on the set of outcomes of a random experiment and it takes on a numerical value corresponding to each outcome of the experiment. Sometimes we use the abbreviation r.v. for a random variable. Now these set of possible values is a numerical representation of the original space. The following figure may help you to see this representation.

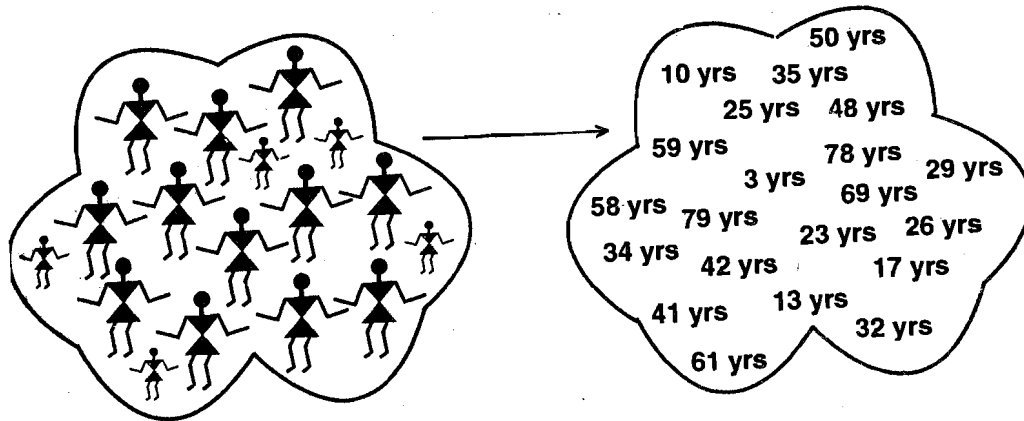


Fig. 1 Representation of a r.v.

Because of this representation, instead of working with an abstract space, we can work with a set of numbers, and this simplifies our problems considerably.

Now let us consider another example.

**Example 2:** You are sitting in a plane waiting for its take off. The pilot announces a delay until some incoming planes land. Suppose you want to find the following:

- i) How long will it be before take off.
- ii) How many incoming planes are there.

Let us discuss the random variables for (i) and (ii) of the above example.

We first take (i) In this case we want to find the 'duration of time' before the plane takes off. Note that the variable takes values continuously along a line as given below, say from time duration 'a' to time duration 'b'. No values in between a and b are left out. In other words there is no break in the values assumed by this random variable.



Fig.2

Now let us consider (ii). Here the random variable is the number of planes. This variable can only take the values 0 or 1 or 2 etc. as shown in Fig. 3. There is no continuity, (see Fig.3) since only non-negative integer values can be assumed.



Fig.3

\* \* \*

The above examples show that random variables can be of different types. There are mainly two types of random variables:

- 1) **discrete random variable**
- 2) **continuous random variable**

The random variable shown in (ii) of Example 2 is discrete and that of (i) is continuous.

In the next subsection we shall discuss discrete random variables. Before that why don't you try an exercise.

- 
- E1) Suppose you take a 50-question multiple-choice exam., guessing every answer, and are interested in the number of correct answers obtained. Then
- a) What is the random variable you will consider for this situation?
  - b) What values might this random variable have?
  - c) What would  $P[X = 40]$  means?
- 

### 3.2.1 Discrete Random Variables

We first formally define a discrete random variable and familiarise you with some of the properties/ aspects of a discrete r.v.

**Definition 1:** A random variable  $X$  is said to be discrete if the number of values that  $X$  can take is finite or countably infinite. These values may be listed as  $x_0, x_1, \dots$ , where say,  $x_0 < x_1 < \dots$ ; these  $x$ 's are called the jump points. **They need not be equidistant.**

Now let us consider the events associated with the values assigned by it.

Let the events be denoted by  $[X = x_i], i = 0, 1, 2, \dots$ . Then, as stated earlier, we can assign probability to these events. We denote  $P[X = x_j]$ , the probability of the event  $[X = x_j]$ . For further simplification we denote the probability for each  $j$  as  $P[X = x_j] = p_j, i = 0, 1, 2, 3, \dots$

From the properties of a random variable and by definition of probability it follows that

- i)  $p_i \geq 0$  (for each  $i$ , i.e., each  $p_i$  is a non-negative number),
- ii)  $p_0 + p_1 + \dots = 1$  (The sum of the probabilities is 1).

Now we have another definition.

**Definition 2:** Let  $p : X \rightarrow R$  be defined as

$$p(x) = \begin{cases} p_i & \text{if } x = x_i, \quad i = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

Then  $p$  is called the **probability mass function (p.m.f)** of the random variable  $X$ . The collection of pairs  $(x_i, p_i), i = 0, 1, 2, \dots$  is called the **probability distribution of  $X$** .

For example, suppose  $X$  is the r.v. denoting the number of heads obtained in three tosses of a coin, then the probability mass function  $p$  is the function  $p : X \rightarrow R$  such that

$$p(0) = 1/8, p(1) = p(2) = 3/8, p(3) = 1/8.$$

Note that  $p(x_i) = p_i \geq 0$  for all  $x_i$  and

$$\sum p_i = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1.$$

Therefore, the probability distribution corresponding to this random variable is the set  $\{(0, \frac{1}{8}), (1, \frac{3}{8}), (2, \frac{3}{8}), (3, \frac{1}{8})\}$ . This is also expressed in a tabular form as follows:

Table 1: Probability distribution of number of heads in three tosses of a coin

| The number of heads<br>(Values of the r.v. $X$ ) | Probability |
|--------------------------------------------------|-------------|
| 0                                                | $1/8$       |
| 1                                                | $3/8$       |
| 2                                                | $3/8$       |
| 3                                                | $1/8$       |

Suppose you have a random variable  $X$  assuming values  $x_1, x_2, \dots$  with probabilities  $p_1, p_2, \dots$ , respectively. You may also visualise this as an illustration of a frequency distribution. In fact a frequency distribution tells you how the total probability “one” is distributed over the possible values of the random variable.

Now let us see the graphical representation of this distribution.

Graphically, along the horizontal axis, plot the various possible values  $x_i$  of a random variable and on each value erect a vertical line with height proportional to the corresponding probability  $p_i$ . (See Fig.4) Recall that in the bar diagram of a frequency distribution, the observed frequencies are graphed along the vertical axis; the total frequency (which is the same as the total number of repetitions of the random experiment) is thus distributed over the possible outcomes.

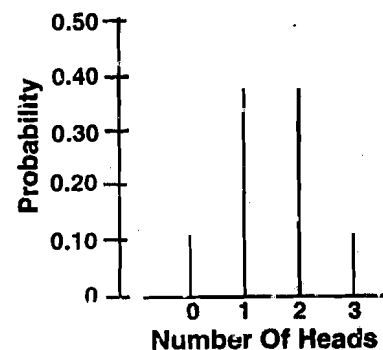


Fig. 4

It is therefore clear that **associated with any random variable, a probability distribution can be defined**. Thus in the study of a random variable it is enough to know the corresponding probability distribution. This is illustrated in the following example.

**Example 3:** Recall the problem of Sunil, the newspaper boy, which was presented in in Section 1, Unit 2. When Sunil mentioned his dilemma about his 10 irregular customers to his sister Sunita, who is doing a course in statistics at the local college, she advised him, as a first step, to start maintaining a record for each of the 10 customers, showing for each day whether the customer has taken the newspaper from him or not. Following her advise, Sunil had at the end of two months, 61 sets of observations, each set corresponding to a day. Each set of observations was written as a sequence of two numbers, 1's and 0's, 1 at position  $i$  showing that customer  $i$  has bought the newspaper on that day and a 0 in that position showing that customer  $i$  has not bought the newspaper on that day. You will also find that the  $k$ th sequence (corresponding to the  $k$ th day) is actually an observed sample point of the sample space corresponding to the random experiment performed on day  $k$  with these 10 customers. Sunil has repeated this random experiment with his 10 customers for 61 days! When Sunil reflected over the mass of observations he has made it suddenly occurred to him that his record may be excessive for solving his problem. Do you also get the same idea as Sunil did? If not, think about it for a while now, and then read ahead.

Sunil reasoned as follows: After all, his daily gain will only depend on how many newspapers he is able to sell on a particular day, irrespective of who among the 10 buys. Therefore, it is enough for his purpose to note down the number of 1's that appear in the  $k$ th sequence corresponding to day  $k$ , to calculate his gain for that day. So why does he have to maintain a sequence? Just the total number of customers buying on a day should serve the purpose. Do you agree with Sunil? When Sunil showed his diary to Sunita and mentioned to her his new idea, she appreciated his line of thought and told him that the variable he wanted to consider, namely the total number of 1's in a sequence, would be called a random variable by a statistician as the sequences were the observed results of a random experiment. Sunil could forget about the sample space containing the sequences, provided he knew the probability distribution of the random variable chosen by him, namely the number of sales on a day to his irregular customers.

We shall consider the problem of finding the probability distribution of this random

\*\*\*

In the following sections you will see that there are some standard distributions and most of the real-life problems can be solved by finding the distribution corresponding to a probability model of the given situation. Let us try some exercises.

E2) Which of the random variables given below are discrete? Give reasons for your answer.

- 1) The daily measurements of snowfall at Shimla
- 2) The number of industrial accidents in each month.
- 3) The number of defective goods in a shipment (lot) of goods from a manufacturer.

E3) A box contains twice as many red marbles as green marbles. One marble is drawn at random from the box and replaced; then a second marble is drawn at random from the box. If both marbles are green, you win Rs. 50; if both are red you lose Rs. 10; and if they are of different colours, you will win or lose nothing. Then what is the probability distribution of the amount you win or lose.

Let us now return to the discussion of the three tosses of an unbiased coin. The r.v.  $X$ , denoting the number of heads obtained, has the following probability distribution:

$$p_0 = \frac{1}{8}, p_1 = \frac{3}{8} = p_2, p_3 = \frac{1}{8}$$

Suppose you want to calculate the probability of the event  $\{X \leq 2\}$ .

First note that the event  $\{X \leq 2\}$  is the same as the event  $\{X = 0\} \cup \{X = 1\} \cup \{X = 2\}$

Then, since the events are disjoint, we can write,

$$\begin{aligned} P[X \leq 2] &= P[X = 0] + P[X = 1] + P[X = 2] \\ &= \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}. \end{aligned}$$

Therefore there are situations in which you are not only interested in the probability  $P[X = x_i] = p_i$ , but are also interested in the probability of the  $X \leq x_i$ , i.e., events of the type  $[X \leq x_i]$  which we denote by  $P[X \leq x_i]$ .

As we already pointed out, the probability distribution of a random variable is analogous to a frequency distribution. It is therefore not surprising that just as we found it useful to calculate the mean of a frequency distribution, we now find it useful to calculate the mean of a probability distribution. Another term often used to denote the mean of a probability distribution is the expected value of the random variable which is defined as follows:

**Definition 3:** For a discrete random variable  $X$ , its **expected value** (or mean), denoted as  $E(X)$ , is defined as:

$$E(X) = x_0p_0 + x_1p_1 + \dots$$

where  $x_0, x_1, x_2$  are the values assumed by  $X$  and  $p_0, p_1, p_2$  are probabilities these values.

Expected value is a fundamental idea in the study of probability distributions. For many years, the concept has been put to considerable practical use in the insurance industry, and in the last twenty years, it has been widely used by many others who must make decisions under conditions of uncertainty.

We shall illustrate this idea with a real-life problem given in Example 4.

**Example 4:** The Director of a breast cancer screening clinic wants to know how many women will be screened on any one day. If past daily records of the clinic indicate that the number of women screened daily ranges between 100 to 115. The following table illustrates the number of times this level, between 100 to 115, has been reached during the last 100 days.

**Table-2 : Number of women screened daily during 100 days**

| Number screened | Number of days<br>this level was observed | Probability that the random variable<br>will take on this value |
|-----------------|-------------------------------------------|-----------------------------------------------------------------|
| 100             | 1                                         | .01                                                             |
| 101             | 2                                         | .02                                                             |
| 102             | 3                                         | .03                                                             |
| 103             | 5                                         | .05                                                             |
| 104             | 6                                         | .06                                                             |
| 105             | 7                                         | .07                                                             |
| 106             | 9                                         | .09                                                             |
| 107             | 10                                        | .10                                                             |
| 108             | 12                                        | .12                                                             |
| 109             | 11                                        | .11                                                             |
| 110             | 9                                         | .09                                                             |
| 111             | 8                                         | .08                                                             |
| 112             | 6                                         | .06                                                             |
| 113             | 5                                         | 0.05                                                            |
| 114             | 4                                         | .04                                                             |
| 115             | 2                                         | .02                                                             |
| Total - 100     |                                           | Total - 1.00                                                    |

Let us see how the Director can use this past record to get information about the long-run pattern of daily patient screenings.

We hope you have understood the real-life problem. We will try to model this in statistical language. For that we shall first describe the 'random variable' of interest in this problem.

The random variable here is the number of patients screened, on any given day. Note that this is a discrete random variable, which can assume only non-negative integer values with positive probability. The past record of the clinic indicates that the values of this random variable range between 100 and 115 patients daily. These values are given in the 1st column of the table. The 2nd column contains the number of days each value was observed. For example the value '103' occurred on 5 days. The last column gives the probability/relative frequency for which a particular value is observed. How do you calculate these probabilities? Note that the total number of days is 100 and that the value '100' was observed only one day.

$$\text{Probability that the value '100' is observed} = \frac{1}{100} = .01$$

In this manner you can calculate the other probabilities. (The thumb rule is 'divide each value in the middle column by 100'). This is how the last column is obtained. Notice that the sum of the values in the last column is one. The relative frequencies are taken as probabilities. This is the statistician's empirical approach to assigning probabilities.

Now plot the 'observed values' (i.e. numbers in the 1st column) against the probabilities in a graph. Then you get a graph as given in the next page.

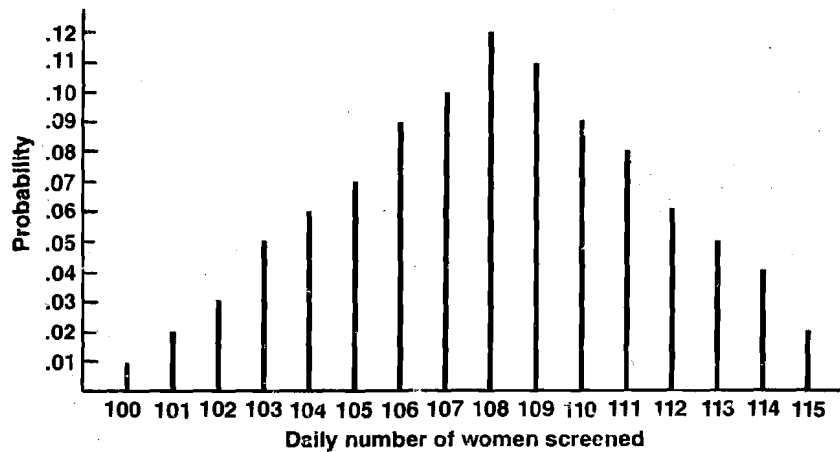


Fig.5: Probability distribution for the discrete random variable 'number screened'

So, this is the graph of the probability distribution of the r.v.

Once we have a frequency distribution, we know that we can find its mean, variance etc. Let us calculate the mean. Here is the mean

$$\text{Mean} = \frac{\sum x_i p_i}{\sum p_i}$$

where  $x_i$ 's are the observed values in the first column and  $p_i$ 's are the frequencies

$$\text{Mean} = \frac{100 \times 1 + 101 \times 2 + \dots + 115 \times 2}{100}$$

This we can rewrite as

$$\begin{aligned} \text{Mean} &= 100 \times \frac{1}{100} + 101 \times \frac{2}{100} + \dots + 115 \times \frac{2}{100} \\ &= 100 \times .01 + 101 \times .02 + \dots + 115 \times .02 \\ &= 108.02 \end{aligned}$$

So, essentially what we got is that 'if we multiply each number in the 1st column by the corresponding number in the 3rd column' we will get the mean of the probability distribution. This mean equals the expected value.

The above computation tells us that the expected value of the discrete random variable "number screened" is 108 women. What does this mean? It means that over a long period of time, the number of daily screenings should average about 108.02. This does not mean that 108 women will visit the clinic per day; this only says that over the **long run** and on an average 108 women can visit the clinic. This value is a long run average.

Now based on this expected value (or the mean) the director can decide on what resources/infrastructure is required to get ready for dealing with the expected number of people.

\*\*\*

Why don't you try an exercise now.

- E4) A second-hand car dealer has sold as many as five cars in one day, and as few as one. The dealer has tabulated sales records for a large number of days and found that on 5 percent of the days no cars were sold. The dealer took 0.05 as the probability of zero sales in a day, as shown in Table 3 below. Probabilities for sales of 1,2,3,4 and 5 cars were assigned in the same manner (see Table below).

| Table 3                     |      |      |      |      |      |      |
|-----------------------------|------|------|------|------|------|------|
| Number of cars sold per day | 0    | 1    | 2    | 3    | 4    | 5    |
| Probability                 | 0.05 | 0.15 | 0.35 | 0.25 | 0.12 | 0.18 |

He wants to find how many cars per day will be sold on the average over a long



### 3.2.2 Continuous Random Variable

The variables, we discussed in the last section such as 'number of women screened', 'number of heads obtained', etc. take on values 0, 1, 2, 3, .... These are discrete random variables. We saw that the values of a discrete random variable are graphed as separated points and probabilities as lengths of vertical line segments (see Fig.4). We also saw that a probability distribution of such a random variable contains all possible values of the random variable, so the sum of all the probabilities must be 1.

On the contrary, a continuous random variable can take on any value in an interval.

For example, it can take all values  $x$  in the interval  $0 \leq x \leq 1$  of the form 0, 0.01, 0.0002, etc. ...., 0.98, 0.99, 1.00. As we have seen in the discrete case, we have to assign probabilities to each value of the variable. How do we do this? Since the possible values of  $X$  are uncountable, we cannot really speak of the  $i^{\text{th}}$  value of  $X$ , hence  $p(x_i)$  becomes meaningless. So, what we shall do is to replace  $x_i$  by any interval of the type of  $(x_{i-1}, x_i)$  where  $0 \leq x_{i-1} < x_i \leq 1$ , and define probability for such intervals. To define this we consider a function defined on this interval  $0 \leq x \leq 1$ , which assumes non-negative values such that the total area under the graph of this function and over the interval is 1. Then we define the probability of any interval  $(x_{i-1}, x_i)$  as the area over this interval and under the graph of  $f$ . (See Fig.6(b)) At present you don't have to worry about these functions. Wherever the need arises we will specify these functions.

The above discussion may appear a little vague to you. You need not worry about this at this stage. The ideas will be clear to you once we discuss particular cases of continuous random variables and their probability distributions.

Now let us compare the graphs of probability distributions drawn for a discrete and continuous random variable.

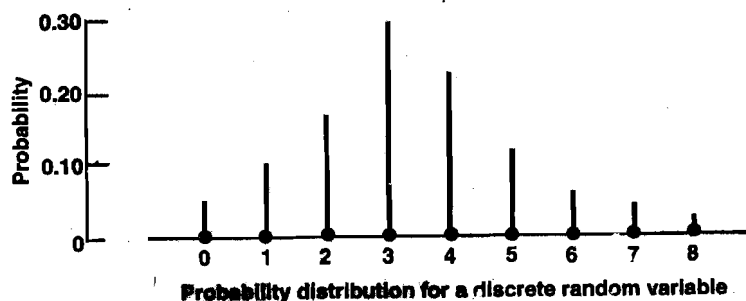


Fig.6(a)

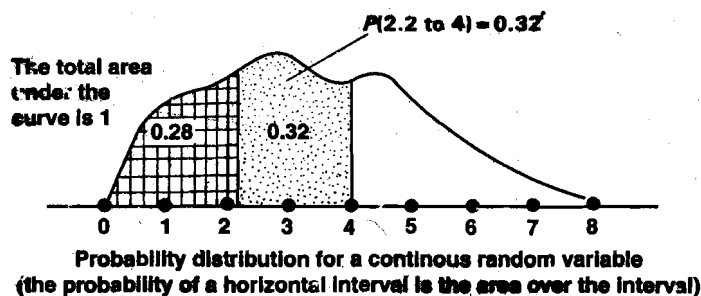


Fig.6(b)

Note that in Fig.6(b) we did not put vertical lines because a vertical distance tells us what the height of the curve is, and that is not what we want. We want areas under the portion of curves. You will see later that there are tables available for certain continuous random variables where these areas are given for different segments. For

examples in Fig. 6(b), the probability of the line segment  $2.2 \leq x \leq 4$  is

$$P[2.2 \leq x \leq 4] = 0.32$$

and that of  $0 \leq x \leq 2.2$  is

$$P[0 \leq x \leq 2.2] = 0.28$$

It is important to remember that a line segment has no area (zero area), because a line has no width. Thus, the vertical line segment at 4 in the distribution of Figure 6 (b) has an area of zero. This means the probability of a single value 4 is zero. In general, the **probability for an exact (single) value of a continuous random variable is zero**. Consequently, the probability of an interval is the same whether the endpoints are included or not — because the endpoints have probability zero.

Now we formalise the above discussion and make the following definition.

**Definition 4:** Let  $X$  be a continuous random variable which takes on values in the interval  $(a,b)$ . [i.e. all values between  $a$  and  $b$ ,  $a < b$ ]. A function  $f(x)$  defined on  $X$  is called the **probability density function of  $X$**  if

- (i)  $f(x)$  is nonnegative for  $a \leq x \leq b$  i.e.,  $f(x) \geq 0$  for all  $x$  lying between  $a$  and  $b$ .
- (ii) the area under the graph and above the interval  $(a, b)$  is 1.
- (iii) For any two real numbers  $c$  and  $d$  between  $a$  and  $b$ , the probability that the random variable **adopt** a value between  $c$  and  $d$  is equal to the area under the graph and above the interval  $(c, d)$  i.e.

$$P[c \leq x \leq d] = \text{area under the graph over the interval}(c, d)$$

**Note:** Those who are familiar with the mathematical concept of integration can easily see that the above mentioned area is given by the integral of the function i.e.

$$P[c \leq x \leq d] = \int_c^d f(x)dx$$

and (ii) says that

$$\int_a^b f(x)dx = 1$$

We shall now see how we can use the graph of the distribution of a continuous random variable to study real-life problems.

**Problem 1:** Suppose the Director of a training programme wants to conduct a programme to upgrade the supervisory skills of production line supervisors. Because the programme is self-administered, supervisors require different numbers of hours to complete the programme. Based on a past study of participants, the following distribution (see Fig.7) showing the time spent by candidates is available which shows that average time spent is 500 hours.

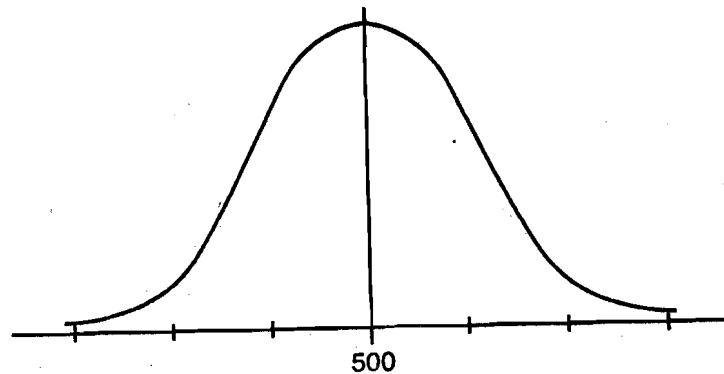


Fig.7

How can she use the graph of distribution, to find the following: What is the chance that a participant selected at random will require

- i) more than 500 hours to complete the programme?
- ii) less than 500 hours to complete the programme?

**Solution:**

- i) From the figure, we see that half of the area under the curve is located on either side of the mean of 500 hours. Thus, we get that the probability that the random variable will take on value higher than 500 is one half, or 0.5.
- ii) A similar argument shows that the chance is 0.5.

————— × —————

Why don't you try some exercise now.

E5) Suppose  $X$  is a continuous random variable defined on  $[2, 4]$  and  $f$  is a function of  $[2, 4]$  such that

$$f(x) = \begin{cases} \frac{1}{2} & \text{for } 2 < x < 4 \\ 0, & \text{elsewhere} \end{cases}$$

- i) Draw a rough sketch of  $f(x)$ ,
- ii) Does  $f$  define a probability density function of  $f$ ? if so why?

E6) Classify the r.v.'s given at the beginning of Sec. 3.1 as discrete or continuous.

In the following sections we shall discuss some standard distribution.

### 3.3 BINOMIAL DISTRIBUTION

One of the important discrete random variables (or, discrete distributions) is the binomial variable. In this section we shall discuss this random variable and its probability distribution.

Many times we have to deal with experiments where there are only two possible outcomes. For example, when a coin is tossed, either head or tail comes up, seed either generates or fails to generate, a newborn is either a girl or boy.

Let us consider such an experiment. For example, consider the experiment of tossing a fair coin 3 times. This experiment has certain characteristic. First of all, it involves **repetition of three identical experiments (trials)**. Each trial has **only two possible outcomes** - a head or tail. We call outcome "head" success and outcome "tail" a failure. All trials are independent of each other. We also know that probability of getting a head in a trial and probability of getting a tail in a trial are both  $\frac{1}{2}$ .

$$P(H) = P(\text{success}) = \frac{1}{2}$$

and

$$P(T) = P(\text{failure}) = \frac{1}{2}$$

This shows that the **probability of a "success" and of a "failure" do not change from one trial to another.**

If  $X$  denotes the total number of heads, obtained in 3 trials, then  $X$  is a random variables which takes values from  $\{0, 1, 2, 3\}$ .

Suppose that  $p$  denote the probability of a success (i.e. getting a head) and  $q$  denote the probability of failure (i.e. getting a tail).

Then regarding the above experiment, we have observed the following:

- 1) It involves a repetition of  $n$  identical trials (Here  $n = 3$ ).
- 2) The trials are independent of each other.
- 3) Each trials has two possible outcomes
- 3) The probabilities of a "success" ( $p$ ) and of a "failure" ( $q$ ) do not change.

If you go back for a moment to Sec. 3.1, you will see that we have already obtained the probability distribution of this in Example 3. Let us look at the probabilities once again.

$$\begin{aligned}
 P[X = 0] &= P[\text{getting three tails}] \\
 &= P[T, T, T] = q \times q \times q \\
 &= q^3 = \frac{1}{8}
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 P[X = 1] &= P[[TTH], [THT], [HTT]] \\
 &= P[TTH] + P[THT] + P[HTT] \\
 &= q^2p + q^2p + q^2p = 3q^2p
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 P[X = 2] &= P[THH] + P[TTH] + P[HHT] \\
 &= 3p^2q
 \end{aligned}$$

and

$$P[X = 3] = p^3$$

In fact the probability  $P[X = r]$ ,  $r = 0, 1, 2, 3$  gives that if we toss a coin three times, how many ways, or combinations, will yield  $r$  heads and  $n - r$  tails.

Now you recall from your school mathematics that the number of combinations of  $n$  objects taken  $r$  at a time is calculated by the formula

$$c(n, r) = \frac{n!}{r!(n-r)!}$$

In the case of tossing of three coins,  $n=3$ . Therefore, we rewrite the probabilities as

$$\begin{aligned}
 P[X = 0] &= C(3, 0)p^0q^{3-0} = q^3 \\
 P[X = 1] &= C(3, 1)p^1q^{3-1} = 3pq^2 \\
 P[X = 2] &= C(3, 2)p^2q^{3-2} = 3p^2q \\
 P[X = 3] &= C(3, 3)p^3q^0 = p^3
 \end{aligned}$$

This suggests that the probability  $p[X = r] = p_r$  for a given  $r$  can be calculated using the formula

$$p_r = C(n, r)p^r q^{n-r} \quad (3)$$

where  $r$  = number of successes

$n$  = number of trials made

$p$  = probability of success in a trial

$q = 1 - p$  = probability of failure in a trial.

Why don't you check this formula for  $n=5$ , i.e. tossing of a coin 5 times. For example, try this exercise.

- 
- E7) In the experiment of tossing a coin 5 times find the probability of getting 3 heads and 2 tails. Verify that this probability is given by the formula given in Equation (3).
- 

Let us now sum up the points we have observed in the example, above.

**An experiment consisting of  $n$  trials is performed such that**

- i) each trial has two possible outcomes, viz., a 'success'(p) and a "failure"(q);
- ii) the probability of success, p, is the same for any trial;
- iii) the outcomes of different trials are statistically independent (i.e. the trials are independent).

These trials are called Bernoulli trials.

The **sample space** of this experiment consists of elements like "SSFSSF....." of length n of 'S's and 'F's where S stands for the success and F stands for the failure.

Let X represent the number of successes (in any order whatsoever) in the set of n trials. Then X is a **discrete random variable taking integral values 0, 1, ..., n**. The **probability of the event  $P[X = r]$  is given by**

$$P[X = r] = p_r = C(n, r)p^r q^{n-r} \quad (4)$$

where r = exact number of successes

n = number of trials made

p = probability of success on a trial

q = 1 - p = probability of failure on a trial and

$$C(n, r) = \frac{n!}{(n-r)!r!} \text{ (The earlier example illustrates how we got this formula for } p_r \text{).}$$

Such a random variable X is called a **binomial random variable** and its probability distribution is called **binomial distribution** and is given by Eqn.(2).

**Problem 2:** A sales representative, calls on four potential clients. The probability that she will obtain an order from each of them is  $\frac{1}{2}$  and whether or not she obtains an order from one of them is statistically independent of whether or not she obtains an order from any of the others. What is the probability distribution of the number of orders she will receive?

**Solution:** We note that there are two mutually exclusive events (obtaining an order or no order) each time she makes a call and the probability of an order  $\frac{1}{2}$  each time. Also the outcomes of the calls are statistically independent. Therefore this is a situation where there are four Bernoulli trials and where the probability of a success (an order) equals  $\frac{1}{2}$ . Substituting  $n = 4$  and  $p = \frac{1}{2}$  in Eqn.(4), we get that

$$P(0) = \frac{4!}{0!4!} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^4 = \frac{1}{16},$$

$$P(1) = \frac{4!}{1!3!} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^3 = \frac{1}{4},$$

$$P(2) = \frac{4!}{2!2!} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{3}{8},$$

$$P(3) = \frac{4!}{3!1!} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^1 = \frac{1}{4},$$

$$P(4) = \frac{4!}{4!0!} \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right)^0 = \frac{1}{16},$$

Thus, the probability of no orders is  $\frac{1}{16}$ , of one order is  $\frac{1}{4}$ , of two order is  $\frac{3}{8}$ , of three orders is  $\frac{1}{4}$ , and of four orders is  $\frac{1}{16}$ .

————— x —————

**Problem 3:** It has been claimed that in 60% of all solar heat installations, the utility bill is reduced by at least one-third. Accordingly, what are the probabilities that the utility bill will be reduced by one-third in

- i) four of five installations?
- ii) at least four of the five installations?

#### Binomial Distribution



James Bernoulli was a seventeenth century Swiss mathematician who performed some of the early work on binomial distribution.

**Solution** Here the random variable follows binomial distribution with  $p = 0.6$ ,  $r = 4$  and  $n=5$ .

To find (i), we have to calculate  $P[X = 4]$ , which is given by

$$\begin{aligned} P[X = 4] &= C(5, 4)(0.6)^4(0.4) \\ &= 0.259 \end{aligned}$$

Now to find (ii), we have to find the probability that  $X$  is at least 4. This probability is the sum of the probabilities that  $X = 4$  and  $X = 5$  because 'at least 4 means 4 or more'. Thus we have to find  $p[X = 4] + P[X = 5]$ .

$$\begin{aligned} P[X = 5] &= (5, 5)(0.6)^5 \\ &= 0.078 \end{aligned}$$

$\therefore$  the required probability  $= 0.259 + 0.078 = 0.337$ .

————— × —————

Binomial distribution is very applicable in situations where we have to decide whether to accept a lot of goods (items) coming out of a manufacturing process. This decision is based on how many defective items are in the lot. Companies (or firms) will generally return the entire items if there is evidence that more than certain items is defective. To make such decision, let us see how we can make use of the binomial distribution.

An item coming out of a manufacturing process can either be defective or non defective. Consider a lot of  $N$  items produced by the manufacturing process. Let  $m$  of these be defective. Suppose a quality control inspector draws a random sample of  $n$  items from the lot, one by one, with replacement (i.e. an item drawn is put back in the lot, after noting down whether it is defective or non defective, before the next item is drawn at random). Let  $X$  be the number of defective items drawn by the inspector. Note that there are  $n$  trials and in each trial the probability that a defective item is picked remains the same, namely  $\frac{m}{N}$ , as the drawing is done with replacement and at random. Also note that the trials are independent. Therefore, the random variable  $X$  defined above is distributed as a Binomial  $(n, p)$  where  $p = \frac{m}{N}$ .

By now you must have got some idea for recognising those situations where we can apply binomial formula. If we can apply binomial distribution to study a situation, then we say that the situation can be modelled by binomial distribution.

Here are some exercises for you.

- 
- E8) A farmer buys a quantity of cabbage seeds from a company that claims that approximately 80 % of the seeds will germinate if planted properly. If four seeds are planted, what is the probability that exactly two will germinate?
- E9) Consider again the data collected by Sunil, the newspaper boy. When Sunita, the statistics student, saw the data, she started wondering if the number of customers from among his ten irregular customers, who actually buy from him on a given day, will follow a binomial distribution? What do you think? Under what conditions will this random variable follows a binomial distribution.
- E10) Sunita was still glancing through Sunil's diary wondering to herself if she could think of the 10 customers as 'ten identical coins', when she noticed something significant. She noticed that a lot more of the sequences had a 1 in the third position than in the 8th position. Sunil remembered that customer 3 was the management trainee whom he called 'Alka Didi'. She was from a neighbouring town and was undergoing training in a software company. She was interested in news about software companies, science and environmental issues. She would often buy from Sunil but not always. Customer 8, Sunil told his sister, was a mysterious young man by name Kapil, who was rumoured to be working for a detective agency. One could rarely find him in the morning hours and if he was at

home in the morning hours, he would certainly buy from Sunil.

Given the situation above, do you still think that the number of sales on a day can be modelled as a binomially distributed random variable? Give reasons for your answer.

Once we have the probability distribution, we naturally ask what is the 'expected value'. We shall see that now.

### Expected Value of a Binomial Variable

We have already seen in Sec.3.2 that for a discrete random variable  $X$ , the 'Expected Value'  $E(X)$  is

$$E(X) = x_0p_0 + x_1p_1 + \dots$$

where  $x_0, x_1, \dots$  are the values assumed by  $X$  and  $p_0, p_1, \dots$  are the probabilities associated with these values i.e.

$$P[X = x_i] = p_i, \quad i = 0, 1, 2, \dots$$

If  $X$  is a binomial r.v., taking values  $0, 1, \dots, n$ , then, we know that

$$P[X = i] = C(n, i)p^i(1-p)^{n-i}$$

$$\therefore E(X) = n \times C(n, n)p^n(1-p)^0 + (n-1)C(n, n-1)p^{n-1}(1-p)^1 + \dots + 0 \times C(n, 0)p^0(1-p)^n$$

We rewrite this expression in the sum notation  $\sum$  (called sigma) as

$$\begin{aligned} E(X) &= \sum_{j=0}^n j C(n, j)p^j(1-p)^{n-j} \\ &= np \sum_{j=1}^{n-1} C(n-1, j-1)p^{j-1}(1-p)^{n-j} \end{aligned}$$

Those who are familiar with binomial expansion can recognise that the second expression on the R.H.S. is  $1 - p + p)^{n-1}$ . Therefore we have

$$\begin{aligned} E(X) &= np[1 - p + p]^{n-1} \\ &= np. \end{aligned}$$

This means that the expected number of successes is  $np$ .

Let us do a problem.

**Problem 4:** An oil exploration firm plans to drill six holes. It is believed that the probability that each hole will yield oil is 0.1. Since the holes are in quite different locations, the outcome of drilling one hole is statistically independent of that of drilling any of the other holes.

- If the firm will be able to stay in business only if two or more holes produce oil, what is the probability of its staying in business?
- Give the expected value of the number of holes that result in oil.

**Solution:** (a) If the firm can stay in business only if two or more holes produce oil, it follows that the probability that it will stay in business equals 1 minus the probability that the number of holes resulting in oil is 0 or 1. Each hole drilled can be viewed as a Bernoulli trial where the probability of success is .1. Thus, the probability that the number of successes is 0 or 1 equals:

$$\begin{aligned} P(0 \text{ or } 1) &= P(0) + P(1) = \frac{6!}{0!6!}(.9)^6 + \frac{6!}{1!5!}(.1)(.9)^5 \\ &= .531 + .354 = .885. \end{aligned}$$

Consequently, the probability that the firm will be able to stay in business is  $1 - .885 = .115$ .

(b) The expected value of the number of holes yielding oil is  $6 \times 0.1 = 0.6$ , since  $n = 6$  and  $p = .1$ .

————— × —————

A problem with the binomial distribution is that if the number trials 'n' is very large and probability 'p' is very small, computation of  $P[X = r]$  is cumbersome.

The distribution which we introduce in the next section may be useful in such a situation.

### 3.4 POISSON DISTRIBUTION

In this section we introduce you to another discrete distribution called 'Poisson distribution'. We will familiarise you with different situations where we can apply this distribution. Let us try to understand this distribution through an example.

Suppose it is the the busy Friday noon hour at a bank, and we are interested in the number of customers who might arrive during that hour, or during a 5-minute or a 10-minute interval in that hour;

In statistical terms, we want to find the probabilities for the number of arrivals in a time interval.

As in the case of binomial , here also we make some assumptions.

- 1) The average arrival rate at any unit time remains the same over the entire noon hour.
- 2) The number of arrivals in a time interval does not depend on what happened in previous time intervals.
- 3) It is extremely unlikely that there will be more than one arrival in a very short interval of time. That means that it is impossible for more than one customer to get through the revolving entrance door in a fraction of a second.

Under these assumption we find the required probability. For this we make use of the following formula known as **Poisson formula**, given by

$$p(x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

where  $\lambda$  is the Greek letter lambda which denotes the average arrival rate per unit of time and  $t$  is the number of units of time is the number of arrivals in  $t$  units of time

Also we know that  $\lambda = 72$  arrivals per hour is a constant for this situation. Since in the question  $\lambda$  is given in 'hour', to standardise the unit, we have to find 't' in hour.

$$\begin{aligned} \text{i.e. 60 minutes} &= 1 \text{ hour} \\ \therefore 3 \text{ minutes} &= \frac{1}{20} \text{ hours} \\ \therefore t &= \frac{1}{20} \text{ hours} \end{aligned}$$

Then

$$p(4) = \frac{e^{-72 \times \frac{1}{20}} (72 \times \frac{1}{20})^4}{4!} = \frac{e^{-3.6} (3.6)^4}{4!}$$

To find  $P(4)$ , we use the Table 2, given in the Appendix. This table shows  $p(x)$  for selected values of  $\lambda$ .

Poisson  
A nineteenth century swiss  
mathematician.



$$p(4) = 0.191$$

What does this value 0.191 specify? This tells us that if the arrivals are arrivals of customers at a bank, there is 19.1% chance that exactly four customers will arrive in the next 3 minutes.

If we vary the values of  $x$  and  $t$ , we can get different probabilities. This gives the probability distribution which is called Poisson probability distribution.

In the above discussion we saw that the Poisson formula is applicable only if certain conditions are specified. We re-state the formula now.

### Poisson Formula

The Poisson Formula is given by

Poisson Distribution

$$p(x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}$$

where  $\lambda$  is used to compute probabilities for the number of occurrences in an interval of time, if the occurrences have the following characteristic.

- 1) the average occurrence rate per unit of time is constant
- 2) occurrence in an interval is independent of what happened previously.
- 3) It rarely happens that there will be more than one occurrence in a very short time interval

A distribution having probabilities given by Poisson Formula is called Poisson distribution.

Now let us see some situations where we can apply Poisson distribution . Here is an example.

**Problem 5:** Calls at a telephone switch board occur at an average rate of six calls per 10 minutes. Suppose the operator leaves for a 5-minute coffee break, what is the probability that exactly two calls come in (and so go unanswered) while the operator is away?

**Solution :** Here you can check that the conditions 1,2 and 3 of the Poisson formula are satisfied in this case. Therefore we can use the formula. Now that here  $\lambda = \frac{6}{10}$ , In this case  $t = 5$  so that  $\lambda t = 3$ . Hence the required probability  $P(2)$  is given by

$$p(2) = \frac{e^{-3} 3^2}{2!} = 3e^{-3} = 0.2240$$

That means there is 0.2240 chance that two calls go unanswered.

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Here are some exercises for you.

E11) If a bank receives on an average  $\lambda = 6$  bad checks per day, what is the probability that it will receive 4 bad checks on any given day.

E12) A hospital has 20 kidney dialysis machines and that the chance of any one of them malfunctioning during any day is .02. We want to find the probability that exactly 3 machines will be out of service on the same day. Then,

- i) can we use the binomial formula to find this probability? If yes, calculate the probability.

In the above exercise we have seen that the difference between the two calculations is very small.

The Poisson formula can be used to approximate the binomial probability of  $r$  successes in  $n$  binomial trials in the situations where  $n$  is large and probability of success ' $p$ ' is small.

For instance, suppose we are interested in number of road accidents in a metropolitan city or daily number of machine breakdown in a work shop etc., during a specified interval of time. Each of these subintervals is so small that at best one and no more occurrence happens within it. Thus we may look upon each subinterval as a trial. Each trial leads to a "success" if the occurrence happens during that subinterval and to a "failure" if the occurrence does not happen. Assume that the occurrences are independent of each other. Hence, the total number of occurrences can be constructed to be distributed binomially, the total number of trials being equal to the number of subintervals which we have ensured to be large; also, the length for each subinterval being small, the probability of an arrival (success) is likely to be small.

Thus we have seen that there are situations where both binomial and Poisson are applied. The rule of thumb followed by most statisticians is that if  $n \geq 20$  and  $p \leq 0.05$ , then Poisson formula can be used to calculate binomial probability.

It is clear that the Poisson calculation is simpler than the binomial calculation. An advantage of the Poisson distribution, if it is applicable, is that it has only one parameter,  $\lambda$ , whereas the binomial distribution has two parameters,  $n$  and  $p$ ; consequently, Poisson probabilities can be tabulated more compactly than binomial probabilities. For example, the Poisson probability  $P(3)$  is the same for  $n = 200$ ,  $p = 0.01$  as it is for  $n = 100$ ,  $p = 0.02$ , and for any other pair of  $n$  and  $p$  values whose product is  $\lambda = np = 2$ .

By now you must have got a fairly good idea where the Poisson formula can be used.

In all the situation we have considered so far, we have calculated the probability over an interval of time. But there are situations where we need to calculate probability over a region (or space) or something else as our physical reference. In the following example we give such a situation and illustrate how to use Poisson distribution to calculate the probability.

**Example 5:** During second world war, a V-2 rocket hit in South London. Later a study was conducted on 'what are regions not affected by the rocket hit. Let us see how they used Poisson distribution for this study.

They took  $\lambda$  as the average number of hits per unit area (Note that earlier in the formula  $\lambda$  was average rate per unit time). Instead of the variable ' $t$ ' they replace the variable ' $v$ ', and  $x$  denotes the number of hits per unit area. Then they assumed that all the conditions to satisfy the Poisson formula holds in this case. With all these assumptions, they calculated the probability using the formula

$$P(x) = \frac{e^{-\lambda v} (\lambda v)^x}{x!}$$

According to the problem stated, they have to calculate the probability of 'no hit' per unit area. That is, the  $x = 0$  and  $v = 1$ , so that  $\lambda v = \lambda$ . Now, to calculate  $\lambda$ , what they did was, they divided the area into 576 areas of equal size (the number 576 is chosen based on some other study and they found that they were 537 hits).

$$\therefore \text{the average number of hits per unit area } \lambda = \frac{537}{576} = 0.9323$$

Then the required Probability is

$$P[x = 0] = e^{-0.9323} = 0.3936$$

This means that if we take one region, then the probability that the region is not hit by the rocket is 0.3936. Hence, out of 576 regions, the number of regions not hit by the rocket is given by

$$576 \times 0.3936 = 226$$

\* \* \*

Now, the actual number got from the record was that there are 229 regions not hit by the rocket. This number is quite close to 226. This shows that the values got using Poisson formula are very close to the actual values.

Thus we saw that the Poisson distribution is very effective in studying various real-life problems where the occurrence is very rare.

One of the main disadvantages of this distribution is that it is applicable only in situation where the outcomes are independent i.e. each outcome is independent of what happened previously.

In the next section we shall discuss another standard distribution.

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### 3.5 UNIFORM DISTRIBUTION

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The uniform distribution is the simplest of a few well-known continuous distributions which occur often.

As we have seen in Sec.3.2 in the continuous case we are interested in behaviour of the variable in the subintervals of the sample space, rather than at single points. If for example, the sample space is what we call the unit interval  $[0, 1]$ , and we set the random variable  $X$  as a value selected from this interval, then we are no longer interested in the outcome of the kind  $\{X = a\}$ , but rather outcome of events of the kind  $\{a < x < b\}$  i.e. values lying between the two numbers  $a$  and  $b$ , where  $0 \leq a \leq b \leq 1$ .

Suppose  $X$  is a random variable such that if we take any subinterval of the sample space, then the probability of this interval is the same as the probability of any other subinterval of the same length. The distribution corresponding to this r.v. is called a uniform distribution. As the name suggests the probability is uniform along subintervals.

Let us see some examples of such sample spaces.

**Example 6:** A train is likely to arrive at a station at any time between 6.10 p.m. and 6.40 p.m. The time the train reaches, measured in minutes, after 6 p.m. is a random variable  $X$ . Here  $X$  can take any value between 10 and 40 minutes. Therefore the sample space is the interval  $(10, 40)$ . It is reasonable to assume that the likelihood for  $X$  taking any value between 10 and 40 is equal. So if we take subintervals of equal lengths, then the probability will be the same. The distribution corresponding to this r.v. is uniform over the interval  $(10, 40)$ .

\* \* \*

**Example 7:** An office fire drill is scheduled for a particular day, and the fire alarm is likely to ring at any time between 9 a.m. and 5 p.m. The time the fire alarm starts, measured in minutes, after 9 a.m. is therefore a random variable which takes any value between 0 and 480 ( $= 8 \text{ hours} = 8 \times 60 = 480 \text{ minutes}$ ) equally. The distribution corresponding to this r.v. is uniform.

\* \* \*

Now, why don't you look for such sample spaces on your own. Try this exercise now.

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E13) Verify whether the following situations can be described by uniform distribution or not?

- a) The average life span of a life bulb produced by a manufacturing company.
- b) The number of defective items produced by an assembly process.

Next we will see how we can define (calculate) the probabilities for this distribution. As we have seen in Sec. 3.2, in the case of a continuous distribution, the probabilities are calculated using a function called 'probability density function' (p.d.f.). The p.d.f. for uniform distribution is given as follows.

**Definition 5:** The **pd.f. of a random variable X** which is distributed uniformly in the interval  $[a, b]$ , where  $a < b$  is given by

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

We can easily draw the graph of this distribution. It is given in Fig.8.

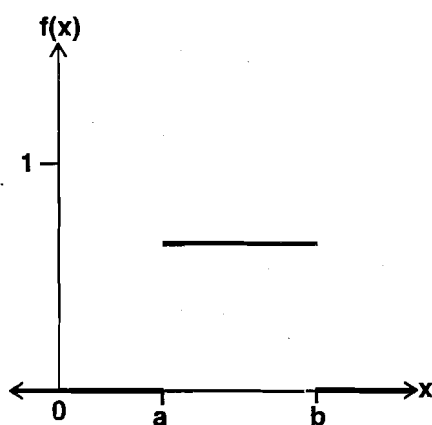


Fig. 8: Graph of P.d.f. of a uniform distribution

Now let us see how we calculate different probabilities for this distribution. As stated in Sec.3.2, for a continuous r.v., we calculate the probability of an interval rather than a point. For example, what will be  $P[c < X < d]$  where  $a < c < d < b$ ? We have seen that it is given by the area above this interval and under the graph. The area is shown in Fig.9.

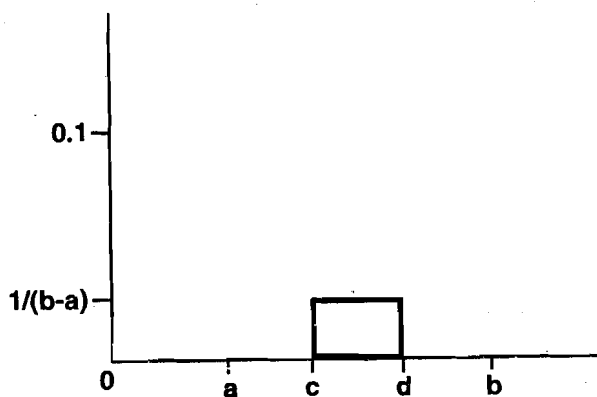


Fig.9  $P[c < X < d] = \text{Area of the rectangle shown}$

So, essentially it is the area of the rectangle with length  $d - c$  and height  $= \frac{1}{b-a}$  i.e.

$$P[c < X < d] = (d - c) \times \frac{1}{b - a}$$

For example, if we take the situation in Example 4, let us find the probability that the

alarm sounds between 1 p.m. and 2 p.m. Here the pdf,

$$f(x) = \frac{1}{480}, 0 \leq x \leq 480$$

$$= 0, \text{ otherwise}$$

To find the required probability, you have to find time elapsed in minutes between 9 a.m. and 1 p.m. and between 9 a.m. and 2 p.m.

For 1 p.m. this is  $4 \times 60 = 240$  minutes.

Similarly, for 2 p.m., it is  $5 \times 60 = 300$  minutes.

Therefore you have to calculate the probability  $P[240 < X < 300]$ . This is given by the shaded area given below in Fig.10.

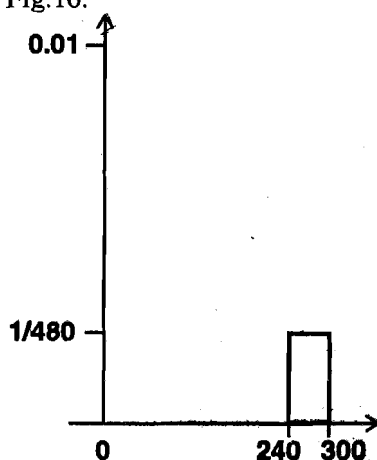


Fig.10

This area is the rectangle with base  $60 (= 300 - 240)$  and height  $\frac{1}{480}$ .

$$P[240 < X < 300] = \frac{1}{8} = 0.125$$

That is there is 12.5% chance that the alarm sounds between 1 p.m. and 5 p.m. [Some of you may think that this fact was rather obvious from the statement of the problem itself. But we have given this situation as an illustrative example. There are situations which are complicated, where we can easily calculate the probability using this distribution.]

Next we state below the **expected value of this distribution**.

$$E(X) = \frac{a + b}{2}$$

You can try this exercise now.

E14) Suppose that the weight of sugar obtained by processing a tank of sugar cane juice is uniformly distributed with a mean of 10 kg. and range of 1.8 kg. Then

- i) What are the largest and smallest weights of sugar obtained from a tank of sugar cane juice?
- ii) What is the probability that a tank of juice will yield sugar weighing between 9 kg. and 10.5 kg.?

E15) A train is due to arrive at 5.30 p.m. but in practise is equally likely to arrive at any time between 2 minutes early and 30 minutes late. Let the time of arrival (expressed as minutes from due time) be  $X$ . Sketch the pdf  $f(x)$  of the r.v.  $X$  and shade the areas given below

- 1) The probability that the train is less than 10 minutes late.

Next we shall discuss another continuous distribution which is widely used in statistical problems.

### 3.6 NORMAL DISTRIBUTION

Normal distribution' is a class of distribution which can be used to study the probability distribution occurring frequently in real-life situations, of biology, manufacturing machines, psychology etc.

A particular form of this distribution was found by seventeenth - eighteenth century mathematicians Abraham De Moivre and Pierre Laplace, while they were working on various problems in probability. They found that the distribution corresponding to certain random variables had got special property that when graphed, a bell-shaped curve is obtained and came to be called the normal pattern. The graph of the pattern became known as normal curve. Later this class of distribution was studied extensively by another mathematician Karl Friedrich Gauss and therefore this became known as 'Gaussian distribution'.

We shall now state the distribution.

**Definition 6:** We say that a random variable  $X$  is normally distributed with parameters  $\mu$  and  $\sigma$  if the probability density function  $f(x)$  of  $X$  is given by,

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} \text{ where } -\infty < x < \infty$$

where  $\mu$  is a real number lying between  $-\infty$  and  $\infty$  and  $\sigma$  is a real number lying between 0 and  $\infty$ .

The function  $f(x)$  may look rather formidable to you at first sight. At this stage we just ask you to notice that it involves two parameters,  $\sigma$  and  $\mu$ . Corresponding to each pair  $(\mu, \sigma)$ , we get a distribution. Therefore there is a whole family of distributions, each one specified by a particular pair of values for  $\sigma$  and  $\mu$ .

The most important characteristic of this distribution is that the graph of pdf,  $f(x)$  for a particular value of  $\mu$  and  $\sigma$  is bell-shaped as shown in Fig.11.

The probability density function, pdf is also symmetrical about the mean  $\mu$ . The word symmetrical means that the two halves of the curve are mirror images (see Fig.11). In Fig.11 you note that if we place a mirror on the dashed vertical line ( which occurs at 75 in Fig.11) then the mirror image of the portion on the left is the same as the portion on the right side.

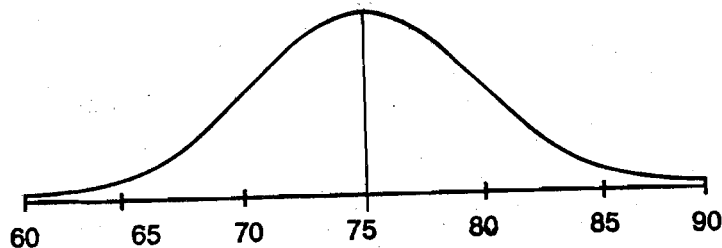


Fig.11

Both  $\mu$  and  $\sigma$  have a 'nice' interpretation. We have already said that the pdf is symmetric about  $\mu$ , so it is no surprise that  $\mu$  is the mean of the distribution. The other constant,  $\sigma^2$  dictates how spread out and flat the 'bell-shape' is and in fact  $\sigma^2$  is the variance of the normal distribution.

As an illustration, the following figure shows that the normal pdfs for  $\mu$  and  $\sigma$  are given as follows:

- A  $\mu = 10, \sigma = 1$
- B  $\mu = 10, \sigma = 2$
- C  $\mu = 10, \sigma = 3$
- D  $\mu = 15, \sigma = 1$

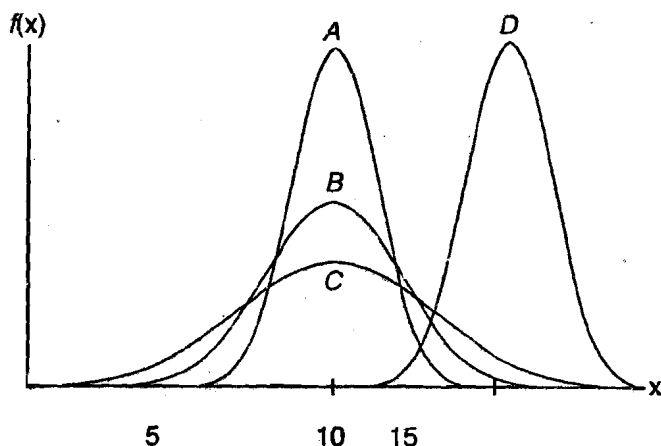


Fig.12

Pdfs A, B and C all have the mean 10 and so they are all centred at  $x = 10$ . Of these three curves, C has the largest variance and so is the most 'spread out'. Curve B has a smaller variance and so is less spread out, and curve A has the smallest variance and so is the most 'squeezed in'. Curves A and D have the same variance and so they have exactly the same shape, but they have different means so they are centred at  $x = 10$  and  $x = 15$  respectively.

### Some notation

As a normal distribution is entirely specified by its parameters  $\mu$  and  $\sigma$  we denote such distribution by  $N(\mu, \sigma^2)$  where  $\mu$  is the mean and  $\sigma^2$  is the variance. So, for instance, the curve shown in (A) above is the pdf  $N(10, 1)$  the curve in (B) is the pdf  $N(10, 4)$  and so on.

### The standard normal distribution

The normal distribution with mean  $\mu = 0$  and variance  $\sigma^2 = 1$ , is called the standard normal distribution.  $Z$  is the notation usually used for a random variable which has this distribution. A graph of the standard normal pdf,  $p(z)$  is shown in Fig.13.

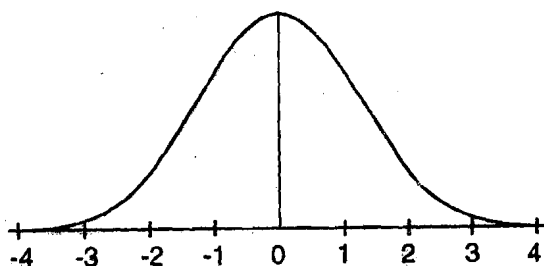


Fig.13

Notice that most of the area under the standard normal curve lies between  $-3$  and  $+3$ .

### Calculating Probabilities

The normal distribution is continuous and so the probability that the random variable  $X$  lies between the interval  $(a, b)$  is calculated by obtaining the area under the pdf curve between  $a$  and  $b$ .

For example, suppose an individual's IQ score,  $X$  has a normal distribution with mean  $\mu = 100$  and standard deviation  $\sigma = 15$ . Fig. 14 shows the areas under the pdf which correspond to  $P(X < 85)$  and  $P(115 < X < 120)$ .

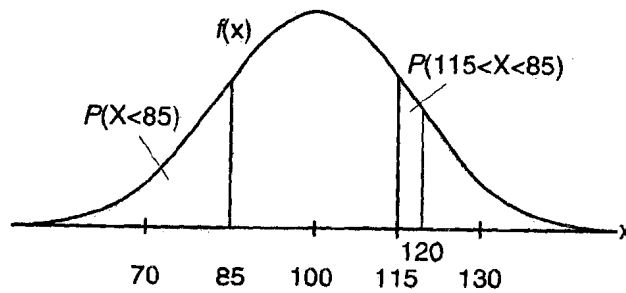


Fig.14

Unfortunately there are no 'nice' formulae for calculating such areas. But there are tables available from which we can find out the area. Statistical software are also available by which we can calculate the area.

Because the number of possible values for  $\mu$  and  $\sigma$  is unlimited, the number of different normal distributions is unlimited. However, probabilities for every normal distribution can be obtained from a table of probability for standard normal distribution.

We shall first discuss how to use the table for calculating probabilities for a standard normal distribution. Then we shall discuss how to use this to find the probability for any normal distribution.

#### Using tables to calculate normal probability

We denote by  $F(a) = P[Z \leq a]$ , the probability that the standard normal variable  $Z$  takes values less than or equal to 'a'. The values of  $F$  for different values of  $a$  are calculated and listed in a table. One such table is given in Sec. 3.9 Appendix.

Note that the entries in the table are the values of  $z$  for  $z=0.00, 0.01, 0.02, \dots, 3.49$ . To find the probability that a random variable having the standard normal distribution will take on a value between  $a$  and  $b$ , we use the equation

$$P[a < z < b] = F(b) - F(a),$$

and, if either  $a$  or  $b$  is negative, we also make use of the identity

$$F(-z) = 1 - F(z).$$

In the following example we illustrate how we use the table to calculate different probabilities.

**Example 8:** Suppose want to find the following probability

- i)  $P[0.87 < Z < 1.28]$
- ii)  $P[-0.34 < Z < 0.62]$
- iii)  $P[Z \geq 0.85]$
- iv)  $P[Z \geq -0.65]$

We proceed as follows.

- i) We know that

$$P[0.87 < Z < 1.28] = F(1.28) - F(0.87)$$

To find  $F(1.28)$ , we find the row where  $Z=1.2$ , then move across that row to the column headed 0.08 and found the entry 0.8997. Similarly we can find that  $F(0.87) = 0.8078$ . Then the required Probability is

$$\begin{aligned} P[0.87 < Z < 1.28] &= 0.8997 - 0.8078 \\ &= 0.0919 \end{aligned}$$



ii) Similarly,

$$\begin{aligned} P[-0.34 < Z < 0.62] &= F(0.62) - F(-0.34) \\ &= F(0.62) - [1 - F(0.34)] \end{aligned}$$

by the identity  $F(z) = 1 - F(-z)$ .

$$\begin{aligned} &= 0.7324 - (1 - 0.6331) \\ &= 0.3655. \end{aligned}$$

iii) From the previous unit (Unit 2), you have already learnt that

$$P[Z \geq a] = 1 - P[Z \leq a]$$

Hence we have

$$\begin{aligned} P[Z > 0.85] &= 1 - P[Z \leq 0.85] \\ &= 1 - F(0.85) \\ &= 0.1977. \end{aligned}$$

v) As in (iii), we can write

$$\begin{aligned} P[Z > 0.65] &= 1 - P[Z \leq -0.65] \\ &= 1 - F(-0.65) \\ &= 1 - F(1 - F(0.65)) \\ &= F(0.65) \\ &= 0.7422. \end{aligned}$$

\* \* \*

In the following exercise we ask you to find certain probabilities using the normal distribution table.

E16) If a random variable has the standard normal distribution, find the probability that it will take on a value

- i) less than 1.50
- ii) less than -1.20
- iii) greater than -1.75

E17) A filling machine is set to pour 952 ml (millimetres) of oil into bottles. The amounts of fill are normally distributed with a mean of 952 ml. and a standard deviation of 4 ml. Use the standard normal table to find the probability that a bottle contains oil between 952 and 956 ml.

Next we shall see that how to use the standard normal probability table to calculate probability of any normal distribution.

### Standardising

Any normal random variable  $X$ , which has mean  $\mu$  and variance  $\sigma^2$  can be standardised as follows.

Take the variable  $X$ , and

- i) subtract its mean,  $\mu$  and then
- ii) divide by its standard deviation,  $\sigma$ .

We will call the result,  $Z$ , so

$$Z = \frac{X - \mu}{\sigma}$$

For example, suppose, as earlier, that  $X$  is an individual's IQ score and that it has a normal distribution with mean  $\mu = 100$  and standard deviation  $\sigma = 15$ . To standardise

an individual's IQ score,  $X$ , we subtract  $\mu = 100$  and divide the result by  $\sigma = 15$  to give,

$$Z = \frac{X - 100}{15}$$

In this way every value of  $X$ , has a corresponding value of  $Z$ . For instance, when  $X = 130$ ,  $Z = \frac{130-100}{15} = 2$  and when  $X = 90$ ,  $Z = \frac{90-100}{15} = -0.67$ .

### The distribution of standardised normal random variables

The reason for standardising a normal random variable in this way is that a standardised normal random variable

$$Z = \frac{X - \mu}{\sigma}$$

has a standard normal distribution.

That is,  $Z$  is  $N(0, 1)$ . So if we take any normal random variable, subtract its mean and then divide by its standard deviation, the resulting random variable will have a standard normal distribution. We are going to use this fact to calculate (non-standard) normal probabilities.

### Calculating probabilities

With reference to the problem of IQ score, suppose we want to find the probability that an individual's IQ score is less than 85, i.e.  $P[X < 85]$ . The corresponding area under the pdf  $N(100, 15^2)$  is shown in Fig.15.

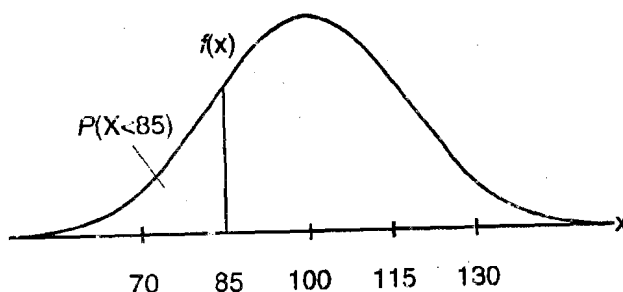


Fig.15

We cannot use normal tables directly because these give  $N(0, 1)$  probabilities. Instead, we will convert the statement  $X < 85$  into an equivalent statement which involves the standardised score,  $Z = \frac{X-100}{15}$  because we know it has a standard normal distribution.

We start with  $X = 85$ . To turn  $X$  into  $Z$  we must standardise the  $X$ , but to ensure that we preserve the meaning of the statement we must treat the other side of the inequality in exactly the same way. (Otherwise we will end up calculating the probability of another statement, not  $X < 85$ ). 'Standardising' both sides gives,  $\frac{X-100}{15} < \frac{85-100}{15}$ .

The left hand side is now a standard normal random variable and so we can call it  $Z$ , and we have

$$Z < \frac{85 - 100}{15}$$

which is

$$Z < -1.$$

So we have established that the statement we started with,  $X < 85$  is equivalent to  $Z < -1$ . This means that whenever an IQ score,  $X$ , is less than 85 the corresponding standardised score,  $Z$  will be less than  $-1$  and so the probability we are seeking,  $P[X < 85]$  is the same as  $P[Z < -1]$ .

$P[Z < -1]$ , is just a standard normal probability and so we can look it up in Table 1 in the usual way, which gives 0.1587. We get that  $P[X < 85] = 0.1587$ .

This process of rewriting a probability statement about  $X$ , in terms of  $Z$ , is not difficult if you are systematically writing down what you are doing at each stage. We would lay out the working we have just done for  $P[X < 85]$  as follows.

$X$  has a normal distribution with mean 100 and standard deviation 15. Let us find the probability that  $X$  is less than 85.

$$\begin{aligned} P[X < 85] &= P\left[\frac{X - 100}{15} < \frac{85 - 100}{15}\right] \\ &= P[Z < -1] = 0.1587 \end{aligned}$$

Let us do some problems now.

**Problem 6:** For each of these write down the equivalent standard normal probability.

- The number of people who visit a historic monument in a week is normally distributed with a mean of 10,500 and a standard deviation of 600. Consider the probability that fewer than 9000 people visit in a week.
- The number of cheques processed by a bank each day is normally distributed with a mean of 30,100 and a standard deviation of 2450. Consider the probability that the bank processes more than 32,000 cheques in a day.

**Solution:** Here we want to find the standard normal probability corresponding to the probability  $P[X < 9000]$ .

$$\text{a) We have } P[X < 9000] = P\left[\frac{X - 10500}{600} < \frac{9000 - 10500}{600}\right] = P[Z < -2.5].$$

- Here we want to find the standard normal probability corresponding to the probability  $P[X > 32000]$ .

$$P[X > 32000] = P\left[\frac{X - 30100}{2450} > \frac{32000 - 30100}{2450}\right] = P[Z > 0.78]$$

————— × —————

**Note** Probabilities like  $P[a < X < b]$  can be calculated in the same way. The only difference is that when  $X$  is standardised, similar operations must be applied to both  $a$  and  $b$ . That is,  $a < X < b$  becomes

$$\frac{a - \mu}{\sigma} < \frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}$$

which is

$$\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}$$

**Problem 7:** An individual's IQ score has a  $N(100, 15^2)$  distribution. Find the probability that an individual's IQ score is between 91 and 121.

**Solution:** We require  $P[91 < X < 121]$ . Standardising gives

$$P\left[\frac{91 - 100}{15} < \frac{X - 100}{15} < \frac{121 - 100}{15}\right]$$

The middle term is a standardised normal random variable and so we have,

$$P\left[\frac{-9}{15} < Z < \frac{21}{15}\right] = P[-0.6 < Z < 1.4] = 0.9192 - 0.2743 = 0.6449.$$

————— × —————

- E18) A flight is due at Palam airport at 1800 hours. Its arrival time has a normal distribution with mean 1810 hours and standard deviation 10 minutes.
- What is the probability that the flight arrives before its due time?
  - Passengers must check in for a connecting flight by 1830 at the latest. What is the probability that passengers from the first flight arrive too late for the connecting flight? (Assume no travelling time from aircraft to check-in.)
- E19) The length of metallic strips produced by a machine has mean 100 cm. and variance 2.25 cm. Only strips with a weight between 98 and 103 cm. are acceptable. What proportion of strips will be acceptable? You may assume that the length of a strip has a normal distribution.

With this we come to an end of this unit.

Let us now summarise the points we have covered in this unit.

### 3.7 SUMMARY

In this unit we have covered the following points

- A random variable is a variable that takes on different numerical values according to chance outcomes
- There are two types of random variables - discrete and continuous;
- A probability distribution gives the probabilities with which the random variables take an various values in their range.
- We have discussed three standard distributions:

- Binomial Distribution.** The probabilities of an event  $P[X = r]$  in this distribution is given by

$$P[X = r] = C(n, r)P^r q^{n-r}$$

- Poisson distribution:** The probability of an event  $P[X = x]$  in this distribution is given by

$$P[X = x] = \frac{e^{-\lambda} \lambda^x}{x!}$$

where  $\lambda$  is a constant for a particular situation.

- Uniform Distribution** The probability density function is defined by

$$f(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$$

$$\text{The probability } P[c < X < d] = \frac{d-c}{b-a}$$

- Normal distribution.** The probability for this distribution is calculated by finding the area, under the curve of a function called probability density function defined by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} \quad -\infty < x < \infty.$$

### 3.8 SOLUTIONS/ANSWERS

- E1) a) If X denote the number of correct answers, then X is the random variables for this situation.

b)  $X$  can take values  $0, 1, 2, \dots$  up to  $50$

c)  $P[X = 40]$  means the probability that the number of correct answers is  $40$ .

E2) 1 and 2 is not discrete. 2 and 3 are discrete.

(1) is not discrete because it takes values in an interval.

(2) is discrete because the number of accidents is finite. Similarly, you argue for the situation in (3).

E3) Let  $X$  denote the amount you win or lose. Then  $X$  takes values Rs.  $.50, 0$  or  $-10$  (loss in Rs.  $10$ ). The probability that both the marbles are green is  $1/9$ . The i.e.

$P[X = 50] = 1/9$ . The probability that both the marbles are red is  $4/9$  i.e.

$P[X = -10] = 4/9$ .

The probability that the marbles are of different colour is  $4/9$  is i.e.  $P[X = 0] = 4/9$ .

Thus the probability distribution is as given in the following table.

| Amount (in Rs. won (+) or lost(-)) | Probability |
|------------------------------------|-------------|
| 50                                 | $1/9$       |
| 0                                  | $4/9$       |
| -10                                | $4/9$       |

E4) He has to calculate the mean. It is given by

$$\begin{aligned} \text{Mean} &= \frac{0 \times 0.5 + 1 \times 0.15 + 2 \times 0.35 + 4 \times 0.12 + 5 \times 0.18}{0.05 + 0.15 + 0.35 + 0.25 + 0.12 + 0.18} \\ &= \frac{2.48}{1} = 2.48. \end{aligned}$$

This means that he can expect that on an average 2.48 cars will be sold per day over long run (or more precisely 5 cars will be sold over (2 days)).

E5) i)

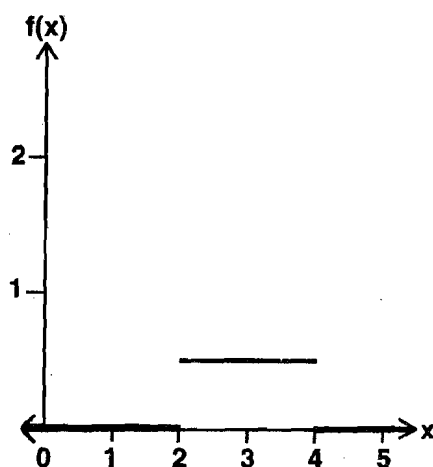


Fig.16

ii) The area under the graph and above the interval  $[2, 4]$  is the area of the rectangle shown in Fig.16 which is given by

$$\text{Area} = 2 \times \frac{1}{2} = 1.$$

$\therefore f$  defines a probability density function of  $X$ .

E6) 1,3,4,5,6 are discrete. 2 and 7 are continuous.

E7) Here we have to calculate Probability of 3 heads and 2 tails. That is  $P[X = 3]$ .

The possibility of getting 3 heads and 2 tails in five tosses of a coin is given by

[HHHTT], [HHTHT], [HHTTH], [HTHHT], [HTTHH], [THTHH], [TTHHH], [THHHT], [THHTH], [THTHH], [HTHTH].

Each of these events are having probability  $p^3q^2$  and there are 10 such events.

Therefore we get

$$P[X = 3] = 10p^3q^2.$$

If we apply the formula in Equation (3) to find  $P[X = 3]$ , then we substitute  $r = 3, n = 5$  in the formula, and we get

$$\begin{aligned} P[X = 3] &= C(5, 3)p^3q^2 \\ &= \frac{5!}{2! \times 3!}p^3q^2 \\ &= 10p^3q^2. \end{aligned}$$

- E8) This situation follows binomial distribution with  $n=4$  and  $p = \frac{80}{100} = \frac{4}{5}$ . The random variable  $X$  is the number of seeds that germinate. We have to calculate the probability that exactly two of the four seeds will germinate. That is  $P[X = 2]$ . By applying binomial formula, we get

$$\begin{aligned} P[X = 2] &= C(4, 2) \left(\frac{4}{5}\right)^2 \times \left(\frac{1}{5}\right)^2 \\ &= 6 \times \frac{16}{625} \\ &= 0.154 \end{aligned}$$

Therefore the required probability is 0.154.

- E9) If  $X_i$  denote the random variable that the  $i$ th customer buys the paper on a given day, then  $X_i$ 's may not be identically distributed. Therefore  $X_i$ 's may not be binomially distributed. But if the customers are having the same business activities or same kind of habits or working nature, then we can expect that  $X_i$ 's will be identically distributed. In such situation we can expect that  $X_i$ 's will follow binomial distribution.
- E10) The number of sales in a day is actually  $X_1 + X_2 + \dots + X_{10}$  where each  $X_i$  is either 0 or 1 depending on whether customer  $i$  buys the paper or not on the given day. Now since customer 8 is more likely to buy on a day, than customer 3,  $X_3$  and  $X_8$  are not identically distributed. That is,  $P[X_8 = 1] > P[X_3 = 1]$ . Therefore  $X_1 + X_2 + \dots + X_{10}$  cannot be thought of as binomially distributed random variable.
- E11) Since the problem deals with the receipts of bad cheques which is an event with rare occurrence over an interval of time (a day, in this case), we can apply Poisson distribution.

Since on an average 6 bad cheques are received per day,

Substituting  $\lambda = 6$  and  $x = 4$  in the Poisson Formula, we get

$$\begin{aligned} P[X = 4] &= \frac{6^4 e^{-6}}{4!} = \frac{1296 \times (0.0025)}{24} \\ &= 0.135. \end{aligned}$$

- E12) Note that here the experiment or trial is 'checking the machine for its functioning'. There are 20 trials and each trial is identically distributed with probability 0.2.
- i) The trials are independent also. Therefore we can apply binomial formula. We are required to calculate  $P[X = 3]$ . Then

$$\begin{aligned} P[X = 3] &= \frac{20!}{3! \times 17!} (0.02)^3 (0.98)^{17} \\ &= 0.0065. \end{aligned}$$

- ii) Here we have to check whether we can apply Poisson distribution.

Note that here the occurrences are 'function of the dialysis machines'. Then the average rate of machines that go out of service in a day is a constant  $\lambda = 20 \times 0.02 = 0.4$ .

Also note that we can make the subintervals so small that at best only one machine go out of service. Thus conditions (2) and (3) are satisfied. Therefore we

can apply Poisson Formula to calculate the required probability  $P(3)$ . Then

$$\begin{aligned} P(3) &= \frac{(0.4)^3 e^{-0.4}}{3!} \\ &= \frac{(0.64)(.67032)}{6} \\ &= .00715. \end{aligned}$$

E13) a) We can model it as uniform.

b) We cannot model it as uniform.

E14) i) The mean, 10 is the centre point of a line segment whose length is the range, 1.8 kg. Hence, the line segment extends  $\frac{1}{2} \times (1.8) = 0.9$  kg. to the left and to the right of 10 i.e. 9.1 to 10.9 kg. Hence the smallest weight is 9.1 kg. and the largest weight is 10.9 kg.

ii) We are required to calculate  $p[9 < X < 10.5]$ .

$$\begin{aligned} p[9 < X < 10.5] &= (10.5 - 9) \times \frac{1}{1.8} \\ &= 0.833. \end{aligned}$$

That is the probability that the weight lies between 9.1 kg. and 10.9 kg. is 0.833.

E15) Here the pdf  $f(x)$  is given by

$$\begin{cases} \frac{1}{32}, & 5.28 < x < 6 \\ 0, & \text{otherwise} \end{cases}$$

The sketch of  $f(x)$  is as given below.

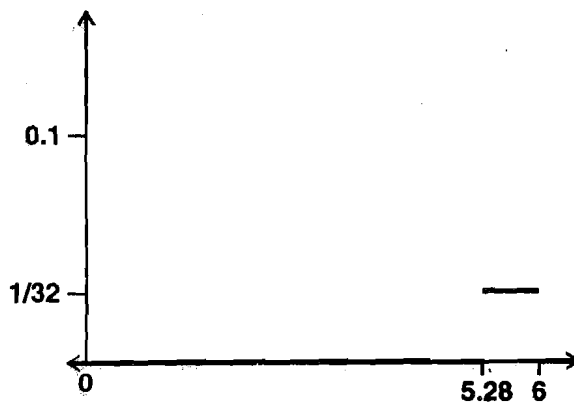


Fig.17

i) To find the required probability we note that  $X$  can take values in the interval  $(5.28, 5.40)$ . Hence the required probability is

$$P[5.28 < X < 5.40] = 0.2 \times \frac{1}{32} = \frac{0.003}{8} = 0.0037$$

E16) a) 0.9332

b) 0.1151

c) 0.9599.

E17) The standard normal probability corresponding to this probability is given by

$$\begin{aligned} P[952 < X < 956] &= P\left[\frac{952 - 952}{4} < \frac{X - 952}{4} < \frac{956 - 952}{4}\right] \\ &= P[0 < Z < 1] \\ &= F(1) - F(0) \\ &= 0.8413 - 0.5 \\ &= 0.343. \end{aligned}$$

E18) Let the time of arrival in minutes past 1800hrs be X. Then X follows normal distribution  $N(10, 10^2)$ .

- a) The required probability is  $P[X < 18]$ . The standard probability corresponding to this is

$$\begin{aligned} P\left[Z < \frac{18 - 10}{10}\right] &= P[Z < .8] \\ &= F(0.8) \\ &= 0.7881. \end{aligned}$$

- b) The required probability is  $P[X > 30]$ . Then

$$\begin{aligned} P[X > 30] &= P\left[\frac{X - 10}{10} > \frac{30 - 10}{10}\right] = P[Z > 2] \\ &= 1 - P[Z \leq 2] \\ &= 1 - 0.9772 = 0.0228. \end{aligned}$$

E19) We have to find the probability  $P[98 < X < 103]$ . The standard probability is

$$\begin{aligned} P\left[\frac{-2}{\sqrt{2.25}} < Z < \frac{3}{\sqrt{2.25}}\right] \\ \text{i.e. } P\left[\frac{-2}{1.5} < Z < \frac{3}{1.5}\right] &= P[-1.33 < Z < 2] \\ &= F(2) - F(-1.33) \\ &= F(2) - (1 - F(1.33)) \\ &= 0.9772 - (1 - 0.9049) \\ &= 0.9772 - 0.0951 \\ &= 0.8821 \\ &= \frac{88.21}{100} \text{ approximately} \end{aligned}$$

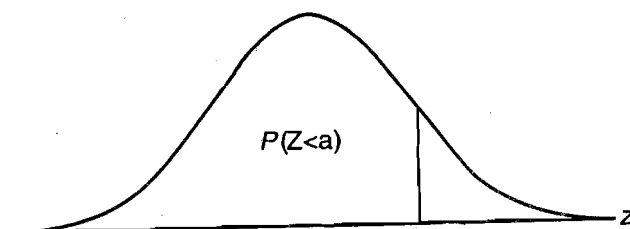
So, only 88.21 % will be acceptable.



### 3.9 APPENDIX

Cumulative Standard Normal Probabilities  $P[Z < a]$  where  $Z \sim N(0, 1)$ .

| a   | 0.00   | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |
|-----|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7703 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| 3.4 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |
| 3.5 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 | 0.9998 |
| 3.6 | 0.9998 | 0.9998 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| 3.7 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| 3.8 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 | 0.9999 |
| 3.9 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |



# POISSON PROBABILITY DISTRIBUTION

This table shows the value of

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}$$

for selected values of  $x$  and for  $\mu = .005$  to  $8.0$ .

| $x$ | $\mu$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|     | .005  | .01   | .02   | .03   | .04   | .05   | .06   | .07   | .08   | .09   |
| 0   | .9950 | .9900 | .9802 | .9704 | .9608 | .9512 | .9418 | .9324 | .9231 | .9139 |
| 1   | .0050 | .0099 | .0192 | .0291 | .0384 | .0476 | .0565 | .0653 | .0738 | .0823 |
| 2   | .0000 | .0000 | .0002 | .0004 | .0008 | .0012 | .0017 | .0023 | .0030 | .0037 |
| 3   | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 | .0001 | .0001 |

| $x$ | $\mu$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|     | 0.1   | 0.2   | 0.3   | 0.4   | 0.5   | 0.6   | 0.7   | 0.8   | 0.9   | 1.0   |
| 0   | .9048 | .8187 | .7408 | .6703 | .6065 | .5488 | .4966 | .4493 | .4066 | .3679 |
| 1   | .0905 | .1637 | .2222 | .2681 | .3033 | .3293 | .3476 | .3595 | .3659 | .3679 |
| 2   | .0045 | .0164 | .0333 | .0536 | .0758 | .0988 | .1217 | .1438 | .1647 | .1839 |
| 3   | .0002 | .0011 | .0033 | .0072 | .0126 | .0198 | .0284 | .0383 | .0494 | .0613 |
| 4   | .0000 | .0001 | .0002 | .0007 | .0016 | .0030 | .0050 | .0077 | .0111 | .0153 |
| 5   | .0000 | .0000 | .0000 | .0001 | .0002 | .0004 | .0007 | .0012 | .0020 | .0031 |
| 6   | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 | .0002 | .0003 | .0005 |
| 7   | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 |

| $x$ | $\mu$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|     | 1.1   | 1.2   | 1.3   | 1.4   | 1.5   | 1.6   | 1.7   | 1.8   | 1.9   | 2.0   |
| 0   | .3329 | .3012 | .2725 | .2466 | .2231 | .2019 | .1827 | .1653 | .1496 | .1353 |
| 1   | .3662 | .3614 | .3543 | .3452 | .3347 | .3230 | .3106 | .2975 | .2842 | .2707 |
| 2   | .2014 | .2169 | .2303 | .2417 | .2510 | .2584 | .2640 | .2678 | .2700 | .2707 |
| 3   | .0738 | .0867 | .0998 | .1128 | .1255 | .1378 | .1496 | .1607 | .1710 | .1804 |
| 4   | .0203 | .0260 | .0324 | .0395 | .0471 | .0551 | .0636 | .0723 | .0812 | .0902 |
| 5   | .0045 | .0062 | .0084 | .0111 | .0141 | .0176 | .0216 | .0260 | .0309 | .0361 |
| 6   | .0008 | .0012 | .0018 | .0026 | .0035 | .0047 | .0061 | .0078 | .0098 | .0120 |
| 7   | .0001 | .0002 | .0003 | .0005 | .0008 | .0011 | .0015 | .0020 | .0027 | .0034 |
| 8   | .0000 | .0000 | .0001 | .0001 | .0001 | .0002 | .0003 | .0005 | .0006 | .0009 |
| 9   | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 | .0001 | .0001 | .0002 |

| $x$ | $\mu$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|     | 2.1   | 2.2   | 2.3   | 2.4   | 2.5   | 2.6   | 2.7   | 2.8   | 2.9   | 3.0   |
| 0   | .1225 | .1108 | .1003 | .0907 | .0821 | .0743 | .0672 | .0608 | .0550 | .0498 |
| 1   | .2572 | .2438 | .2306 | .2177 | .2052 | .1931 | .1815 | .1703 | .1596 | .1494 |
| 2   | .2700 | .2681 | .2652 | .2613 | .2565 | .2510 | .2450 | .2384 | .2314 | .2240 |
| 3   | .1890 | .1966 | .2033 | .2090 | .2138 | .2176 | .2205 | .2225 | .2237 | .2240 |
| 4   | .0992 | .1082 | .1169 | .1254 | .1336 | .1414 | .1488 | .1557 | .1622 | .1680 |

Continued .....

|    |       |       |       |       |       |       |       |       |       |       |
|----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 5  | .0417 | .0476 | .0538 | .0602 | .0668 | .0735 | .0804 | .0872 | .0940 | .1008 |
| 6  | .0146 | .0174 | .0206 | .0241 | .0278 | .0319 | .0362 | .0407 | .0455 | .0504 |
| 7  | .0044 | .0055 | .0068 | .0083 | .0099 | .0118 | .0139 | .0163 | .0188 | .0216 |
| 8  | .0011 | .0015 | .0019 | .0025 | .0031 | .0038 | .0047 | .0057 | .0068 | .0081 |
| 9  | .0003 | .0004 | .0005 | .0007 | .0009 | .0011 | .0014 | .0018 | .0022 | .0027 |
| 10 | .0001 | .0001 | .0001 | .0002 | .0002 | .0003 | .0004 | .0005 | .0006 | .0008 |
| 11 | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 | .0001 | .0001 | .0002 | .0002 |
| 12 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 |

| $\mu$ |       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $x$   | 3.1   | 3.2   | 3.3   | 3.4   | 3.5   | 3.6   | 3.7   | 3.8   | 3.9   | 4.0   |
| 0     | .0450 | .0408 | .0369 | .0334 | .0302 | .0273 | .0247 | .0224 | .0202 | .0183 |
| 1     | .1397 | .1304 | .1217 | .1135 | .1057 | .0984 | .0915 | .0850 | .0789 | .0733 |
| 2     | .2165 | .2087 | .2008 | .1929 | .1850 | .1771 | .1692 | .1615 | .1539 | .1465 |
| 3     | .2237 | .2226 | .2209 | .2186 | .2158 | .2125 | .2087 | .2046 | .2001 | .1954 |
| 4     | .1734 | .1781 | .1823 | .1858 | .1888 | .1912 | .1931 | .1944 | .1951 | .1954 |
| 5     | .1075 | .1140 | .1203 | .1264 | .1322 | .1377 | .1429 | .1477 | .1522 | .1563 |
| 6     | .0555 | .0608 | .0662 | .0716 | .0771 | .0826 | .0881 | .0936 | .0989 | .1042 |
| 7     | .0246 | .0278 | .0312 | .0348 | .0385 | .0425 | .0466 | .0508 | .0551 | .0595 |
| 8     | .0095 | .0111 | .0129 | .0148 | .0169 | .0191 | .0215 | .0241 | .0269 | .0298 |
| 9     | .0033 | .0040 | .0047 | .0056 | .0066 | .0076 | .0089 | .0102 | .0116 | .0132 |
| 10    | .0010 | .0013 | .0016 | .0019 | .0023 | .0028 | .0033 | .0039 | .0045 | .0053 |
| 11    | .0003 | .0004 | .0005 | .0006 | .0007 | .0009 | .0011 | .0013 | .0016 | .0019 |
| 12    | .0001 | .0001 | .0001 | .0002 | .0002 | .0003 | .0003 | .0004 | .0005 | .0006 |
| 13    | .0000 | .0000 | .0000 | .0000 | .0001 | .0001 | .0001 | .0001 | .0002 | .0002 |
| 14    | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 |

| $\mu$ |       |       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $x$   | 4.1   | 4.2   | 4.3   | 4.4   | 4.5   | 4.6   | 4.7   | 4.8   | 4.9   | 5.0   |
| 0     | .0166 | .0150 | .0136 | .0123 | .0111 | .0101 | .0091 | .0082 | .0074 | .0067 |
| 1     | .0679 | .0630 | .0583 | .0540 | .0500 | .0462 | .0427 | .0395 | .0365 | .0337 |
| 2     | .1393 | .1323 | .1254 | .1188 | .1125 | .1063 | .1005 | .0948 | .0894 | .0842 |
| 3     | .1904 | .1852 | .1798 | .1743 | .1687 | .1631 | .1574 | .1517 | .1460 | .1404 |
| 4     | .1951 | .1944 | .1933 | .1917 | .1898 | .1875 | .1849 | .1820 | .1789 | .1755 |
| 5     | .1600 | .1633 | .1662 | .1687 | .1708 | .1725 | .1738 | .1747 | .1753 | .1755 |
| 6     | .1093 | .1143 | .1191 | .1237 | .1281 | .1323 | .1362 | .1398 | .1432 | .1462 |
| 7     | .0640 | .0686 | .0732 | .0778 | .0824 | .0869 | .0914 | .0959 | .1002 | .1044 |
| 8     | .0328 | .0360 | .0393 | .0428 | .0463 | .0500 | .0537 | .0575 | .0614 | .0653 |
| 9     | .0150 | .0168 | .0188 | .0209 | .0232 | .0255 | .0280 | .0307 | .0334 | .0363 |

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|     | $\mu$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $x$ | 4.1   | 4.2   | 4.3   | 4.4   | 4.5   | 4.6   | 4.7   | 4.8   | 4.9   | 5.0   |
| 10  | .0061 | .0071 | .0081 | .0092 | .0104 | .0118 | .0132 | .0147 | .0164 | .0181 |
| 11  | .0023 | .0027 | .0032 | .0037 | .0043 | .0049 | .0056 | .0064 | .0073 | .0082 |
| 12  | .0008 | .0009 | .0011 | .0014 | .0016 | .0019 | .0022 | .0026 | .0030 | .0034 |
| 13  | .0002 | .0003 | .0004 | .0005 | .0006 | .0007 | .0008 | .0009 | .0011 | .0013 |
| 14  | .0001 | .0001 | .0001 | .0001 | .0002 | .0002 | .0003 | .0003 | .0004 | .0005 |
| 15  | .0000 | .0000 | .0000 | .0000 | .0001 | .0001 | .0001 | .0001 | .0001 | .0002 |

|     | $\mu$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $x$ | 5.1   | 5.2   | 5.3   | 5.4   | 5.5   | 5.6   | 5.7   | 5.8   | 5.9   | 6.0   |
| 0   | .0061 | .0055 | .0050 | .0045 | .0041 | .0037 | .0033 | .0030 | .0027 | .0025 |
| 1   | .0311 | .0287 | .0265 | .0244 | .0225 | .0207 | .0191 | .0176 | .0162 | .0149 |
| 2   | .0793 | .0746 | .0701 | .0659 | .0618 | .0580 | .0544 | .0509 | .0477 | .0446 |
| 3   | .1348 | .1293 | .1239 | .1185 | .1133 | .1082 | .1033 | .0985 | .0938 | .0892 |
| 4   | .1719 | .1681 | .1641 | .1600 | .1558 | .1515 | .1472 | .1428 | .1383 | .1339 |
| 5   | .1753 | .1748 | .1740 | .1728 | .1714 | .1697 | .1678 | .1656 | .1632 | .1606 |
| 6   | .1490 | .1515 | .1537 | .1555 | .1571 | .1584 | .1594 | .1601 | .1605 | .1606 |
| 7   | .1086 | .1125 | .1163 | .1200 | .1234 | .1267 | .1298 | .1326 | .1353 | .1377 |
| 8   | .0692 | .0731 | .0771 | .0810 | .0849 | .0887 | .0925 | .0962 | .0998 | .1033 |
| 9   | .0392 | .0423 | .0454 | .0486 | .0519 | .0552 | .0586 | .0620 | .0654 | .0688 |
| 10  | .0200 | .0220 | .0241 | .0262 | .0285 | .0309 | .0334 | .0359 | .0386 | .0413 |
| 11  | .0093 | .0104 | .0116 | .0129 | .0143 | .0157 | .0173 | .0190 | .0207 | .0225 |
| 12  | .0039 | .0045 | .0051 | .0058 | .0065 | .0073 | .0082 | .0092 | .0102 | .0113 |
| 13  | .0015 | .0018 | .0021 | .0024 | .0028 | .0032 | .0036 | .0041 | .0046 | .0052 |
| 14  | .0006 | .0007 | .0008 | .0009 | .0011 | .0013 | .0015 | .0017 | .0019 | .0022 |
| 15  | .0002 | .0002 | .0003 | .0003 | .0004 | .0005 | .0006 | .0007 | .0008 | .0009 |
| 16  | .0001 | .0001 | .0001 | .0001 | .0001 | .0002 | .0002 | .0002 | .0003 | .0003 |
| 17  | .0000 | .0000 | .0000 | .0000 | .0000 | .0001 | .0001 | .0001 | .0001 | .0001 |

|     | $\mu$ |       |       |       |       |       |       |       |       |       |
|-----|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $x$ | 6.1   | 6.2   | 6.3   | 6.4   | 6.5   | 6.6   | 6.7   | 6.8   | 6.9   | 7.0   |
| 0   | .0022 | .0020 | .0018 | .0017 | .0015 | .0014 | .0012 | .0011 | .0010 | .0009 |
| 1   | .0137 | .0126 | .0116 | .0106 | .0098 | .0090 | .0082 | .0076 | .0070 | .0064 |
| 2   | .0417 | .0390 | .0364 | .0340 | .0318 | .0296 | .0276 | .0258 | .0240 | .0223 |
| 3   | .0848 | .0806 | .0765 | .0726 | .0688 | .0652 | .0617 | .0584 | .0552 | .0521 |
| 4   | .1294 | .1249 | .1205 | .1162 | .1118 | .1076 | .1034 | .0992 | .0952 | .0912 |
| 5   | .1579 | .1549 | .1519 | .1487 | .1454 | .1420 | .1385 | .1349 | .1314 | .1277 |
| 6   | .1605 | .1601 | .1595 | .1586 | .1575 | .1562 | .1546 | .1529 | .1511 | .1490 |
| 7   | .1399 | .1418 | .1435 | .1450 | .1462 | .1472 | .1480 | .1486 | .1489 | .1490 |
| 8   | .1066 | .1099 | .1130 | .1160 | .1188 | .1215 | .1240 | .1263 | .1284 | .1304 |
| 9   | .0723 | .0757 | .0791 | .0825 | .0858 | .0891 | .0923 | .0954 | .0985 | .1014 |

Continued .....

| $x$ | <u>6.1</u> | <u>6.2</u> | <u>6.3</u> | <u>6.4</u> | <u>6.5</u> | <u>6.6</u> | <u>6.7</u> | <u>6.8</u> | <u>6.9</u> | <u>7.0</u> |
|-----|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 10  | .0441      | .0469      | .0498      | .0528      | .0558      | .0588      | .0618      | .0649      | .0679      | .0710      |
| 11  | .0245      | .0265      | .0285      | .0307      | .0330      | .0353      | .0377      | .0401      | .0426      | .0452      |
| 12  | .0124      | .0137      | .0150      | .0164      | .0179      | .0194      | .0210      | .0227      | .0245      | .0264      |
| 13  | .0058      | .0065      | .0073      | .0081      | .0089      | .0098      | .0108      | .0119      | .0130      | .0142      |
| 14  | .0025      | .0029      | .0033      | .0037      | .0041      | .0046      | .0052      | .0058      | .0064      | .0071      |
| 15  | .0010      | .0012      | .0014      | .0016      | .0018      | .0020      | .0023      | .0026      | .0029      | .0033      |
| 16  | .0004      | .0005      | .0005      | .0006      | .0007      | .0008      | .0010      | .0011      | .0013      | .0014      |
| 17  | .0001      | .0002      | .0002      | .0002      | .0003      | .0003      | .0004      | .0004      | .0005      | .0006      |
| 18  | .0000      | .0001      | .0001      | .0001      | .0001      | .0001      | .0001      | .0002      | .0002      | .0002      |
| 19  | .0000      | .0000      | .0000      | .0000      | .0000      | .0000      | .0000      | .0001      | .0001      | .0001      |

| $x$ | <u>7.1</u> | <u>7.2</u> | <u>7.3</u> | <u>7.4</u> | <u>7.5</u> | <u>7.6</u> | <u>7.7</u> | <u>7.8</u> | <u>7.9</u> | <u>8.0</u> |
|-----|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 0   | .0008      | .0007      | .0007      | .0006      | .0006      | .0005      | .0005      | .0004      | .0004      | .0003      |
| 1   | .0059      | .0054      | .0049      | .0045      | .0041      | .0038      | .0035      | .0032      | .0029      | .0027      |
| 2   | .0208      | .0194      | .0180      | .0167      | .0156      | .0145      | .0134      | .0125      | .0116      | .0107      |
| 3   | .0492      | .0464      | .0438      | .0413      | .0389      | .0366      | .0345      | .0324      | .0305      | .0286      |
| 4   | .0874      | .0836      | .0799      | .0764      | .0729      | .0696      | .0663      | .0632      | .0602      | .0573      |
| 5   | .1241      | .1204      | .1167      | .1130      | .1094      | .1057      | .1021      | .0986      | .0951      | .0916      |
| 6   | .1468      | .1445      | .1420      | .1394      | .1367      | .1339      | .1311      | .1282      | .1252      | .1221      |
| 7   | .1489      | .1486      | .1481      | .1474      | .1465      | .1454      | .1442      | .1428      | .1413      | .1396      |
| 8   | .1321      | .1337      | .1351      | .1363      | .1373      | .1382      | .1388      | .1392      | .1395      | .1396      |
| 9   | .1042      | .1070      | .1096      | .1121      | .1144      | .1167      | .1187      | .1207      | .1224      | .1241      |
| 10  | .0740      | .0770      | .0800      | .0829      | .0858      | .0887      | .0914      | .0941      | .0967      | .0993      |
| 11  | .0478      | .0504      | .0531      | .0558      | .0585      | .0613      | .0640      | .0667      | .0695      | .0722      |
| 12  | .0283      | .0303      | .0323      | .0344      | .0366      | .0388      | .0411      | .0434      | .0457      | .0481      |
| 13  | .0154      | .0168      | .0181      | .0196      | .0211      | .0227      | .0243      | .0260      | .0278      | .0296      |
| 14  | .0078      | .0086      | .0095      | .0104      | .0113      | .0123      | .0134      | .0145      | .0157      | .0169      |
| 15  | .0037      | .0041      | .0046      | .0051      | .0057      | .0062      | .0069      | .0075      | .0083      | .0090      |
| 16  | .0016      | .0019      | .0021      | .0024      | .0026      | .0030      | .0033      | .0037      | .0041      | .0045      |
| 17  | .0007      | .0008      | .0009      | .0010      | .0012      | .0013      | .0015      | .0017      | .0019      | .0021      |
| 18  | .0003      | .0003      | .0004      | .0004      | .0005      | .0006      | .0006      | .0007      | .0008      | .0009      |
| 19  | .0001      | .0001      | .0001      | .0002      | .0002      | .0002      | .0003      | .0003      | .0003      | .0004      |
| 20  | .0000      | .0000      | .0001      | .0001      | .0001      | .0001      | .0001      | .0001      | .0001      | .0002      |
| 21  | .0000      | .0000      | .0000      | .0000      | .0000      | .0000      | .0000      | .0000      | .0001      | .0001      |

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## UNIT 4 SAMPLING DISTRIBUTIONS

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### 4.1 INTRODUCTION

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In the last block we emphasised the techniques used to describe data. To illustrate these techniques we organised the data into a frequency distribution and computed various averages and measures of dispersion. Measures such as mean and standard deviation were computed to describe the central tendency of the data and the extent of its spread. In this unit we start discussing the procedures of statistical inference which is the core of statistical analysis. We have already discussed the concepts of probability and probability distributions which lays a foundation stone for statistical inference.

As we have seen in Unit 1, in statistical inference we use the sample to make inference about the unknown population characteristic. As you know, it is a common practice for every body to draw inference about a large population by examining a sample from it. This technique is used in decision making based on statistical analysis. There if we want to make decision about a large population, may be finite or infinite, we take representative samples from the population and get results for these samples which is then used to make decisions about the population. But the results we obtain from the samples may vary from sample to sample. To make valid judgements about the population, it is necessary to study this variation. In this unit we introduce you to the concept of sampling distribution. Further we discuss how to construct a sampling distribution by selecting all samples of size, say,  $n$  from a population and how this is used to make inferences about the population. To start with we study the sampling distribution of means in Sec.4.3.

The fact that different samples are likely to give different results shows that there is a possibility of error which may affect the final decision. The sampling distribution allows us to calculate this error also.

Very often, it is not easy to determine the sampling distribution exactly. In such cases we make use of a fundamental theorem in statistics known as the Central Limit Theorem. We shall only state this theorem and discuss its utility. In Sec.4.3, we also discuss sampling distribution of proportions. You will see that this distribution is useful when we take samples from a binomial population.

In Sections 4.4, 4.5 and 4.6 we shall discuss three important sampling distributions,  $t$ ,  $\chi^2$  and  $F$ .

The examples and exercises in this unit are focused on how sampling techniques can assist us in making decision about various real-life problems.

Here is a list of what you should be able to do by the end of this unit.

### Objectives

After reading this unit, you should be able to

- explain the need to study the sampling distribution of a statistic,
- use sampling distribution of mean and proportion to draw inferences about the population mean and population proportion,
- calculate the error in sampling using sampling distribution,
- use central limit theorem to make inferences,
- use  $\chi^2$ , F and t-distributions to solve some problems of statistical inference.

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## 4.2 POPULATION AND SAMPLES

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The term population in statistics is applied to sets or collection of objects, actual or conceptual, and mainly to sets of numbers, measurements, or observations. For example, if we collect data on the number of television sets for each household in Mumbai, then a listing of number of television sets per household constitutes the population.

In some cases, such as concerning the number of television sets per household, the population is **finite**; in other cases, such as the determination of some characteristic of all units past, present, and future, that might be manufactured by a given process, it is convenient to think of the population as **infinite**. Similarly, we look upon the results obtained in a series of flips of a coin as a sample from the hypothetically infinite population consisting of all conceivably possible flips of the coin

In the last block we have seen that populations are often described by the distributions of the values, and it is a common practice to refer to a population in terms of this distribution. (For finite populations, we are referring here to the actual distribution of its values; for infinite populations, we are referring to the corresponding probability distribution or probability density.). For example, we may refer to a number of flips of a coin as a sample from a “binomial population” or to certain measurements as a sample from a “normal population.” Hereafter, when referring to a “population  $f(x)$ ” we shall mean a population such that its elements have a frequency distribution, a probability distribution, or a density with values given by  $f(x)$ .

If a population is infinite it is impossible to observe all its values, and even if it is finite it may be impractical or uneconomical to observe it in its entirety. Thus, it is usually necessary to use a **sample**, a part of a population, and infer from it results pertaining to the entire population. Clearly, such results can be useful only if the sample is in some way “representative” of the population. It would be unreasonable, for instance, to expect useful generalisations about the population of 1984 family incomes in India on the basis of data pertaining to home owners only. Similarly, we can hardly expect reasonable generalisations about the performance of a tyre if it is tested only on smooth roads.

Here we notice that the elements in the population may not be always uniform in character. For example in the case of population of 1984, all of them may not be home owners. Like that all of them may not be having other characteristics like educated etc. Such population where the elements of the population are not having uniform

characteristic is called **heterogeneous population**. Otherwise the population is called **homogeneous**. So we have to take some care while taking samples from a heterogeneous population. To assure that a sample is representative of the population from which it is obtained, and to provide a framework for the application of probability theory to problems of sampling, we shall limit our discussion to what is called a random sample. For sampling from finite populations, they are defined as follows:

**Definition 1 :** A set of observations  $x_1, x_2, \dots, x_n$  constitutes a random sample of size  $n$  from a finite population of size  $N$ , if it is chosen so that each subset  $S$  containing  $n$  of the  $N$  elements of the population has the same probability of being selected. 7

Note that this definition of randomness pertains essentially to the manner in which the samples are selected. This holds also for the following definition of a random sample from a theoretical (possibly infinite) population:

**Definition 2:** A set of observations  $x_1, x_2, \dots, x_n$  constitutes a random sample of size  $n$  from a population with density or mass function  $f(x)$  if

- 1) each  $x_i$  is a realisation of a random variable whose frequency/ density function is given by  $f(x)$ .
- 2) these  $n$  random variables are independent.

There are several ways of selecting a random sample. For relatively small sample, this can be done by drawing lots, or equivalently by using a table of “random numbers” specially constructed for such purposes. We shall discuss the utility of this table in and other selection procedures in detail in Unit 12, Block 4.

Before proceeding further, why don't you try some exercises now.

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- E1) Suppose we want to know the average age of female students enrolled in IGNOU BDP programme. What will be the population for this study? Is this population finite or infinite?
- E2) Which of the following is an appropriate sample for studying the situation given in E1.
- i) IGNOU female of BDP students selected from Delhi region.
  - ii) Randomly selected female students from the list of BDP students.
  - iii) Female students selected from two study centres of each regional centres.
  - iv) Randomly selected students from all students registered with IGNOU.
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A population is completely determined (or characterised) by certain fixed quantities (often unknown). These fixed quantities are called **parameters** and the problem consists of “inferring” about these characteristics based on a sample data.

A function of a sample observation is called a **statistic**.

For example in the case of the problem determining the average of income daily wagers in a particular city, we find average income of a sample of workers chosen. Such measures (or quantities) calculated from a sample is called a statistic.

Try this exercise, now.

- 
- E3) Give an example to show the difference between a parameter and a statistic.
- 

Now, suppose we want to study the population using the “mean” as given in the earlier example. When we are referring to the mean of a population, we call it **population mean** and when we are referring to the mean of a sample, we call it **sample mean**.

**Population mean is usually denoted by  $\mu$  and sample mean is denoted by  $\bar{x}$**



Similarly for other measures like standard deviation, proportion etc, we have population standard deviation, sample standard deviation and population proportion, sample proportion respectively.

To understand the terms parameters and statistic in a better way, let us consider an example.

Assume that we want to draw inferences regarding the accuracy of the quantity of Milk being packed by a leading milk processing company in Western Region of India, AMUL, in 500 ml. packets.

Here the population consists of all milk packets of 500 ml. packets by AMUL company in a day. By finding out the mean of the measurements obtained from all the packets of a day's production, we get population mean. We pick up a random sample of size  $n$ , and take the measurements. We denote the measurements as  $x_1, x_2, x_3, \dots, x_n$ . Note that these measurements vary with different samples. We use  $X_1, X_2, \dots, X_n$  to denote the random variables whose particular observations are  $x_1, x_2, \dots, x_n$ .

To determine the average quantity of milk being packed, we calculate the mean of the random observations,  $x_1, x_2, x_3, \dots, x_n$ . This is the sample mean for this sample. We denote this mean by  $\bar{x}$ .  $\bar{x}$  is a particular value of  $\bar{X}$ , where  $\bar{X}$  is

$$\bar{X} = t(X_1, X_2, \dots, X_n)$$

From the above discussion, we observe that the sample statistic, like sample mean, is a random variable. Next we shall study these random variables in detail.

### 4.3 WHAT IS A SAMPLING DISTRIBUTION

Let us start with situation of 'quantity of milk' discussed in the above section.

Now to study the average quantity of milk, suppose we decide that we take a sample of 10 packets without replacement and observe the quantity of milk in each packets. So here the sample size is  $n=10$ . The observed values are given in Column 2 Table 1 in the next page.

The mean of these 10 observations  $\bar{x}_1$  is 496.01 for example 1. That is the average of the quantities of milk obtained from the sample is 496.01. Now, if we are using this sample to make judgements about the population, then we say that, the sample mean value is 496.01 gives an estimate of the average quantity of milk for the whole population.

Table 1

| No. | Quantity of Milk in mls |          |
|-----|-------------------------|----------|
|     | Sample 1                | Sample 2 |
| 1   | 502                     | 501      |
| 2   | 501                     | 493.9    |
| 3   | 499.5                   | 499.6    |
| 4   | 501.05                  | 490.03   |
| 5   | 499.05                  | 500.09   |
| 6   | 497.56                  | 500      |
| 7   | 501.06                  | 500      |
| 8   | 459.3                   | 499.3    |
| 9   | 499.6                   | 497.5    |
| 10  | 500                     | 502.09   |

But, if we take another sample from the population and observe the values as given in Column 3 Table 1, the mean of these values is 498.35 for sample 2. This is different

from the mean of the first sample. Now, if we are using this sample to make inference about the population, then we get a different estimate of the average quantity of milk for the whole population. Like this we can take many different samples of size 10 and each case we get sample means which may or may not be distinct. From all these values we try to estimate the mean of the whole population. In this case it is the average amount of milk in any carton.

To give you a better understanding of generalising from sample statistic to the value of the parameter, let us look at the following example in which the size of the population is very small.

**Example 1:** Suppose we have a population of  $N=4$  incomes of four business firms and we want to find the average return of these firms. The incomes (in Lakhs) are 100, 200, 300 and 400.

We first note that in this case the (population) mean income is 250 lakhs. Now we use this situation to illustrate how sample means differ from the population mean.

Suppose we select a sample of  $n = 2$  observations in order to estimate the population mean  $\mu$ . Now, there are  $C(4, 2) = 6$  possible samples of size 2 and we will randomly be selecting one sample from this. We shall now calculate the means of these 6 different samples. These six different samples and their means are given in the following table.

Table 2

| Sample | Sample elements<br>$X_i$ | Sample means<br>$\bar{X}$ |
|--------|--------------------------|---------------------------|
| 1      | 100,200                  | 150                       |
| 2      | 100,300                  | 200                       |
| 3      | 100,400                  | 250                       |
| 4      | 200,300                  | 250                       |
| 5      | 200,400                  | 300                       |
| 6      | 300,400                  | 350                       |

\* \* \*

Now, from the table above, you can find that each sample has a different mean, with the exception of third and fourth samples. Therefore four of the six samples will result in some error in the estimation process. This sampling error is the difference between the population mean  $\mu$  and the sample mean we use to estimate it.

Let us now consider the possible sample means and calculate with their probability. We assume that each sample is equally likely to be chosen. Then the probability of selecting a sample is  $\frac{1}{6}$

Then we list every possible sample means and their respective possibilities in a table. (See Table 3 given below).

Table 3

| Sample mean<br>$\bar{X}$ | Number of<br>samples yielding $\bar{X}$ | Probability<br>$P(\bar{X})$ |
|--------------------------|-----------------------------------------|-----------------------------|
| 150                      | 1                                       | 1/6                         |
| 200                      | 1                                       | 1/6                         |
| 250                      | 2                                       | 2/6                         |
| 300                      | 1                                       | 1/6                         |
| 350                      | 1                                       | 1/6                         |
|                          |                                         | Total 1                     |

The table obtained above is called the sampling distribution of mean.

**Definition 3:** A list of all possible values for a sample statistic and the probability associated with each value is called a **sampling distribution of the statistic**.

In the above example if we take the sample mean  $\bar{x}$  as the random variable, then Table 3 is nothing but the probability distribution of the means. That means, here the observed values are the means. Like any other list of numbers, these sample means have a mean. It is called '**mean of the sample means**' or the **grand mean**. The mean of the sample means is calculated in the usual fashion: the individual observations are summed, and the result is divided by the number of observation. Therefore if  $\bar{\bar{X}}$  denotes this mean, then we have

$$\bar{\bar{X}} = \frac{\sum_{i=1}^k \bar{x}_i}{k}$$

where  $k$  is the number observation (samples). For the given situation of incomes, we can calculate  $\bar{\bar{X}}$  from table as

$$\begin{aligned}\bar{\bar{X}} &= \frac{150 + 200 + 250 + 250 + 300 + 350}{6} \\ &= 250\end{aligned}$$

Notice that this equals the population mean  $\mu = 250$ . This is no coincidence. The grand mean  $\bar{\bar{X}}$  will always equal the population mean. Thus we have arrived at an important observation.

**If we were to take every possible sample of size  $n$  from a population and calculate each sample mean, the mean of those sample means would equal the population mean. That means  $\bar{\bar{X}}$  is the population mean.**

Now, why don't you calculate the mean  $\bar{\bar{X}}$  for the income problem discussed, in Example 1, by taking samples of size 3 (see E3).

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E4) Construct the sampling distribution for the income problem discussed in Example 1, by taking samples of size 3. Calculate the grand mean and compare it with the population mean.

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After doing E4, you must have noticed that in the case of samples of size 3 also, we get that the mean of the sample means is equal to the population mean.

**Note:** You should not confuse  $n$ , the number of observations in a single sample, with  $k$ , the number of possible samples. In the situation of 4 incomes, the sample size is  $n = 2$ , while the number of possible samples is  $k = 4C_2 = 6$ .

Let us do a problem now.

**Problem 1:** A Statistics professor has given five tests. A student scored 70, 75, 65, 80 and 95 respectively in five tests. The professor decides to determine his grade by randomly selecting a sample of three test scores. Construct the sampling distribution for this process. What observations might you make?

**Solution:** There are  ${}^5C_3 = 10$  possible samples. The samples and their means are given in Table 4.

The sampling distribution is given in Table 5. Note that the population mean is  $\mu = 77$ . None of these samples gives the mean as 77. Five of the ten possible samples produce values of the sample mean  $\bar{X}$  in excess of the population mean, while the other five samples underestimate it.

Table 4

| Sample number | Sample elements $X_i$ | Sample mean $\bar{X}$ | Sample number | Sample elements $X_i$ | Sample mean $\bar{X}$ |
|---------------|-----------------------|-----------------------|---------------|-----------------------|-----------------------|
| 1             | 70,75,65              | 70.0                  | 6             | 70,80,95              | 81.7                  |
| 2             | 70,75,80              | 75.0                  | 7             | 75,65,80              | 73.3                  |
| 3             | 70,75,95              | 80.0                  | 8             | 75,65,95              | 78.3                  |
| 4             | 70,65,80              | 71.7                  | 9             | 75,80,95              | 83.3                  |
| 5             | 70,65,95              | 76.7                  | 10            | 65,80,95              | 80.0                  |

Table 5

| $\bar{X}$ | $P(\bar{X})$ |
|-----------|--------------|
| 70.0      | 1/10         |
| 71.7      | 1/10         |
| 73.3      | 1/10         |
| 75.0      | 1/10         |
| 76.7      | 1/10         |
| 78.3      | 1/10         |
| 80.0      | 2/10         |
| 81.7      | 1/10         |
| 83.3      | 1/10         |
|           | 1.00         |

You can try some exercises now.

E5) The ages of six executives of a company are

| Name        | Age |
|-------------|-----|
| Mr. Ravi    | 54  |
| Mrs. Veena  | 50  |
| Mrs. Shanti | 52  |
| Mr. Suresh  | 48  |
| Mr. Rajiv   | 50  |
| Mr. Anil    | 52  |

- How many samples of size 2 are possible?
- Construct the sampling distribution of means by taking samples of size 2 and organise the data.
- Calculate the mean of the sampling distribution and compare it with the population mean.

As we observed in the example above, it will be interesting to note how much these sample means vary from the population mean. From your knowledge of frequency distribution (see Unit 1, Block 1), you may think that this variance can be obtained by calculating the variance of the sample means. You are right. The sample means also have a variance. It measures the dispersion of the individual observations (sample means) around their population means. Furthermore this variance is calculated like any other variance. We can obtain this by performing the following.

- the amount by which each of the observations (sample means) differs from the population mean.
- squaring these deviations.

We denote by  $\sigma_{\bar{X}}$ , the standard deviation of the sampling distribution of sample means. Then we have

$$\sigma_{\bar{X}}^2 = \frac{\sum (\bar{x}_i - \bar{\bar{X}})^2}{k} \quad (1)$$

Note that  $\bar{\bar{X}}$  is the mean of the sampling distribution which is the same as the population mean.

Now that we have seen too many "standard deviations. One is the standard deviation of the entire population which we usually denote by  $\sigma$ . then the standard deviation of a single sample, and now we have the standard deviation  $\sigma_{\bar{X}}$  of an entire set of sample means. **Since  $\sigma_{\bar{X}}$  measures the dispersion of the sample means around  $\mu$ , it gives a measure for the error in sampling. Because of this,  $\sigma_{\bar{X}}$  is called sampling error or standard error. We formally define this now.**

**Definition 3:** The standard error is the standard deviation of the sample means around the population mean and is denoted by SE ( $\bar{x}$ ).

Let us now calculate the standard errors of the sampling distribution of mean obtained in Problem 1.

**Problem 2:** Compute the standard error of the sampling distribution of sample scores obtained in Problem 1.

**Solution:** Here

$$\begin{aligned} \bar{\bar{X}} &= \frac{\sum_{i=1}^k \bar{x}_i}{k} = 77 \\ \sigma_{\bar{X}}^2 &= \frac{\sum_{i=1}^k (\bar{x}_i - \bar{\bar{X}})^2}{k} \\ \sigma_{\bar{X}}^2 &= \frac{(70 - 77)^2 + (75 - 77)^2 + (80 - 77)^2 + (71.7 - 77)^2 + \dots + (80 - 77)^2}{10} \\ &= 17.6 \\ \sigma_{\bar{X}} &= \sqrt{17.6} = 4.20 \end{aligned}$$

This shows that the mean of all 10 possible sample means is 77. The square root of the average square deviation of the sample means from the population mean of 77 is 4.2.

You will study later in Unit 5 that the standard error of any statistic gives us an idea of how good a statistic is in estimating the parameters.

You may have realised that the computation of the standard deviation from the sampling distribution is a tedious process. There is an alternative method to compute standard error of the means, SE( $\bar{x}$ ), from a single sample if we know the population standard deviation. By this method, we have the following formula for obtaining the standard error of the mean for a finite population.

$$SE(\bar{x}) = \sqrt{\frac{(N - n)}{N - 1} \frac{(\sigma^2)}{n}}$$

where  $N$  = population size = the total number of individuals,  $n$  = sample size = number of individuals selected in the random sample.  $\sigma$  = standard deviation of the individuals in the population.

We will not derive the formula here since the process is too technical for the scope of this course. The factor  $\sqrt{\frac{N-n}{N-1}}$  is called the finite population correction factor. As a rule of thumb, when  $\frac{n}{N}$  is less than 0.1, this correction factor can be ignored. We use the above formula for computing  $SE(\bar{x})$  when  $N-n$  is not very large. When  $N$  is large, relative to  $n$ , we use the formula,

$$SE(\bar{x}) = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

This formula can also be used for calculating  $SE(\bar{x})$  for infinite population.

Let us see an example.

**Problem 3:** The U.S. Bureau of census wishes to estimate the birth rates per 1,00,000 people in the nation's largest cities. It is known that the standard deviation in the birth rates for these 100 urban centres is 12 births per 1,00,000 people. Then

- calculate the variance and standard error of the sampling distribution of i)  $n = 8$  cities ii)  $n = 15$  cities.
- compare the values obtained in both the cases.

**Solution:**

- i) Here  $N = 100$  and  $n = 8$  and the population variance is 12. Therefore  $\frac{n}{N}$  is less than 0.1. Then we use the formula for calculating the variance as

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} = \frac{12^2}{8} = 18$$

and the standard error is  $\sigma_{\bar{x}} = \sqrt{18} = 4.24$

- ii) In this case  $N = 100$  and  $n = 15$  and therefore  $\frac{n}{N}$  is greater than 0.1. Also  $\sigma = 12$ . Therefore we use the formula for variance as

$$\begin{aligned}\sigma_{\bar{x}}^2 &= \left( \frac{N-n}{N-1} \right) \frac{\sigma^2}{n} \\ &= \left( \frac{100-15}{100-1} \right) \left( \frac{12^2}{15} \right) \\ &= 8.24\end{aligned}$$

$$\text{Hence, } \sigma_{\bar{x}} = \sqrt{8.24} = 2.87$$

- On comparing both the values we observe that, the larger sample has a smaller standard error and will tend to result in less sampling error in estimating the birth rates in the 100 cities.

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Here is an exercise for you.

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E6) From a population of 200 observations, a sample of  $n=50$  is selected. Calculate the standard error if the population standard deviation equals 22.

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Thus we have seen that we can compute the standard error of the sample means if we know the population standard deviation. Here you can note one thing. Usually, we do not know  $\sigma$ . But it is possible to estimate  $\sigma$  from the sample. We will talk about this later in the next section. Using that estimate we compute the standard error by the shortcut formula given earlier.

Now we summarise our discussion about mean and the standard deviation of a sampling distribution. We state the following Theorem.

**Theorem 1:** If a random sample of size  $n$ , say  $X_1, X_2, \dots, X_n$  is taken from a population having the mean  $\mu$  and variance  $\sigma^2$ , then  $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$  is a random variable whose distribution has the mean  $\mu$ . For samples from infinite populations the variance of the distribution is  $\sigma^2/n$ , for samples from finite population of size  $N$ , the variance is  $\frac{\sigma^2}{n} \frac{N-n}{N-1}$

So far we have been discussing about the mean and standard deviation of the sampling distribution. Next we shall see some interesting properties regarding the shape of the sampling distribution. For that let us consider the following situation.

Suppose in an experiment 50 random samples of size  $n = 10$  are taken from a population having discrete uniform distribution.

Sampling is with replacement, so that we are sampling from an infinite population. The sample means of these 50 samples are given below.

4.4 3.2 5.0 3.5 4.1 4.4 3.6 6.5 5.3 4.4  
 3.1 5.3 3.8 4.3 3.3 5.0 4.9 4.8 3.1 5.3  
 3.0 3.0 4.6 5.8 4.6 4.0 3.7 5.2 3.7 3.8  
 5.3 5.5 4.8 6.4 4.9 6.5 3.5 4.5 4.9 5.3  
 3.6 2.7 4.0 5.0 2.6 4.2 4.4 5.6 4.7 4.3

We group these means into a distribution with the classes 2.0-2.9, 3.0-3.9, ..., and 6.0-6.9, then we get the following table.

Table 6

| $\bar{x}$ | Frequency |
|-----------|-----------|
| 2.0-2.9   | 2         |
| 3.0-3.9   | 14        |
| 4.0-4.9   | 19        |
| 5.0-5.9   | 12        |
| 6.0-6.9   | 3         |
|           | 50        |

It is clear from this distribution as well as its histogram shown in Figure 1 that the distribution of the means is fairly **bell-shaped** even though the population itself has a discrete uniform distribution.

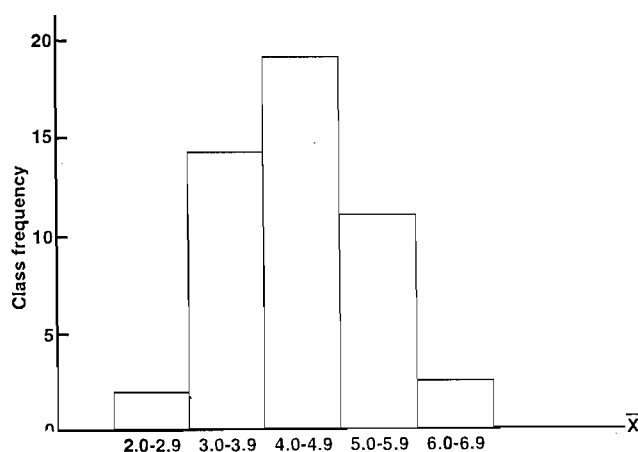


Fig. 1

This raises the question whether our result is typical of this population or any other!!

Let us consider the other situation which was discussed in the earlier section (where we want to estimate the average quantity of Milk).

Suppose in that experiment we are taking 10 random samples having size  $n=10$  (In Block 4 we shall see how we can select random samples). The following table illustrates the 10 samples.

Table 7

| 1      | 2      | 3      | 4     | 5      | 6      | 7     | 8      | 9      | 10    |
|--------|--------|--------|-------|--------|--------|-------|--------|--------|-------|
| 502.0  | 501    | 493.9  | 497.8 | 502.09 | 502.09 | 502.9 | 493.9  | 459.3  | 497.3 |
| 501.0  | 493.9  | 499.6  | 499.3 | 499.6  | 501    | 501.9 | 493.09 | 497.56 | 493   |
| 499.5  | 499.6  | 502    | 499   | 490.9  | 500.09 | 501   | 499    | 493.0  | 499   |
| 501.05 | 490.03 | 501.3  | 501   | 496.8  | 500    | 500   | 499.6  | 499.05 | 499   |
| 499.05 | 500.09 | 500    | 502.9 | 500    | 500    | 500   | 499.9  | 499.5  | 499.3 |
| 497.56 | 500    | 499.9  | 501.9 | 503    | 499.3  | 499.3 | 500    | 499.6  | 500   |
| 501.06 | 500    | 493.99 | 500   | 502    | 499.6  | 499   | 500    | 500.0  | 500   |
| 459.3  | 499.3  | 500    | 499   | 501    | 493.9  | 499   | 501.3  | 501.0  | 501   |
| 499.6  | 497.5  | 502    | 500   | 493.9  | 497.5  | 493   | 502    | 501.03 | 501.9 |
| 500.0  | 502.09 | 499    | 493   | 503    | 490.03 | 497.8 | 502    | 501.06 | 502.9 |

If we plot these samples means, then we get the following graph.

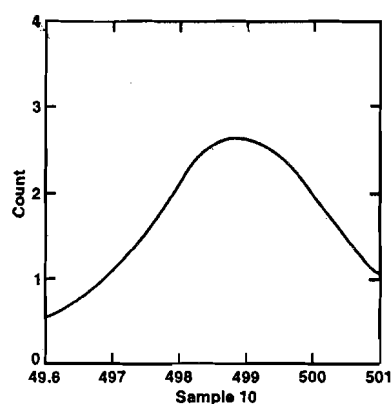


Fig.2

Here also you can observe that it is approximately bell-shaped.

Note that unlike in the earlier situation, in this case we do not know the distribution of the population.

Does the above observations give an indication that irrespective of the nature of the distribution of the population, the sampling distribution of means is approximately normal? This idea will be more clear to you if you look at the figure given at end of the unit (see Appendix 1). The figure shows graphs of the distribution function of 4 different populations see Fig. (a). The graphs in each of the figures in (b), (c) and (d) shows the sampling distribution for the sample sizes  $n = 2$ ,  $n = 5$  and  $n = 30$  for the respective population. You note that the parent distribution for population 4 is normal and sampling distributions are also normal. This is not surprising. But the surprising fact is that for population 1, 2 and 3 parent distributions are not normal, still, as the sample size increases, the sampling distribution in each case approaches a normal distribution. The remarkable normality of the distribution of the sample mean is the substance of the famous central limit theorem. We state a simple version of the theorem.

**Theorem 2: Central Limit Theorem :** -  $X_1, X_2, \dots, X_n$  be  $n$  independent and identically distributed random variables having the same mean  $\mu$  and variance  $\sigma$ . Let  $S_n = X_1 + X_2 + \dots + X_n$ . Then  $\frac{S_n - n\mu}{\sqrt{n}\sigma}$  has approximately the standard normal distribution  $(0, 1)$ .

Now we shall illustrate this theorem in the case of sampling distribution of means ( $\bar{X}$ ).

Suppose we take a sample. Let  $X_1, X_2, \dots, X_n$ , are independent random variables by the sampling scheme and also they are identically distributed.  $X_i$ 's have the same mean  $\mu < \infty$  and variance  $\sigma^2 < \infty$  (also called the population mean and population variance) and therefore we can apply the central limit theorem to  $X_1, X_2, \dots, X_n$  and



conclude that  $\frac{X_1 + X_2 + \dots + X_n - n\mu}{\sqrt{n}\sigma}$  is distributed approximately as standard normal. But note that  $\frac{X_1 + \dots + X_n}{n}$  is nothing but the sample mean  $\bar{X}$ , therefore according to the central limit theorem we have  $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$  follows standard normal distribution  $N(0, 1)$ .

So far we have been discussing the sample means and their distribution. Our interest has been in some variable which might be measured and averaged. However there are many instances where we may need some other statistic, other than mean, to make inference about the population. In the next section we shall discuss such a statistic known as 'proportion' and discuss its sampling distribution.

### Sampling Distribution of Proportion

There are many instances where we are interested in determining the number or proportion of observations falling in a particular category. For example, a doctor wants to know how many of his patients can survive after administering a particular drug. A politician may want to know how many voters will not vote for him. In all these situations there are two possible outcomes of the observation - whether an observation falls in a particular category or not. You might recall that such situations are related to the problems dealing with binomial distribution. The politician may not be interested in the actual number of people who are going to vote for him, she is interested on what percentage of the people will do so. In such a situation we deal with sample proportions instead of sample means. The population parameter in this case is the proportion. We denote this by  $\pi$ . In general, for a finite population, we define the population proportion  $\pi$  as

$$\pi = \frac{k}{n}$$

where  $k$  is the number of observations that fall in a particular category and  $n$  is the total number of observation. When the population is very large, we may take samples to study the population and for each sample we calculate the sample proportion,  $p$ , as

$$p = \frac{s}{n}$$

where  $s$  denotes the number of observations in the sample which meet the particular characteristic, under study and  $n$  is the sample size.

For example, assume that a politician surveys 1,000 voters and finds that only 350 are going to vote for him. Then

$$p = \frac{350}{1000} = 0.35$$

You note that, a different sample of  $n = 1,000$  voters may yield a different  $p$ . If we calculate the possible sample proportions then a list of these observations is called the **sampling distribution of proportions**.

Let us consider an example. As, in the case of sample means, we are considering a simple situation where the population size is very small.

**Example 2:** In our discussion about the quantification of milk packets, let us put the condition that if the quantity in any packet measured is less than 495ml, it will be rejected by the consumer, hence call them as defective and other wise they are non-defective.

For the data given in Table 7 we consider the first five columns (five cartons) as population then the population has 50 milk packets.

From the population we get the proportion of defective as

$$P(\text{Defective}) = \pi = \frac{7}{50} = 0.14$$

and the proportion of non-defective is

$$P(\text{non-defective}) = 1 - \pi = 0.86$$

Let us try to use the sampling method to find this proportion. Assume that random samples of size 2 out of 5 (that is two cartons out of five cartons) have been selected and the possible samples and their defective proportions are as given below

Table 8

| Sample     | Sample proportions ( $p_i$ ) |
|------------|------------------------------|
| $C_1, C_2$ | 2/20                         |
| $C_1, C_3$ | 3/20                         |
| $C_1, C_4$ | 2/20                         |
| $C_1, C_5$ | 3/20                         |
| $C_2, C_3$ | 3/20                         |
| $C_2, C_4$ | 2/20                         |
| $C_2, C_5$ | 3/20                         |
| $C_3, C_4$ | 3/20                         |
| $C_3, C_5$ | 4/20                         |
| $C_4, C_5$ | 3/20                         |

Let us calculate the mean of the sample proportion  $\bar{p}$

$$\begin{aligned}\bar{p} &= \frac{\sum p_i}{k} \\ &= \frac{28}{20 \times 10} \\ &= .14\end{aligned}$$

This is same as the population proportion.

\* \* \*

From the earlier example we conclude the following fact:

**The mean of the sampling distribution of proportions is equal to the population proportion**

Now the standard error of the sample proportions, by definition, is the standard deviation of this sampling distribution. As we have mentioned earlier, in the case of mean, the computation of standard deviation from the table is a tedious process. So we make use of a short cut formula by which we can compute the **standard error** which we **denote by SE(p)**. If we know the population proportion, population size and sample size, the formula is

$$SE(p) = \sqrt{\frac{(N - n)\pi(1 - \pi)}{(N - 1).n}}$$

where  $\pi$  is the population proportion and  $N$  and  $n$  are the population size and sample size, respectively. If the population is infinite or if the size of the population, relative to the sample size is extremely large, then

$$SE(p) = \sqrt{\frac{\pi(1 - \pi)}{n}}$$

Let us now calculate the standard error for the situation in Example 3. In this case  $\pi = 0.4$ ,  $N = 5$  and  $n=2$ . Therefore,

$$\begin{aligned} SE(p) &= \sqrt{\frac{(0.14)(0.86)}{2}} \sqrt{\frac{5-2}{5-1}} \\ &= 0.2454 \times 0.866 \\ &= 0.2125. \end{aligned}$$

Here is an exercise for you.

- 
- E7) a) In a manufacturing process of electric bulbs, company gives an output of 250 electric bulbs per day. To test the defectiveness of the manufactured bulbs, a random sample of 50 has selected in a particular day's production. Inspection of the samples showed 4 as defective and the rest as non-defective. Calculate the standard error of sample proportion of defective.
- b) If the process is considered to be a continuous one (observed for many days) and a random sample of 60 has been selected from the produced electric bulbs and observed that 7 are defective. What will be the standard error of sample proportion of defective.
- 

So far we have discussed sampling distributions of two statistics  $\bar{x}$  and  $\bar{p}$ . We also emphasised the role of Central limit theorem and the idea that distributions of  $\bar{x}$  and  $\bar{p}$ , will be approximately normal even when the data in the parent population are not distributed normally, provided the sample sizes are large.

But there are many situations where we may have to deal with small samples, may be due to limited availability of items or by other factors such as time of cost. We may also have to deal with certain statistics other than  $\bar{x}$  and  $\bar{p}$ . For example in some situations we may be interested in the distribution of  $\bar{x}/s$ , where  $s$  is the sample standard deviation. In some other situations, we may be interested in the distribution of  $\sum_{i=1}^n (x_i - \bar{x})^2$ . The important exact sampling distributions are chi-square, student t- and F-distributions. These distributions are widely used in statistical analysis. In the following sections we shall discuss these distributions.

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## 4.4 t-DISTRIBUTION

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William G. Gosset, a brew master for Guinness Brewarier, developed a family of distributions each of which corresponds to a parameter  $\nu$ , (called nu-a positive integer). The density function of a random variable whose distribution belongs to this family is given by

$$f_{\nu}(y) = \frac{1}{\Gamma(\pi\nu)} \frac{\Gamma(\nu + 1/2)}{\Gamma(\nu/2)} \left(1 + \frac{y^2}{\nu}\right)^{-\nu/2}$$

where  $\Gamma(x)$  for any  $x > 0$  is called the gamma function. The values of  $\Gamma(x)$  for different values of  $x > 0$  are tabulated. The three members of the family of this distribution for  $\nu = 9$ ,  $\nu = 14$  and  $\nu > 30$  are shown in the accompanying figure.(See Fig. 3 in the next page.)

Now we shall illustrate the importance of this distribution as distribution in the case of sampling distribution.

Suppose we take a sample from a population having normal distribution  $N(\mu, \sigma^2)$ , where  $\mu$  and  $\sigma^2$  are unknown. If  $X_1, X_2, \dots, X_n$  are  $n$  observations in the sample, then  $X_1, \dots, X_n$  are independently and identically distributed(i.i.d) random variables. and

we have  $\frac{\bar{X} - \mu}{s/\sqrt{n}}$  has t-distribution with parameter  $\nu = n - 1$ , where  $s$  is the sample variance. The parameter  $\nu$  is called the number of degrees of freedom (in short d.f) of the distribution.

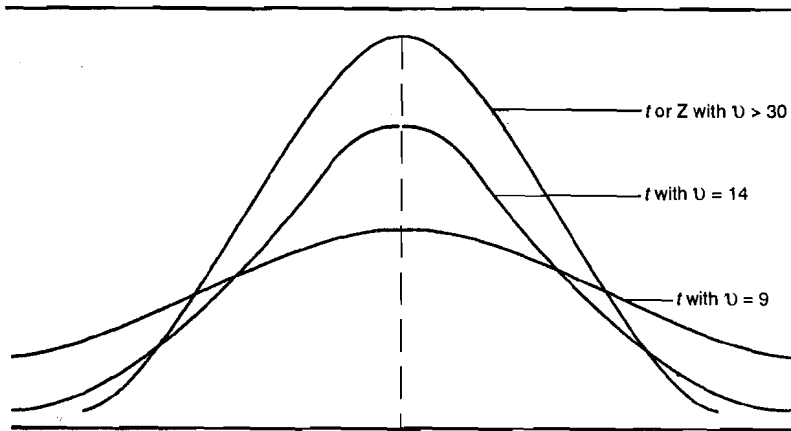


Fig. 3 t-distributions

Now that, you have been introduced to the distribution, we shall see how we use this distribution to make judgements about the population.

As we have seen in the case of normal distribution (see Unit 3, Block 1), there are tables available using which we can calculate probability for this distribution. One such table is given in the appendix (see Table 2 in appendix of this block). The table contains selected values of  $t_\alpha$  for various values of  $\nu$ , where  $t_\alpha$  is such that the area under the t-distribution to its right is equal to  $\alpha$ . In this table the left-hand column contains values of  $\nu$ , the column headings are areas  $\alpha$  in the right-hand tail of the t-distribution, and the entries are the values of  $t_\alpha$  (See Fig. 4 below)

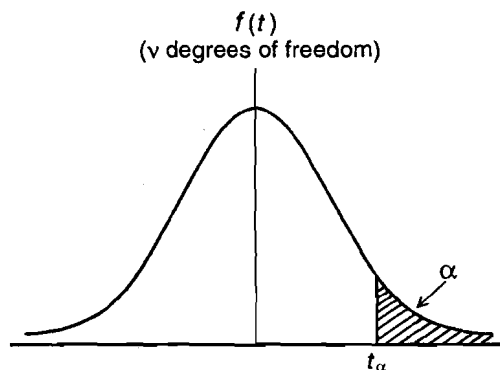


Fig. 4: Tabulated values of  $t$

It is not necessary to tabulate the values of  $t_\alpha$  for  $\alpha > 0.50$ , as it follows from the symmetry of the distribution that  $t_{1-\alpha} = t_\alpha$ ; thus, the value of  $t$  that corresponds to a left-hand tail area of  $\alpha$  is  $t_\alpha$ . You note that the bottom row of the table the entries are the same as the values of the standard normal variate  $Z$ . For example  $t_{0.025} = 1.96 = z_{0.025}$  for large values of  $\nu$  i.e.  $\nu \geq 30$ .

**Remark:** Please note that the tables for the distribution can vary from book to book because of the definitions of parameters involved. Therefore, for the purposes of this course, please only refer to the table given at the end of the block.

**Example 3:** The graph of a t-distribution with 9 degrees of freedom is given below. Let us find the values of  $t_1$ , for which

- i) the shaded area of the right = 0.05
- ii) the total shaded area = 0.05

iii) the total unshaded area = 0.99

iv) the shaded area on the left = 0.01

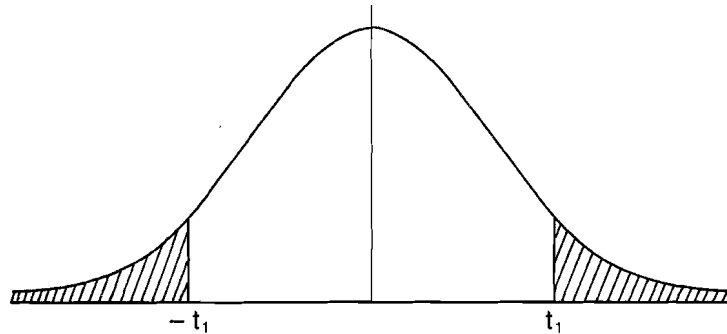


Fig. 5

Let us try (i) to (iv) one by one.

- i) If the shaded area on the right is 0.05, comparing the figure above with Fig.4, we get that  $\alpha = 0.05$ . It is already given that  $\nu = 9$ . Referring to the table given in the Appendix, we proceed downward under column headed  $\nu$  until entry 9 is reached. Then proceed right to the column headed 0.05. and get the required value of  $t$  as 1.833.
- ii) Given that the total shaded area is 0.05. Then by symmetry, the shaded area on the right is  $\frac{0.05}{2} = 0.025$  i.e.  $\alpha = 0.025$ . Also  $\nu = 9$ . Therefore from the table we get the required value as 2.262.
- iii) If the total unshaded area is 0.99, then the total shaded area is  $1 - 0.99 = 0.01$ . Therefore the shaded area on the right of  $t_1$  is  $\frac{0.01}{2} = 0.005$ . Then we get the required value from the table as 3.250.
- iv) If the shaded area on the left is 0.01, then by symmetry the shaded area on the right is also 0.01. Then from the table we get the value as 2.821

\* \* \*

**Problem 4:** For the given sample sizes and the  $t$ -values, find the corresponding probability  $\alpha$

i)  $n = 26, \quad t = 2.485$

ii)  $n = 14, \quad t = 1.771$

**Solution:**

- i) Referring to the  $t$ -table given in the Appendix, we find the row corresponding to  $\nu = 25$  and look for the entry  $t = 2.485$ . Then we observe that this  $t$ -value lies in the column headed  $t_{0.01}$ . This gives that the corresponding  $\alpha$ -value is 0.01.
- ii) Similarly you can see that  $\alpha$ -value corresponding to  $\nu = 13$  and  $t = 1.771$  is 0.05.

————— × —————

**Example 4:** A manufacturer of fuses claims that with a 20 % overload, his fuses will blow in 12.40 minutes on the average. To test this claim, a sample of 20 of the fuses was subjected to a 20 % overload, and the times it took them to blow had a mean of 10.63 minutes and a standard deviation of 2.48 minutes. if it can be assumed that the data constitute a random sample from a normal population, do they tend to support or refute the manufacturer's claim? Let us see how we can make use of the  $t$ -distribution to find out an answer for this.

We first note that the sample is taken from a normal population and the size is small,  $n = 20$ . Then as we have stated in the discussion above, the random variable  $\frac{\bar{x} - \mu}{s/\sqrt{n}}$  follows t-distribution with  $n - 1$  degrees of freedom.

Here  $\bar{x} = 10.63$ ,  $\mu = 12.40$  and  $s = 2.48$  and  $n = 20$ . Therefore,

$$t = \frac{10.63 - 12.40}{2.48/\sqrt{20}} = -3.19$$

Now we compare this t-value with the values of t given in the table for the parameter,  $\nu = 20 - 1 = 19$ . We look for the t-value which is nearest to 3.19 and less than 3.19. Then we find that the required t-value is 2.861. Also from the table, we find that the  $\alpha$ -value corresponding to 2.861 is 0.005. That means the probability that t will exceed 2.861 is 0.005. Now we note that  $-3.19$  is less than  $-2.861$  therefore the  $\alpha$ -value corresponding to  $-3.19$  will be less than 0.005, which is very small (see Fig.6).

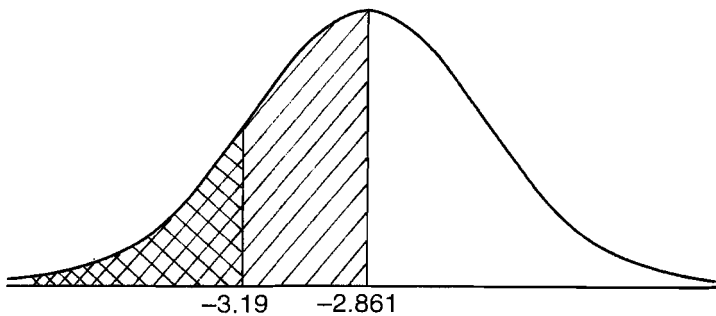


Fig. 6

Hence, we can conclude that the data tend to refute the manufacturer's claim.

\* \* \*

Try these exercises now.

---

E8) Find the values of t for given values of  $\nu$  and  $\alpha$

- i)  $\alpha = 0.01$  and  $\nu = 19$
- ii)  $\alpha = 0.05$  and  $\nu = 6$
- iii)  $\alpha = 0.025$  and  $\nu = 23$
- iv)  $\alpha = 0.1$  and  $\nu = 10$
- v)  $\alpha = 0.005$  and  $\nu = 29$

E9) A process for making certain bearings is under control if the diameters of the bearings have a mean of 0.5 cm. What can we say about this process if a sample of 10 of these bearings has a mean diameter of 0.5060 cm. and a standard deviation of 0.0040 cm. Assume that the data constitute a random sample from a normal population.

E10) Find the values of t for which the area of the right-hand tail of the t-distribution is 0.05 if the number of degrees of freedom is equal to  
(a) 16, (b) 27, (c) 200

---

Note that for applying t-distribution we made the assumption that the samples should come from a normal population and population standard deviation is not known. Studies have been shown later that the first assumption can be relaxed. It has been shown that the distribution of random variable with the values  $t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$  is fairly close

to a t-distribution even for samples from certain non-normal population. In practice, it is necessary to make sure primarily that the population from which we are sampling is approximately bell-shaped and too skewed.

Next we shall consider another exact sampling distribution.

## 4.5 CHI-SQUARE DISTRIBUTION

Another exact sampling distribution which is very useful in statistical problems is the chi-square distribution. In this section we shall introduce you to this distribution.

We say that a random variable  $Y$  has  $\chi^2$  (called chi-square) distribution with  $\nu$  degrees of freedom if the density function  $f_Y(y)$  of  $Y$  is given by

$$f_Y(y) = \frac{y^{(\nu/2-1)}e^{-y}}{\Gamma(\nu/2)}, \text{ if } y > 0$$

$$= 0, \text{ if } y \leq 0$$

where  $\Gamma(x)$  is the gamma function.

Note that this family of distributions is parametrised by  $\nu$ , called the degrees of freedom. The graph of the density function for some members of this family ( $\nu = 1, 3, 8, 10$ ) are shown in Fig.7 below

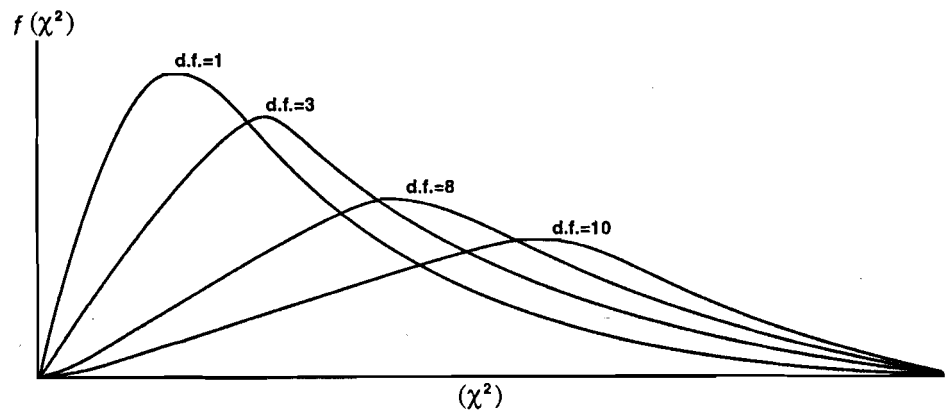


Fig.7 Various Chi-square distributions

As in the case of normal and t-distributions, a table containing selected values of  $\chi^2_\alpha$  for various values of  $\nu$ , again called the number of degrees of freedom, is given in the Appendix (see Table 3).  $\chi^2_\alpha$  is the value such that the area under the chi-square distribution to its right is equal to  $\alpha$ . (i.e. the probability that any value is greater than equal to or  $\chi^2_\alpha$  is  $\alpha$ ). In this table the left-hand column contains values of  $\nu$ , the column headings are areas  $\alpha$  in the right-hand tail of the chi-square distribution, and the entries are the values of  $\chi^2_\alpha$ . (see Fig.8 below).

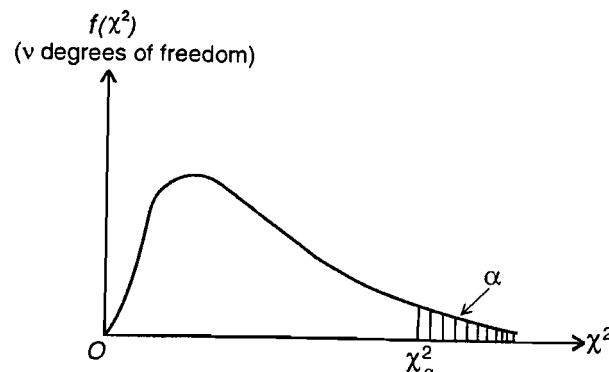


Fig.8

Note that unlike t-distribution, chi-square distribution is not symmetrical. Therefore, it is necessary to tabulate values of  $\chi^2_\alpha$  for  $\alpha > 0.50$  also.

Let us see some examples

**Example 5:** Let us find the values of  $\chi^2_\alpha$  of the  $\chi^2$ -distribution with 5 degrees of freedom for  $\alpha = 0.05$  and  $0.01$ .

Let us first consider  $\alpha = 0.05$ . To find the values of  $\chi^2_\alpha$  for 5 degrees of freedom and  $\alpha = 0.05$ , we refer to Table 3 in the Appendix. We proceed downward under column headed  $\nu$  until entry 5 is reached, then proceed right to column headed  $\alpha = 0.05$  which gives the required value as  $\chi^2_\alpha = 11.070$ .

Similarly for  $\alpha = 0.01$ , we get the required value as  $\chi^2_\alpha = 15.086$ .

\* \* \*

Now we shall illustrate the importance of  $\chi^2$ -distribution in the case of sampling distribution.

Suppose we take a random sample of size  $n$  when  $n$  is small ( $n \leq 30$ ) from a normal population having the variance  $\sigma^2$  and if  $s^2$  is the variance of the random sample, then

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

is a value of a random variable having  $\chi^2$ -distribution with  $\nu = n - 1$ .

Let us see how we use this result for solving the following problem.

**Problem 5:** An optical firm purchases glass to be ground into lenses, and it knows from past experience that the variance of the refractive index of this kind of glass is  $1.26 \cdot 10^{-4}$ . As it is important that the various pieces of glass have nearly the same index of refraction, the firm rejects such a shipment if the sample variance of 20 pieces selected at random exceeds  $2.00 \cdot 10^{-4}$ . Assuming that the sample values may be looked upon as a random sample from a normal population, what is the probability that a shipment will be rejected even though  $\sigma^2 = 1.26 \cdot 10^{-4}$ ?

**Solution:** We first note that the sample is taken from normal population and the size is  $n$  small,  $n = 20$ . Then as (we stated in the discussion above), the random variable

$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$  follows  $\chi^2$ -distribution with  $(n-1)$  degrees of freedom.

Here  $\sigma^2 = 1.26 \times 10^{-4}$ ,  $s^2 = 2 \times 10^{-4}$  and  $n = 20$ . Therefore

$$\chi^2 = \frac{(20-1) \times 2 \times 10^{-4}}{1.26 \times 10^{-4}} = 30.2.$$

Now we compare this  $\chi^2$ -value with the values of  $\chi^2$  given in the table for the parameter  $\nu = 20 - 1 = 19$ . Then we find that the  $\chi^2$ -value 30.1 is close to the calculated value 30.2 and less than the calculated value. The corresponding  $\alpha$ -value is 0.05. That means that the probability that  $\chi^2$ -value exceeds 30.1 is 0.05. Therefore we can conclude that the probability that a good shipment will erroneously be rejected is less than 0.05.

————— × —————

Here is an exercise for you.

E11) Find the values of  $\chi^2$  for which the area of the right-hand tail of the  $\chi^2$ -distribution is 0.5, if the number of degrees of freedom  $\nu$  is equal to (a) 15, (b) 21.

E12) A random sample of 10 observations is taken from a normal population having the variance  $\sigma^2 = 42.5$ . Find the approximate probability of obtaining a sample standard deviation between 3.14 and 8.94.



## 4.6 F-DISTRIBUTION

R.A. Fisher, a British statistician, developed this distribution in the early 1920's. It is defined as follows.

**Definition :** If a random variable  $Y_1$  has a  $\chi^2$ -distribution with  $\nu_1$  degrees of freedom, and if a random variable  $Y_2$  has a  $\chi^2$ -distribution with  $\nu_2$  degrees of freedom and if  $Y_1$  and  $Y_2$  are independent, then  $Y_1/\nu_1 \div Y_2/\nu_2$  has an F-distribution with  $\nu_1$  and  $\nu_2$  degrees of freedom.

In other words, the ratio of two independent  $\chi^2$  random variables, each divided by its number of degrees of freedom, is an F random variable.

Like the t and  $\chi^2$  distributions, the F distribution is in reality a family of probability distributions, each corresponding to certain numbers of degrees of freedom. But unlike the t and  $\chi^2$  distributions, the F distribution has two numbers of degrees of freedom, not one. Figure shows the F distribution with 2 and 9 degrees of freedom. As you can see, the F distribution is skewed to the right. However, as both numbers of degrees of freedom become very large, the F distribution tends toward normality. As in the case of the  $\chi^2$  distribution, the probability that an F random variable is negative is zero. This must be true since an F random variable is a ratio of two nonnegative numbers. ( $Y_1/\nu_1$  and  $Y_2/\nu_2$  are both nonnegative.) Once again, it should be emphasised that any F random variable has two numbers of degrees of freedom. Be careful to keep these numbers of degrees of freedom in the correct order, because an F distribution with  $\nu_1$  and  $\nu_2$  degrees of freedom is not the same as an F distribution with  $\nu_2$  and  $\nu_1$  degrees of freedom.

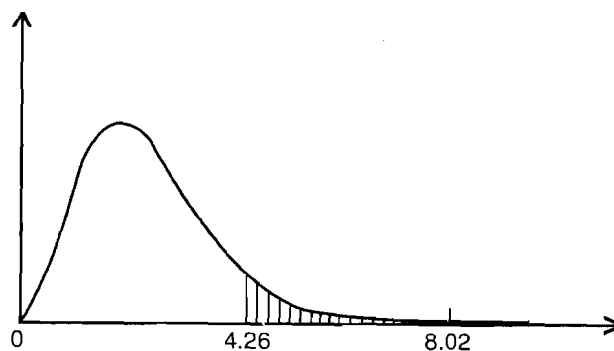


Fig.9

Tables are available which show the values of F that are exceeded with certain probabilities, such as .05 and .01. Appendix Table Q shows, for various numbers of degrees of freedom, the value of F that is exceeded with probability equal to .05. For example, if the numbers of degrees of freedom are 2 and 9, the value of F that is exceeded with probability equal to .05 is 4.26. (See Fig.9.) Similarly, Appendix Table 10 shows, for various numbers of degrees of freedom, the value of F that is exceeded with probability equal to .01. For example, if the numbers of degrees of freedom are 2 and 9, the value of F exceeded with probability equal to .01 is 8.02. (See Fig.9.)

Let us see some examples.

**Example 6:** A random variable has F distribution with 40 and 30 degrees of freedom, let us find the probability that it will exceed a) 1.79 b) 2.30.

We first consider the value 1.79. We look at table of F-distribution, Table 4 for 0.05. We proceed downward under column headed  $\nu_2$  until entry 30 is reached, then proceed right to the column headed  $\nu_1 = 40$  which shows the given value 1.79. Therefore we get that the probability is 0.05.

Since the value cannot be in Table 4 we look at Table 5 which is for 0.01. We proceed similarly and find the value 2.30 in the table. Therefore we get that the probability is 0.01.

\* \* \*

We shall now illustrate the importance of this distribution in the case of sampling distribution.

Suppose that we have two independent random samples of sizes  $n_1$  and  $n_2$  respectively, taken from two normal population having the same variance. If  $s_1^2$  and  $s_2^2$  are sample variances, then the function

$$F = \frac{s_1^2}{s_2^2}$$

has F distribution with  $\nu_1$  and  $\nu_2$  degrees of freedom.

F-distribution is used to compare the equality of two sample variances.

Let us consider the following problem.

**Problem 6:** If two independent random samples of size  $n_1 = 7$  and  $n_2 = 13$  are taken from a normal population, what is the probability that the variance of the first sample will be at least three times as large as that of the second sample?

**Solution:** In this we have to compare the variances of the samples. Therefore we apply F-distribution. From Table we find that  $F_{0.05} = 3.00$  for  $\nu_1 = 7 - 1 = 6$  and  $\nu_2 = 13 - 1 = 12$ ; thus, the desired probability is 0.05.

————— × —————

Why don't you try these exercises now.

E13) A random variable has the F distribution with 15 and 12 degrees of freedom.

 What is the value of this random variable that is exceeded with a probability of 0.5 with a probability of .01?

E14) If a random variable F has 15 and 23 degrees of freedom, what is the probability that it will exceed 2.93.

E15) If two independent random samples of size  $n_1 = 9$  and  $n_2 = 16$  are taken from a normal population, what is the probability that the variance of the first sample will be at least four times as large as the variance of the second sample?

E16) If independent random samples of size  $n_1 = n_2 = 8$  come from normal populations having the same variance, what is the probability that either sample variance will be at least seven times as large as the other?

In this unit you have only been introduced to these three specific distributions t,  $\chi^2$  and F. You will realise the utility of these distribution as you study the following units.

This brings us to the end of this unit. We have discussed the sampling distribution of a statistic at length. Let's now briefly recall the various concepts which we have covered here.

## 4.7 SUMMARY

In this unit we have explained the following concepts

- 1) i) Parameter - Values that describe the characteristic of a population.

- ii) Statistic - Measures describing the characteristic of a sample.
  - iii) Random sample - A sample from a population in which all the items in the population have an equal chance of being in the sample.
  - iv) Sampling distribution of a statistic - For a given population, a probability distribution of all the possible values of a statistic may taken as for a given sample size.
  - v) Standard error - The standard deviation of the sampling distribution of a statistic.
- 2) We have discussed the central limit theorem.
- 3) We have introduced the following three specific distributions
- i) t-distribution
  - ii)  $\chi^2$ -distribution
  - iii) F-distribution
- and explained the use of these distributions as sampling distribution.

## 4.8 SOLUTIONS/ANSWERS

- E1) Female students enrolled in IGNOU BDP: Finite population
- E2) (ii) and (iii)
- E3) Suppose there are 500 employees working in a company and we are interested to find the average salary paid to them in a month, the average amount calculated from the enter employees is called parameter. On the other hand it is possible to estimate the average based on selected few employees. The average income calculated from sample is called statistic.
- E4) The sampling distribution is given in the following table:

Table 9

| Sample No. | Sample elements | Sample Means        |
|------------|-----------------|---------------------|
| 1          | 100, 200, 300   | 200                 |
| 2          | 100, 200, 400   | 233 (approximately) |
| 3          | 100, 300, 400   | 267 (approximately) |
| 4          | 200, 300, 400   | 300                 |

The grand mean is 250 and is equal to the population mean.

- E5) i) There are

$$\begin{aligned}
 & {}^6C_2 \\
 &= \frac{6 \times 5}{1 \times 2} \\
 &= \frac{30}{2} \\
 &= 15
 \end{aligned}$$

- ii) samples of size 2 are given in the following table.

Table 10

| Sample | $\bar{X}_i$ | Sample | $\bar{X}_i$ | Sample | $\bar{X}_i$ |
|--------|-------------|--------|-------------|--------|-------------|
| 54, 50 | 52          | 50, 52 | 51          | 52, 50 | 51          |
| 54, 52 | 53          | 50, 48 | 49          | 52, 52 | 52          |
| 54, 48 | 51          | 50, 50 | 50          | 48, 50 | 49          |
| 54, 50 | 52          | 50, 52 | 51          | 48, 52 | 50          |
| 54, 52 | 53          | 52, 48 | 50          | 50, 52 | 51          |

iii) The population mean, the grand mean,

$$\mu = 51.00$$

$$\bar{X} = \frac{\sum \bar{x}_i}{k} = 51.000$$

The grand mean is equal to the population mean.

E6)  $N = 200$

$$n = 50$$

$$\sigma = 22$$

$$SE(\bar{X}) = \sigma / \sqrt{n} \cdot \sqrt{\frac{N-n}{N-1}}$$

$$22 / \sqrt{50} \sqrt{\frac{200-50}{199}}$$

$$= 2.7012$$

$$\begin{aligned} \text{E7) a) } SE(p) &= \sqrt{\frac{250-50}{250-1} \times \frac{4}{50} \times \frac{46}{50}} \\ &= 0.03438. \end{aligned}$$

$$\begin{aligned} \text{b) } SE(p) &= \sqrt{\frac{7}{60} \times \frac{53}{60}} \\ &= 0.04144. \end{aligned}$$

E8)  $t_{0.01} = 3.143$

$$t_{0.025} = 2.069$$

$$t_{0.1} = 1.372$$

$$t_{0.005} = 2.756$$

E9) We find the value of the random variable

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

where  $\bar{x} = 0.5060$ ,  $\mu = 0.5$  and  $s = 0.0040$  and  $n = 10$ . Substituting, we get the value of  $t$  as  $t = 4.7434$ . From the table we get that the  $t$ -value for the parameter  $\nu = 9$  and  $\alpha = .05$  is 2.26 which is less than the calculate value  $t = 4.7434$ . Since the probability is less i.e. 0.05, we conclude that the process is out of control.

E10) From the table we find that corresponding to  $\alpha = 0.5$  and the parameter

a)  $\nu = 16$  the value is 1.75

b)  $\nu = 27$  the value is 1.70

c)  $\nu = 200$  is 1.645 (note that this value corresponds to the entry to the last row marked  $\infty$ ).

E11) From the table in the appendix, we find that the value of  $\chi^2_\alpha$  for  $\alpha = 0.5$  is

a) 24.996 for  $\nu = 15$  and

b) 32.671 for  $\nu = 21$ .

E12) We make use of the formula

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

Note that we are given that  $\sigma^2 = 42.5$  and  $n = 10$ . Let  $s_1$  and  $s_2$  denote the two sample standard deviations, then  $s_1^2 = 3.14$  and  $s_2^2 = 8.94$  corresponding  $\chi^2$  values are

$$\chi_1^2 = \frac{(n-1) \times 3.14}{42.5}$$

$$= \frac{9 \times 3.14}{42.5} = 0.664$$

$$\text{and } \chi_2^2 = \frac{9 \times 8.94}{42.5} = 1.893$$

To find the required approximate probability, it is enough to find the area  $\alpha_0$  given below.

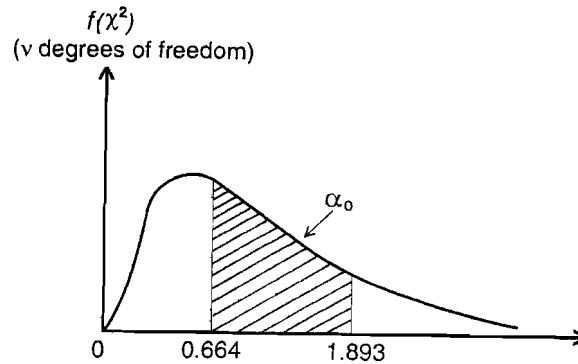


Fig. 10

From the table we find those values of  $\chi^2$  for 9 degrees freedom which are close to 0.664 and 1.893.

E13) Table 4 shows that the answer to the first question is 2.62 and Table 5 shows that the answer to the second question is 3.84.

E14) We first look at Table 4 and find that the given value is not there. Then it must be in Table 5 (check it!). Therefore the probability is 0.01.

E15) Solution: Here  $n_1 = 9$  and  $n_2 = 16 \therefore \nu_1 = 8$  and  $\nu_2 = 15$ . We have

$$F = \frac{s_1^2}{s_2^2} = 4$$

To find the probability we look for value  $F = 4$  in both the tables and find that it is in Table 5. Therefore the probability is 0.01.

E16)  $n_1 = n_2 = 8 \therefore \nu_1 = \nu_2 = 7$  and given that

$$F = \frac{s_1^2}{s_2^2} = 7$$

Proceeding similarly we find the value 6.99 in Table 5 which is almost equal to 7. Therefore the probability is 0.01.